MAT5210 - Galois Theory and Algebraic Number Theory

Preface

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Part of the notes is based on the undergraduate courses (B.3 - Galois Theory and B.9 - Algebraic Number Theory) at Oxford University taught by Prof. Alan Lauder in Michaelmas Term, 2001 and Hilary Term, 2002. Extra references are:

- Joseph Rotman - Galois Theory, Universitext, Springer (1998)
- David Tall and Ian Stewart - Algebraic Number Theory and Fermat's Last Theorem: Third Edition, A K Peters/CRC Press (2001)
- Kiran S. Kedlaya - Notes on class field theory (`https://kskedlaya.org/cft/`)

Please feel free to contact me (danielwong@cuhk.edu.cn) if you find any typos.

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Chapter 0

Preliminaries

0.1 Rings and Fields

In this section, we recall some results on commutative rings $(R, +, \cdot)$ with (multiplicative) identity $1 \in R$.

Definition 0.1.1. *Let R be commutative ring with identity.*

- *R is an* **integral domain (ID)** *if whenever $a \cdot b = 0$ for some $a, b \in R$, then $a = 0$ or $b = 0$.*
- *R is a* **field** *if for all $a \neq 0$ in R, there exists $a^{-1} \in R$ such that $a \cdot a^{-1} = 1$.*

It is obvious that if R is a field, then R is an integral domain. In fact, one has the following inclusion:

$$\text{Integral Domain (ID)} \supset \text{Unique Factorization Domain (UFD)} \supset \text{Principal Ideal Domain (PID)} \supset \text{Euclidean Domain (ED)} \supset \text{Fields}.$$

We refer the reader to basic abstract algebra (e.g. MAT3004) for details. In this course, we only consider commutative rings with identity, and all homomorphism of rings $\phi : R \to S$ must be unital, i.e. $\phi(1_R) = 1_S$.

Definition 0.1.2. *Let L, K be fields. We say $L : K$ is a* **field extension** *if there exists an injective ring homomorphism $\phi : K \hookrightarrow L$.*

In this course, we will often abuse notations and see ϕ as the 'inclusion' map. Note that L is a vector space over K, where the scalar multiplication is defined by

$$k \cdot \alpha := \phi(k)\alpha \qquad (k \in K,\ \alpha \in L)$$

So we can define the **degree** of field extension $L : K$ by

$$[L : K] = \dim_K(L).$$

We will always assume all field extensions $L : K$ are **algebraic**, that is, all $\alpha \in L$ are roots of polynomial in $K[x]$. This is always the case if $[L : K] < \infty$.

Definition 0.1.3. *Let $L : K$ be a field extension. The* **minimal polynomial** *of an element $\alpha \in L$ is the monic polynomial $m_\alpha(x) \in K[x]$ of smallest degree such that $m_\alpha(\alpha) = 0$.*

Since we assume $L : K$ is algebraic, $m_\alpha(x)$ exists for all $\alpha \in L$. Moreover, it is an irreducible polynomial in $K[x]$. Also, for all $f(x) \in K[x]$ such that $f(\alpha) = 0$, one can apply factor theorem to conclude that

$$m_\alpha(x) \mid f(x).$$

Definition 0.1.4. *Let K be a field. Then a field extension $\overline{K} : K$ is called the* **algebraic closure** *of K if every polynomial $p(x) \in K[x]$ splits in $\overline{K}[x]$, that is,*

$$p(x) = c\prod_{i=1}^{n}(x - \beta_i) \in \overline{K}[x].$$

Note that $\overline{K}$ always exists, and is unique up to isomorphism. As an example, the algebraic closure of $\mathbb{R}$ is $\overline{\mathbb{R}} = \mathbb{C}$ by the Fundamental Theorem of Algbera.

We end this section with the Primitive Element Theorem, which gives an effective way to reduce a field extension into a simple extension.

Theorem 0.1.5 (Primitive Element Theorem)**.** *Let K be such that* $\mathrm{char}(K) = 0$*, and let α, β be algebraic over K in a common field extension. Then there exists $d \in K$ such that*

$$K(\alpha + d\beta) = K(\alpha, \beta).$$

Proof. Let $p(x) \in K[x]$ be the minimal polynomial of α, and $q(x) \in K[x]$ be the minimal polynomial of β. Take a field extension $L : K$ containing α and β such that $p(x)$ and $q(x)$ both split in $L[x]$.

Write $\alpha = \alpha_1, \ldots, \alpha_m \in L$ as the distinct roots of $p(x)$ and $\beta = \beta_1, \ldots, \beta_n \in L$ as the distinct roots of $q(x)$. Take $d \in K$ such that for any $1 \leq i \leq m,\ 2 \leq j \leq n$,

$$d \neq \frac{\alpha_i - \alpha}{\beta - \beta_j}, \qquad (*)$$

which is possible since $\mathrm{char}(K) = 0$ so that K has infinitely many elements, while the RHS has only $mn < \infty$ elements (note that the elements on RHS may not even be in K).

Let $c := \alpha + d\beta$. Then one already has

$$K(c) \subseteq K(\alpha, \beta)\ (\subseteq L)$$

As for the other inclusion $K(\alpha, \beta) \subseteq K(c)$, consider

$$r(x) := p(c - dx) \quad \text{and} \quad q(x) \quad \text{in } K(c)[x],$$

so that $r(\beta) = p(c - d\beta) = p(\alpha) = 0$ and $q(\beta) = 0$ in $K(c)[x]$. Therefore, the minimal polynomial $m(x) \in K(c)[x]$ of β satisfies:

$$m(x) \mid r(x), q(x).$$

We study $m(x) \in K(c)[x] \subseteq L[x]$ further: Note that

$$m(x) \mid q(x) \text{ in } K(c)[x] \quad \Rightarrow \quad m(x) \mid q(x) = \prod_{j=1}^{n} (x - \beta_j) \text{ in } L[x],$$

so $m(x)$ can only have roots $\{\beta_1, \ldots, \beta_n\}$.

We *claim* that $\beta_1 = \beta$ is the only root of $m(x) \in K(c)[x]$: Suppose on contrary that there exists $j \neq 1$ such that β_j is a root of $m(x)$. Then it is also a root of $r(x)$, and hence

$$\begin{aligned} r(\beta_j) = p(c - d\beta_j) = 0 \Rightarrow \quad & c - d\beta_j = \alpha_i \\ \Rightarrow \quad & (\alpha + d\beta) - d\beta_j = \alpha_i \\ \Rightarrow \quad & d(\beta - \beta_j) = \alpha_i - \alpha \end{aligned}$$

which contradicts out our choice of d in $(*)$.

Consequently, the only root of $m(x)$ is $\beta_1 = \beta$. Since $m(x) \mid q(x)$ and $q(x)$ is separable, $m(x)$ is separable, so $m(x) = x - \beta \in K(c)[x]$. In other words, $\beta \in K(c)$ and one has

$$\alpha = c - d\beta \in K(c).$$

So one can conclude that $K(\alpha, \beta) \subseteq K(c)$ as we wish to prove. □

Corollary 0.1.6. *Let K be such that* $\mathrm{char}(K) = 0$. *Suppose $L : K$ is finite, then $L = K(\alpha)$ is a simple extension.*

Proof. Suppose $[L : K] = k$ and let $\{\beta_1, \ldots, \beta_k\}$ be a basis of L over K. Then $L = K(\beta_1, \ldots, \beta_k)$. One then applies Theorem 0.1.5 $k - 1$ times to get the result. □

Indeed, the above theorem holds in a more general setting when $L : K$ is a separable extension (Section 1.1) of finite degree. We will not prove it here.

0.2 Group Actions

Definition 0.2.1. *Let X be a set. The set of* ***automorphisms*** *of X is given by*

$$\mathrm{Aut}(X) = \{\phi : X \to X \mid \phi \text{ is bijective}\}$$

Note that $\mathrm{Aut}(X)$ *is automatically a group, whose binary operation is given by composition of functions, and the identity element is the identity map $id(x) := x$ for all $x \in X$.*

We can add some extra structures on the map we study in the above definition. For example, if X is a vector space over a field K, we define

$$\mathrm{Aut}_{\mathrm{Vect}}(X) = \{\phi \in \mathrm{Aut}(X) \mid \phi \text{ is a linear transformation}\}$$

If X is a group,

$$\mathrm{Aut}_{\mathrm{Group}}(X) = \{\phi \in \mathrm{Aut}(X) \mid \phi \text{ is a group homomorphism}\}$$

and one can define $\mathrm{Aut}_{\mathrm{Rings}}(X)$ similarly if X is a ring.

Definition 0.2.2. *Let G be a group. An* ***action*** *of G on X (S is a set) is a group homomorphism*

$$\phi : G \to \mathrm{Aut}(X).$$

In other words, for all $g \in G$, $\phi(g) \in \mathrm{Aut}(X)$ 'permutes' the elements of X. For the rest of the notes, we will write

$$g \cdot x := \phi(g)(x) \in X.$$

Example 0.2.3. *Let $G = S_3$, $X = \{a, b, c, d\}$. Then one can define $\phi : S_3 \to \mathrm{Aut}(X)$ by:*

$$\begin{array}{lllll}
e \cdot a = a & (12) \cdot a = b & (13) \cdot a = c & (23) \cdot a = a & (123) \cdot a = b \\
e \cdot b = b & (12) \cdot b = a & (13) \cdot b = b & (23) \cdot b = c & (123) \cdot b = c \\
e \cdot c = c & (12) \cdot c = c & (13) \cdot c = a & (23) \cdot c = b & (123) \cdot c = a \\
e \cdot d = d & (12) \cdot d = d & (13) \cdot d = d & (23) \cdot d = d & (123) \cdot d = d \\
 & & (132) \cdot a = c & & \\
 & & (132) \cdot b = a & & \\
 & & (132) \cdot c = b & & \\
 & & (132) \cdot d = d & &
\end{array}$$

Definition 0.2.4. *Let $\phi : G \to \mathrm{Aut}(X)$ be a group action.*

1. *The* ***invariants*** *of X over ϕ is the set*

$$X^G := \{x \in X \mid g \cdot x = x \quad \forall g \in G\}$$

2. *The* ***orbit*** *of $x \in X$ over ϕ is the set*

$$\mathrm{orb}(x) := \{g \cdot x \mid g \in G\} \subseteq X$$

3. *The* ***stabilizer*** *of $x \in X$ over ϕ is the set*

$$\mathrm{Stab}_G(x) := \{g \in G \mid g \cdot x = x\} \subseteq G$$

(Exercise: Check that $\mathrm{Stab}_G(x)$ is a subgroup)

Theorem 0.2.5 (Orbit - Stabilizer Theorem)**.** *Let $|G| < \infty$ and $\phi : G \to \mathrm{Aut}(X)$ is a group action. Then for all $x \in X$, one has:*

$$|\mathrm{orb}_G(x)| \cdot |\mathrm{Stab}_G(x)| = |G|$$

The proof is left as an exercise to the reader (Homework 1).

Example 0.2.6. *By our definition of $\phi : S_3 \to \mathrm{Aut}(X)$ with $X = \{a, b, c, d\}$ above, one has*

$$X^{S_3} = \{d\}, \quad \mathrm{orb}(a) = \mathrm{orb}(b) = \mathrm{orb}(c) = \{a, b, c\}, \quad \mathrm{orb}(d) = \{d\}, \quad \mathrm{Stab}_G(a) = \langle (23) \rangle$$

Therefore,

$$|\mathrm{orb}_G(a)| \cdot |\mathrm{Stab}_G(a)| = 3 \cdot 2 = 6, \qquad |\mathrm{orb}_G(d)| \cdot |\mathrm{Stab}_G(d)| = 1 \cdot 6 = 6$$

both are equal to $|S_3| = 6$.

0.3 Symmetric Polynomials

Let K be a field, and $p(x_1, x_2, \ldots, x_n) \in K[x_1, \ldots, x_n]$ be a polynomial with n variables.

Definition 0.3.1. *We say $p(x_1, x_2, \ldots, x_n) \in K[x_1, x_2, \ldots, x_n]$ is a* **symmetric polynomial** *if for all permutations $\sigma \in S_n$,*

$$p(x_1, x_2, \ldots, x_n) = p(x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(n)}) = p(x_1, x_2, \ldots, x_n).$$

For example, for all $k \geq 0$, $x_1^k + \cdots + x_n^k$ is a symmetric polynomial.

We can understand symmetric polynomial using group actions. Namely, let $X = K[x_1, x_2, \ldots, x_n]$, and $G = S_n$ acts on X by

$$\sigma \cdot p(x_1, x_2, \ldots, x_n) := p(x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(n)})$$

Then symmetric polynomials are precisely the polynomials that are fixed by the S_n-action, i.e.

$$K[x_1, x_2, \ldots, x_n]^{S_n} := \{p(x_1, x_2, \ldots, x_n) \mid \sigma \cdot p(x_1, x_2, \ldots, x_n) = p(x_1, x_2, \ldots, x_n)\ \forall \sigma \in S_n\}.$$

The following theorem gives a full description of all symmetric polynomials:

Theorem 0.3.2 (Fundamental Theorem of Symmetric Polynomials)**.** *All symmetric polynomials in $K[x_1, x_2, \ldots, x_n]$ are generated by the* **fundamental symmetric polynomials** $s_1, \ldots, s_n \in K[x_1, x_2, \ldots, x_n]^{S_n}$, *where*

$$\begin{aligned} s_1 &:= x_1 + x_2 + \ldots + x_n \\ &\vdots \\ s_i &:= \sum_{1 \le m_1 < m_2 < \cdots < m_i \le n} x_{m_1} x_{m_2} \ldots x_{m_i} \\ &\vdots \\ s_n &:= x_1 x_2 \ldots x_n. \end{aligned}$$

In other words, $K[x_1, x_2, \ldots, x_n]^{S_n} = K[s_1, s_2, \ldots, s_n]$.

For example, one has

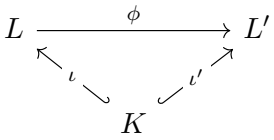
$$\begin{aligned} x_1^2 + \cdots + x_n^2 &= (x_1 + \cdots + x_n)^2 - 2(x_1x_2 + x_1x_3 + \cdots + x_{n-1}x_n) = s_1^2 - 2s_2 \\ x_1^3 + \cdots + x_n^3 &= s_1^3 - 3s_1s_2 + 3s_3. \end{aligned}$$

Chapter 1

Field Extensions

Galois Theory is the study on the roots of polynomials $p(x) \in K[x]$ over a field K. In order to get a sense on what is ahead of us, we begin with the following:

Definition 1.0.1. *Let L, L' be field extensions of K, with $\iota : K \hookrightarrow L$ and $\iota' : K \hookrightarrow L'$ are the corresponding injective ring homomorphisms. A* **homomorphism of K-extensions** *(shorthand as K-***homomorphisms** *from now on) is a ring homomorphism such that the diagram*



commutes, i.e. $\phi|_K = \iota'$.

By abuse of notations, we often treat ι and ι' as identity maps instead of injective maps. Also, we will shorthand the phrase 'homomorphisms of K-extensions' to 'K-homomorphisms' whenever it causes no confusions.

Proposition 1.0.2. *Let $\phi : L \to L'$ be a K-homomorphism. Then ϕ is injective.*

Proof. Suppose $\alpha \in L$ is such that $\phi(\alpha) = 0$. Consider its minimal polynomial $m_\alpha(x) = x^n + \cdots + a_1 x + a_0 \in K[x]$ such that

$$m_\alpha(\alpha) = \alpha^n + \cdots + a_1\alpha + a_0 = 0$$

Suppose on contrary $\alpha \neq 0$. Then the minimal polynomial $m_\alpha(x) \neq x$ must have $a_0 \neq 0$, otherwise it is reducible over K. By applying ϕ in the above equation:

$$\phi(\alpha^n + \cdots + a_1\alpha + a_0) = \phi(0) = 0$$

Since ϕ is a K-homomorphism, $\phi(a_i) = a_i$ for all $a_i \in K$ and one has:

$$0 = \phi(\alpha)^n + \cdots + \phi(a_1)\phi(\alpha) + \phi(a_0) = 0 + \cdots + a_1 \cdot 0 + a_0 = a_0,$$

contradicting the fact that $a_0 \neq 0$. Therefore, $\alpha = 0$, and therefore ϕ is injective. □

In the special case when $L' = L$, we have:

Definition 1.0.3. *Let $L : K$ be a field extension. Define $\mathrm{Aut}_K(L)$ by the collection of bijective K-homomorphisms $\phi : L \to L$, i.e.*

$$\mathrm{Aut}_K(L) := \{\phi : L \to L \mid \phi \text{ is bijective and } \phi|_K = id\}$$

Note that $\mathrm{Aut}_K(L)$ has a natural group structure by letting multiplication as the composition of two (bijective) homomorphisms, and identity element being the identity map $id : L \to L$.

We are now in the position to give a brief overview of Galois Theory – let $L = K(\alpha_1, \ldots, \alpha_n) : K$ be a splitting extension of $p(x) \in K[x]$ so that

$$p(x) = c(x - \alpha_1) \ldots (x - \alpha_n) \in L[x],$$

(See Section 1.3 for more details), one would like to study the roots α_i of $p(x)$ by understanding the structure of $L : K$. To do so, note that for any $\phi \in \mathrm{Aut}_K(L)$,

$$\phi(\alpha_i) = \alpha_{\sigma_\phi(i)}$$

permutes the roots of $p(x)$ (can add proof of it...). Therefore, the group $\mathrm{Aut}_K(L)$ encapsulates the structure of the roots of $p(x)$. In other words,

Study roots of $p(x) \in K[x]$ $\leftrightsquigarrow$ Study the group $\mathrm{Aut}_K(L)$ for splitting field $L : K$ of $p(x)$

1.1 Separable Extensions

Let K be a field and $p(x) \in K[x]$. Technically, we will only study $p(x) \in \overline{K}[x]$ with no repeated roots $\{\alpha_1, \ldots, \alpha_n\}$ with $\alpha_i \neq \alpha_j$ for all $i \neq j$. Under such setting, the permutations by $\mathrm{Aut}_K(L)$ on the α_i's are well-defined. So we need to test if $p(x)$ has no repeated roots.

Definition 1.1.1. *Let $p(x) = a_n x^n + \ldots + a_1 x + a_0 \in K[x]$.*

(a) The **formal derivative** *of $p(x)$ is $p'(x) = (n \cdot a_n)x^{n-1} + \cdots + (2 \cdot a_2)x + a_1 \in K[x]$.*

(b) We say $p(x)$ **has repeated roots** *if $\gcd(p(x), p'(x)) \neq 1$.*

Example 1.1.2. *Let $p(x) = c\prod_{i=1}^{n}(x - \alpha_i) \in K[x]$ be such that $p(x)$ splits in K. Then*

$$(p(x), p'(x)) \neq 1 \iff \exists i \neq j \text{s.t.} \alpha_i = \alpha_j.$$

To see why it is so, recall that the **Leibniz rule** *for formal derivative*

$$(uv)' = u'v + uv'.$$

Then we have

$$p'(x) = c\sum_{j=1}^{n}(x-\alpha_1)\dots\widehat{(x-\alpha_i)}\dots(x-\alpha_n).$$

If $p(x)$ has repeated roots $\alpha_i = \alpha_j$ for $i \neq j$, then each summand of $p'(x)$ in the above equation has a common factor of $(x-\alpha_i) = (x-\alpha_j)$, and hence

$$(x-\alpha_i) | \gcd_{K[x]}(p(x), p'(x)).$$

On the other hand, let all α_i's be all distinct. Suppose on contrary that $\gcd(p(x), p'(x)) \neq 1$. Since the irreducible factors of $p(x)$ are $(x-\alpha_i)$, there exists $1 \leq \ell \leq n$ such that

$$(x-\alpha_\ell) \mid p'(x)$$

and hence $p'(\alpha_\ell) = 0$ by Factor Theorem. However, by the equation of $p'(x)$ again,

$$p'(\alpha_\ell) = c\prod_{i\neq l}(\alpha_\ell - \alpha_i) \neq 0$$

since all α_i's are distinct, which leads to a contradiction.

Lemma 1.1.3. *Let $L : K$ be a field extension, and $p, q \in K[x]$. Then $\gcd_L(p,q) = \gcd_K(p,q)$.*

Proof. The Euclidean algorithm of finding $\gcd(p,q)$ is $K[x]$ and in $L[x]$ are the same. □

Therefore, one has:

Corollary 1.1.4. *Let $L : K$ be any field extension and $p(x) \in K[x]$. Then $p(x)$ has repeated roots in $K[x]$ if and only if $p(x)$ has repeated roots in $L[x]$.*

As a direct consequence of the above corollary, $p(x) \in K[x]$ has repeated roots if and only if in the algebraic closure $\overline{K}$ of K, the factorization

$$p(x) = c\prod_{i=1}^{n}(x-\alpha_i) \in \overline{K}[x]$$

of $p(x)$ satisfies $\alpha_i = \alpha_j$ for some $i \neq j$. For example, let $p(x) = x^2 + 1 \in \mathbb{R}[x]$. Then $(p, p') = (x^2+1, 2x) = 1 \implies p$ has no repeated roots in $\mathbb{R}[x]$. This implies $p(x) = (x-i)(x+i)$ has no repeated roots in $\mathbb{C}[x]$.

Lemma 1.1.5. *Let $p(x), q(x) \in K[x]$ be such that $q(x) \mid p(x)$. Suppose $p(x)$ has no repeated roots, then so is $q(x)$.*

Proof. Suppose on contrary that $\gcd(q, q') = r \neq 1$. Then $r \mid q \Rightarrow q = rw$ and $r \mid q' \Rightarrow q' = rv$. Write $p = s \cdot q$ for some $s \in K[x]$. Then Leibniz rule implies that

$$p' = s'q + sq' = s'(rw) + s(rv) = r(s'w + sv)$$

Then $r \mid \gcd(p, p')$, which contradicts the assumption that p has no repeated roots. □

Proposition 1.1.6. *Let $p(x) \in K[x]$ be irreducible. Suppose $\mathrm{char}(K) \nmid \deg(p)$ (for instance $\mathrm{char}(K) = 0$), then $p(x)$ has no repeated roots.*

Proof. Consider $p(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$, so that $\deg(p) = n$. Then $p'(x) = n a_n x^{n-1} + \cdots + a_1$. By hypothesis, $n a_n \neq 0$ in K, so p' is a nonzero polynomial.

Consequently $\gcd(p, p') = 1$ by irreducibility of p and the fact that $p' \neq 0$. □

Example 1.1.7. *$K = \mathbb{F}_2(t) = \{\frac{a(x)}{b(x)} \mid a(x),\ b(x) \in \mathbb{F}_2[t],\ b(t) \neq 0\}$ be the field of fractions of the polynomial ring $\mathbb{F}_2[t]$ with $\mathrm{char}(K) = 2$.*

Consider $p(x) = x^2 + t \in K[x]$ with $\deg(p) = 2 = \mathrm{char}(K)$. Then $p' = 2x = 0 \in K[x]$, therefore

$$\gcd(p, p') = \gcd(p, 0) = p$$

and hence p has repeated roots.

Definition 1.1.8. *Let $L : K$ be a field extension.*

*(a) A polynomial $p(x) \in K[x]$ is called **separable** if it has no repeated roots.*

*(b) $L : K$ is a **separable extension** if for all $\alpha \in L$, the minimal polynomial $m_\alpha(x) \in K[x]$ is separable.*

Example 1.1.9. *Let $K = \mathbb{F}_2(t)$. Consider $p(x) = x^2 + t \in K[x]$ and $L := K[x]/\langle p(x) \rangle$. Take $\alpha := x + \langle p(x) \rangle \in L$, so that*

$$m_\alpha(x) = p(x).$$

We have seen in Example 1.1.7 that $p(x)$ is not separable. Therefore, $L : K$ is not separable.

Proposition 1.1.10. *Let $M : L$ and $L : K$ be field extensions. Then*

$$M : K \text{ is separable} \quad \Rightarrow \quad \text{both } M : L \text{ and } L : K \text{ are separable.}$$

Proof. $L : K$ is obviously separable, since for any $\alpha \in L(\subseteq M)$, $m_\alpha(x) \in K[x]$ is separable.

As for separability of $M : L$, take $\beta \in M$ and consider $m_\beta^L(x) \in L[x]$ be the minimal polynomial of β in $K[x]$. We want to show $m_\beta^L(x)$ is separable, which has no repeated roots.

Since $M : K$ is separable, $m_\beta^K(x) \in K[x]$ has no repeated root. Moreover, since $m_\beta^K(\beta) = 0$ in $K[x] \subseteq L[x]$, which implies

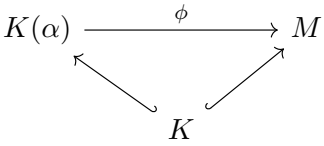
$$m_\beta^L(x) \mid m_\beta^K(x) \in L[x].$$

Then by Lemma 1.1.5, $m_\beta^L(x) \in L[x]$ also has no repeated roots. □

In fact, the converse of the above proposition also holds, but we do not need to use this fact.

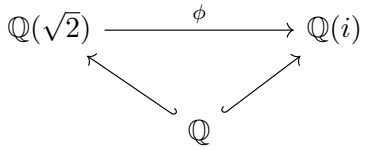
1.2 Simple Extensions

Proposition 1.2.1. *Let α be algebraic over K, with minimal polynomial $m_\alpha(x) \in K[x]$. Then for any field extension $M : K$, the set of K-homomorphisms*



is in one-to-one correspondence with the roots of $m_\alpha(x) \in M[x]$.

Example 1.2.2. *Consider the set of $\mathbb{Q}$-homomorphisms $\phi : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(i)$. By the proposition,*



$$\longleftrightarrow \quad \{\textit{roots of } x^2 - 2 \textit{ in } \mathbb{Q}(i)\}$$

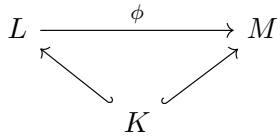
Since right hand side has no roots ($= \pm\sqrt{2} \notin \mathbb{Q}(i)$), there is no such $\phi : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(i)$ on the left hand side.

Proof. Recall that $K(\alpha) \cong K[x]/\langle m_\alpha(x)\rangle$, where the isomorphism maps $\alpha \mapsto \overline{x} := x + \langle m_\alpha(x)\rangle$. Then all K-homomorphisms $\phi : K(\alpha) \cong \frac{K[x]}{\langle m_\alpha(x)\rangle} \xrightarrow{\phi} M$ is uniquely determined by $\phi(\alpha) \in M$ (or equivalently $\phi(\bar{x}) \in M$). Since ϕ is K-homomorphism, one has:

$$m_\alpha(\phi(\bar{x})) = \phi(m_\alpha(\bar{x})) = \phi(0) = 0.$$

Conversely, if $\beta \in M$ is a root of $m_\alpha(x)$, then the map determined by $\alpha \mapsto \beta$ defines a K-homomorphism $K(\alpha) \to M$. Consequently, the correspondence is bijective. □

Theorem 1.2.3. *Let $\alpha_1, \ldots, \alpha_n$ be algebraic over K, and $L := K(\alpha_1, \ldots, \alpha_n)$. Suppose $M : K$ is such that $m_{\alpha_i}(x)$ splits in M for all i (e.g. $M = \overline{K}$), then the number of K-homomorphisms*



is non-zero and bounded above by $[L : K]$.

Moreover, if all $m_{\alpha_i}(x)$ are separable, then the number of K-homomorphisms is **exactly** *equal to $[L : K]$ (the converse of this statement is also true, but we do not need this fact).*

Proof. Put

$$F_i := K(\alpha_1, \ldots, \alpha_i), \qquad F_0 := K.$$

Suppose a K-homomorphism $\phi_{i-1} : F_{i-1} \to M$ has been constructed, and let $f_i(x) \in F_{i-1}[x]$ be the minimal polynomial of α_i over F_{i-1}. Since $f_i(x) \mid m_{\alpha_i}(x)$ in $F_{i-1}[x]$, applying ϕ_{i-1} to the coefficients gives

$$\phi_{i-1}(f_i)(x) \mid m_{\alpha_i}(x)$$

in $M[x]$. Hence $\phi_{i-1}(f_i)$ splits in M. By Proposition 1.2.1, the extensions

$$\phi_i : F_i \to M$$

of ϕ_{i-1} are in one-to-one correspondence with the distinct roots of $\phi_{i-1}(f_i)$ in M. Thus there is at least one extension, and there are at most

$$\deg f_i = [F_i : F_{i-1}]$$

of them.

Iterating this construction gives a non-zero number of K-homomorphisms $L \to M$, bounded above by

$$\prod_{i=1}^{n} [F_i : F_{i-1}] = [L : K].$$

If all $m_{\alpha_i}(x)$ are separable, then every $f_i(x)$, and hence every $\phi_{i-1}(f_i)(x)$, is separable. Therefore at each step the number of extensions is exactly $\deg f_i$, and the total number of K-homomorphisms is $[L : K]$. □

1.3 Splitting Extensions

Let $p(x) \in K[x]$. Recall that $p(x)$ splits in K if $p(x)$ can be factorized into linear terms, i.e.

$$p(x) = c \prod_{i=1}^{n} (x - \alpha_i) \text{ for some } c, \alpha_i \in K.$$

Definition 1.3.1. *A field extension $L : K$ is a **splitting extension** (or a **splitting field**) of $p(x) \in K[x]$ if:*

(a) $p(x)$ splits in L; and

(b) $L = K(\alpha_1, \alpha_2, \ldots, \alpha_n)$, where α_i are roots of $p(x)$.

One can interpret splitting field of $p(x) \in K[x]$ as a minimal field extension of K that splits $p(x)$. The question is whether this 'smallest' field extension is unique, which is addressed by the following:

Theorem 1.3.2. *Let $p(x) \in K[x]$. Then:*

(a) There exists a splitting extension of $p(x)$.

(b) The splitting extension is unique in the following sense: If $L : K$ and $M : K$ are splitting extensions of $p(x)$, then there is an isomorphism of K-extensions $L \cong M$.

Proof. (a) We apply induction on $\deg(p)$: If $\deg(p) = 1$, then one can simply take $M = K$.

By induction, suppose there exists a splitting extension for all polynomials of degree $n-1$. Now consider $p(x)$ with $\deg(p) = n$.

Take any irreducible polynomial $r(x) \mid p(x)$, and consider $M_1 = K[x]/\langle r(x)\rangle$. Then $M_1 : K$ is a field extension such that

- $M_1 = K(\alpha)$; and
- $\alpha := x + \langle r(x)\rangle \in M_1$ is a root of $r(x) \in M_1[x]$ (and hence α is also a root of $p(x) \in M_1[x]$).

Therefore,

$$p(x) = (x - \alpha)\widetilde{p}(x) \text{ in } M_1[x] \quad \text{with } \deg(\widetilde{p}) = n - 1 < n.$$

By induction, there exists the splitting extension $M = M_1(\beta_1, \ldots, \beta_{n-1}) : M_1$ such that:

$$\widetilde{p}(x) = c(x - \beta_1) \cdots (x - \beta_{n-1}) \quad \text{in } M[x]$$

Consequently, $M = M_1(\beta_1, \ldots, \beta_{n-1}) = K(\alpha)(\beta_1, \ldots, \beta_{n-1}) = K(\alpha, \beta_1, \ldots, \beta_{n-1})$ is a field extension of K such that

$$p(x) = c(x - \alpha)(x - \beta_1) \cdots (x - \beta_{n-1}) \text{ splits in } M[x]$$

In other words, $M : K$ is a splitting extension of $p(x) \in K[x]$.

(b) We apply induction on $\deg(p)$. As before if $\deg(p) = 1$, then $M = K$ is the only splitting extension of p. Now assume that the statement holds for all polynomials of degree $< n$, and let p be a polynomial of degree n.

Suppose M, L are splitting extensions of p, and let $\alpha \in M$ be a root of $p(x)$ with minimal polynomial $m_\alpha(x) \in K[x]$. Then $m_\alpha(x) \mid p(x)$. since $p(\alpha) = 0$.

Since $m_\alpha(x) \mid p(x)$ and L is a splitting extension of $p(x)$, L contains all the roots of $m_\alpha(x)$. Let $\beta \in L$ be a root of $m_\alpha(x)$ (which is also a root of $p(x)$). We then have $m_\alpha(x) = m_\beta(x) =: m(x)$, and:

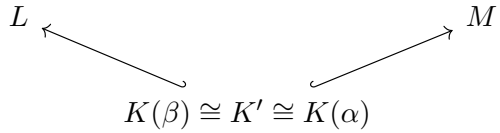
$$K(\alpha) \quad \cong \quad K' := \frac{K[x]}{\langle m(x) \rangle} \quad \cong \quad K(\beta)$$

with

$$\alpha \longleftrightarrow \ \kappa := x + \langle m(x) \rangle \ \longleftrightarrow \beta$$

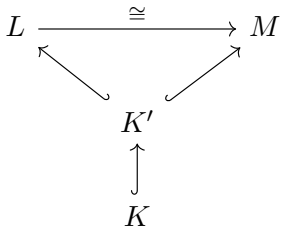
under the isomorphism above.

Now $M : K(\alpha)$ is a splitting extension of $\frac{p(x)}{x-\alpha} \in K(\alpha)[x]$ and $L : K(\beta)$ is a splitting extension of $\frac{p(x)}{x-\beta} \in K(\beta)[x]$. Under the isomorphism $K(\alpha) \cong K' \cong K(\beta)$, we have



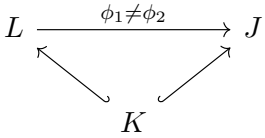
such that both M, L are splitting extensions of $\frac{p(x)}{x-\kappa} \in K'[x]$.

By induction hypothesis (since $\frac{p(x)}{x-\kappa}$ has degree 1 less than $p(x)$), we have:



In other words, the isomorphism $L \cong M$ above is a K-homomorphism, and the result follows. □

Theorem 1.3.3. *Let $L : K$ be a splitting extension of $p(x) \in K[x]$. Then for any extensions $J : K$, the image of any K-homomorphisms $\phi : L \to J$ coincide. In other words, for any*



one has $\phi_1(L) = \phi_2(L)$.

Proof. By Proposition 1.0.2, $\phi : L \hookrightarrow J$ is injective. Therefore, $J : K$ also splits $p(x)$, i.e

$$p(x) = c\prod_{i=1}^{n}(x - \beta_i) \in J[x].$$

On the other hand, since $L : K$ is a splitting extension of $p(x)$, one has $L = K(\alpha_1, \ldots, \alpha_n)$ and $p(x) = c\prod_{i=1}^{n}(x - \alpha_i) \in L[x]$. As before, all K-homomorphisms $\phi : L \to J$ are uniquely determined by the images $\phi(\alpha_1), \phi(\alpha_2), \ldots, \phi(\alpha_k) \in J$, and $\phi(\alpha_i) \in J$ must be the roots of $p(x) \in J[x]$, i.e.

$$\phi(\alpha_i) = \beta_j$$

for some j. Moreover, since ϕ is injective, one must have

$$\{\phi(\alpha_1), \ldots, \phi(\alpha_k)\} = \{\beta_1, \ldots, \beta_k\}$$

for any K-homomorphisms $\phi : L \to J$. Since the right hand side of the above equality $\{\beta_1, \ldots, \beta_k\}$ is independent of the choice of ϕ, the image $\phi(L)$ is independent of the choice of ϕ. □

1.4 Normal Extensions

Definition 1.4.1. *A field extension $L : K$ is normal if for all $\alpha \in L$, the minimal polynomial $m_\alpha(x) \in K[x]$ splits in L.*

Example 1.4.2. *1. $\mathbb{Q}(i)$ is normal: if $b = 0$, then $m_{a+bi}(x) = x - a$; if $b \neq 0$, then $m_{a+bi}(x) = (x - (a + bi))(x - (a - bi))$. In either case the minimal polynomial splits in $\mathbb{Q}(i)$.*

2. $\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}$ is not normal, since $m_{\sqrt[3]{2}}(x) = x^3 - 2 = (x - \sqrt[3]{2})(x^2 + \sqrt[3]{2}x + \sqrt[3]{4})$ and note that the quadratic term does not split in $\mathbb{Q}(\sqrt[3]{2})$.

Theorem 1.4.3. *Let $[L : K] < \infty$. Then*

$$L : K \text{ is normal} \iff L : K \text{ is the splitting extension of some } p(x) \in K[x].$$

Proof. ($\Rightarrow$) Suppose that $L : K$ is normal. Since $[L : K] < \infty$, choose a K-basis

$$\alpha_1, \ldots, \alpha_n$$

of L. For each i, let $m_{\alpha_i}(x) \in K[x]$ be the minimal polynomial of α_i over K. By normality, each $m_{\alpha_i}(x)$ splits in L.

Set

$$p(x) := \prod_{i=1}^{n} m_{\alpha_i}(x) \in K[x].$$

Then $p(x)$ splits in L. Let

$$p(x) = c\prod_{j=1}^{N}(x - \beta_j), \qquad \beta_j \in L.$$

Since each α_i is a root of $m_{\alpha_i}(x)$, every α_i occurs among the roots β_j. Hence

$$L = K(\alpha_1, \ldots, \alpha_n) \subseteq K(\beta_1, \ldots, \beta_N) \subseteq L.$$

Therefore

$$L = K(\beta_1, \ldots, \beta_N),$$

so L is the splitting extension of $p(x)$ over K.

($\Leftarrow$) Suppose that $L : K$ is the splitting extension of some $p(x) \in K[x]$. We must show that $L : K$ is normal.

Fix an arbitrary $\alpha \in L$, and let

$$m_\alpha(x) \in K[x]$$

be its minimal polynomial. Let $\beta \in \overline{K}$ be an arbitrary root of $m_\alpha(x)$. Since α and β have the same minimal polynomial over K, there exists a K-homomorphism

$$\phi : K(\alpha) \longrightarrow \overline{K}$$

such that

$$\phi(\alpha) = \beta.$$

By the extension theorem for field homomorphisms, ϕ extends to a K-homomorphism

$$\psi : L \longrightarrow \overline{K}$$

such that

$$\psi|_{K(\alpha)} = \phi.$$

In particular,

$$\psi(\alpha) = \beta.$$

Now let

$$\iota : L \hookrightarrow \overline{K}$$

be the natural inclusion. Since L is a splitting extension of $p(x) \in K[x]$, Theorem 1.3.3 implies that any two K-homomorphisms from L to $\overline{K}$ have the same image. Therefore

$$\psi(L) = \iota(L) = L.$$

Consequently,

$$\beta = \psi(\alpha) \in \psi(L) = L.$$

Since β was an arbitrary root of $m_\alpha(x)$, every root of $m_\alpha(x)$ lies in L. Hence $m_\alpha(x)$ splits in L.

Finally, since $\alpha \in L$ was arbitrary, $L : K$ is normal. □

Corollary 1.4.4. *Let L:K be splitting extension of a separable polynomial $p(x) \in K[x]$, then the group*

$$\mathrm{Aut}_K(L) := \left\{ \begin{array}{ccc} L & \xrightarrow{\quad\phi\quad} & L \\ & \nwarrow \quad \nearrow & \\ & K & \end{array} \right\}$$

has exactly $[L : K]$ elements.

Proof. Let $L = K(\beta_1, \cdots, \beta_n)$ is the splitting extension of a separable polynomial $p(x) = c\prod_{i=1}^{n}(x - \beta_i)$. Then L is normal by the previous theorem, and hence the minimal polynomial $m_{\beta_i}(x) \in K[x]$ splits in L. Also $m_{\beta_i}(x)$ is separable since $L : K$ is separable. So the result follows immediately from Theorem 1.2.3. □

Theorem 1.4.5. *Let $[L : K] < \infty$, then $|\mathrm{Aut}_K(L)| < \infty$. Moreover, TFAE:*

1. *L:K is the splitting field of a separable polynomial $p(x) \in K[x]$*
2. *$K = L^{\mathrm{Aut}_K(L)} := \{l \in L \mid \sigma(l) = l, \forall \sigma \in \mathrm{Aut}_K(L)\}$*
3. *$L : K$ is normal and separable.*

Proof. For the first statement of the theorem, suppose on contrary that $|\mathrm{Aut}_K(L)| = \infty$, then we have infinitely many $\sigma : L \to L$. By composing σ with the injection $\iota : L \to \overline{K}$, we have infinitely many $\iota \circ \sigma : L \to \bar{K}$

$$\begin{array}{ccc} L & \xrightarrow{\quad\iota\circ\sigma\quad} & \bar{K} \\ & \nwarrow \quad \nearrow & \\ & K & \end{array}$$

However, the number of K-homomorphisms $\phi : L \to \overline{K}$ is bounded above by $[L : K] < \infty$ by Theorem 1.2.3. Which is a contradiction.

For the second part of the theorem

- (1) $\Rightarrow$ (2): First note that by definition of $L^{\mathrm{Aut}_K(L)}$ we immediately have $K \subseteq L^{\mathrm{Aut}_K(L)}$, For the other inclusion, consider the field extension $L : L^{\mathrm{Aut}_K(L)}$. In order to show that $L^{\mathrm{Aut}_K(L)} = K$, it suffices to show that $[L^{\mathrm{Aut}_K(L)} : K] = 1$. By Tower Law, it is equivalent to proving $[L : K] = [L : L^{\mathrm{Aut}_K(L)}]$. Note that

$$[L : K] = |\mathrm{Aut}_K(L)|, \qquad [L : L^{\mathrm{Aut}_K(L)}] = |\mathrm{Aut}_{L^{\mathrm{Aut}_K(L)}}(L)|,$$

then we only need to show that

$$|\mathrm{Aut}_{L^{\mathrm{Aut}_K(L)}}(L)| = |\mathrm{Aut}_K(L)|.$$

Indeed, it is obvious that $\mathrm{Aut}_{L^{\mathrm{Aut}_K(L)}}(L) \subseteq \mathrm{Aut}_K(L)$ since $L^{\mathrm{Aut}_K(L)} \supseteq K$. On the other hand, if $\phi \in \mathrm{Aut}_K(L)$, then for any $\alpha \in L^{\mathrm{Aut}_K(L)}$, $\phi(\alpha) = \alpha$ by definition of invariants. So $\phi \in \mathrm{Aut}_{L^{\mathrm{Aut}_K(L)}}(L)$ as well. Therefore, we have proved that $\mathrm{Aut}_{L^{\mathrm{Aut}_K(L)}}(L) = \mathrm{Aut}_K(L)$ as required.

- (2) $\Rightarrow$ (3): Let $\alpha \in L$. We want to show that $m_\alpha(x)$ is separable and splits in L. Consider

$$\mathrm{orb}(\alpha) := \{\sigma(\alpha) \in L \mid \sigma \in \mathrm{Aut}_K(L)\} \qquad \text{and} \qquad q(x) := \prod_{\beta \in \mathrm{orb}(\alpha)} (x - \beta) \in L[x]$$

Suppose $\mathrm{orb}(\alpha) = \{\beta_1, \cdots \beta_d\}$, then

$$q(x) = x^d - s_1(\beta_1, \cdots, \beta_d)x^{d-1} + \cdots + (-1)^d s_d(\beta_1, \cdots, \beta_d),$$

where $s_i(\beta_1, \cdots, \beta_d)$ are the fundamental symmetric polynomials of degree i (Section 0.3). Note that $\forall \sigma \in \mathrm{Aut}_K(L)$, σ permutes $\{\beta_1, \cdots \beta_d\}$, then

$$\sigma(s_i(\beta_1, \cdots, \beta_d)) = s_i(\sigma(\beta_1), \cdots, \sigma(\beta_d)) = s_i(\beta_1, \cdots, \beta_d)$$

and hence $q(x) \in L^{\mathrm{Aut}_K(L)}[x] = K[x]$ by hypothesis. Since $\alpha \in \mathrm{orb}(\alpha)$, α is a root of $q(x) \in K[x]$ and hence $m_\alpha(x) | q(x)$. Since $q(x)$ has no repeated roots and splits in L, so is $m_\alpha(x)$.

- (3) $\Rightarrow$ (1): Suppose $\alpha_1, \cdots, \alpha_n \in L$ is a basis of L over K, then $L = K(\alpha_1, \cdots, \alpha_n)$. Let $f_1(x), \ldots, f_r(x)$ be the distinct minimal polynomials among $m_{\alpha_1}(x), \ldots, m_{\alpha_n}(x)$, and set

$$p(x) := \prod_{j=1}^{r} f_j(x).$$

Since $L : K$ is separable, the distinct irreducible polynomials f_j are separable, so $p(x)$ is separable. Since $L : K$ is normal, every $f_j(x)$ splits in L. Moreover, every α_i is a root of $p(x)$, so the roots of $p(x)$ generate L. Hence $L : K$ is the splitting field of $p(x)$.

□

Chapter 2

Galois Extensions

2.1 Fundamental Theorem of Galois Theory

We begin this section by defining what it means for a field extension to be **Galois**:

Definition 2.1.1. *A field extension $L : K$ is* **Galois** *if $K = L^{\mathrm{Aut}_K(L)}$, the Galois group of a Galois extension is $\mathrm{Gal}(L/K) := \mathrm{Aut}_K(L)$.*

If $[L : K] < \infty$ (which we will assume to be the case for the rest of this course), then we may apply Theorem 1.4.5 to get three equivalent characterizations of Galois extensions.

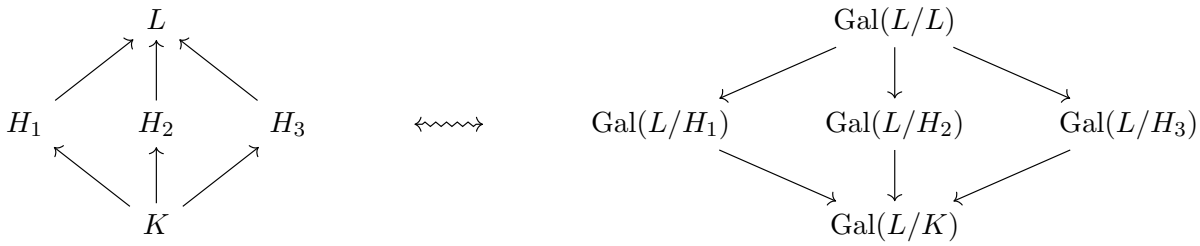
Example 2.1.2. *1. $\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}$ is not Galois, since it is not a normal extension by Example 1.4.2. Note that $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2})) = \{id\}$ (Exercise) has only one element, which is strictly less than $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.*

2. $\mathbb{Q}(i) : \mathbb{Q}$ is a normal extension, since it is the splitting extension of $p(x) = x^2 + 1 \in \mathbb{Q}[x]$. Moreover, by Corollary 1.4.4, $|\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})| = |\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(i))| = [\mathbb{Q}(i) : \mathbb{Q}] = 2$ has two elements, namely: $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q}) = \{id, \sigma\}$, where $id(a + bi) := a + bi$ and $\sigma(a + bi) := a - bi$ for $a + bi \in \mathbb{Q}(i)$.

We begin with an overview of Fundamental Theorem of Galois Theory: Suppose $L : K$ is Galois extension, the Fundamental Theorem of Galois Theory gives an 1-1 correspondence between

$$\{M \mid L \supseteq M \supseteq K\} \text{ and } \{H \mid H \leq \mathrm{Gal}(L/K)\}.$$

The relations between field extensions and Galois groups are shown in the following diagram:



Theorem 2.1.3 (Fundamental Theorem of Galois Theory, Part I)**.** *Let $L : K$ be Galois extension, then there's an order reversing bijection between the sets*

$$\{M \mid L \supseteq M \supseteq K\} \underset{\varphi}{\overset{\phi}{\rightleftarrows}} \{H \mid \{e\} \leq H \leq \mathrm{Gal}(L/K)\}$$

given by

$$\phi(M) := \mathrm{Gal}(L/M)$$
$$L^H =: \varphi(H)$$

Remark 2.1.4. *1. If $L : K$ is Galois, then for any $L \supseteq M \supseteq K$, $L : M$ is also Galois. To see why, $L : K$ is Galois implies $L : K$ is a splitting extension of some separable polynomial $p(x) \in K[x]$. Then the same polynomial is separable in $M[x]$ by Corollary 1.1.4. So L is the splitting extension of separable extension $p(x) \in M[x]$.*

2. For any $H \leq \mathrm{Gal}(L/K)$, if $\phi \in H$, then $\phi(k) = k$ for any $k \in K$. So $K \subseteq L^H$. The fact that L^H is a field is left as an exercise.

3. By order reversing, we mean

$$(M_1 \supseteq M_2) \mapsto (\mathrm{Gal}(L/M_1) \leq \mathrm{Gal}(L/M_2));$$

$$(L^{H_1} \supseteq L^{H_2}) \hookleftarrow (H_1 \subseteq H_2).$$

4. Since $L : M$ is Galois, we have $L^{\mathrm{Aut}_M(L)} = M$ for all $L \supseteq M \supseteq K$. Therefore, $\varphi \circ \phi(M) = M$, i.e. $\varphi \circ \phi = id$. So we just need to show $\phi \circ \varphi = id$ in Theorem 2.1.3.

Proof of Theorem 2.1.3. We only show $H = \mathrm{Gal}(L/L^H)$ for any $H \leq \mathrm{Gal}(L/K)$. We already have $H \leq \mathrm{Gal}(L/L^H)$ since for any $\sigma \in H$, $\sigma : L \to L$ satisfies $\sigma(\gamma) = \gamma$ for any $\gamma \in L^H$. So σ is an element in $\mathrm{Aut}_{L^H}(L) = \mathrm{Gal}(L/L^H)$.

To see it is an equality, we make the following:

Claim. *Let $L : K$ be of finite degree (which is not necessarily Galois), and $H \leq \mathrm{Aut}_K(L)$. Then $[L : L^H] \leq |H|$.*

Proof of Claim. Suppose on the contrary, i.e. $[L : L^H] > |H|$. Let $n = [L : L^H]$ and $d = |H|$. Let $\{\sigma_1, \ldots, \sigma_d\} = H \leq \mathrm{Aut}_K(L)$, and $\{\alpha_1, \ldots, \alpha_n\}$ be a basis of L over L^H. Set

$$v_1 = \begin{pmatrix} \sigma_1(\alpha_1) \\ \sigma_2(\alpha_1) \\ \vdots \\ \sigma_d(\alpha_1) \end{pmatrix}, v_2 = \begin{pmatrix} \sigma_1(\alpha_2) \\ \sigma_2(\alpha_2) \\ \vdots \\ \sigma_d(\alpha_2) \end{pmatrix}, \ldots, v_n = \begin{pmatrix} \sigma_1(\alpha_n) \\ \sigma_2(\alpha_n) \\ \vdots \\ \sigma_d(\alpha_n) \end{pmatrix} \in L^d$$

as n vectors in the d-dimensional vector space L^d over L. Since $n > d$, those vectors are linearly dependent over L. Assume σ_1 is the identity element in H.

Suppose "$r \leq n$" is the smallest integer such that there exists nontrivial linear relations between vectors above, and (after reordering the v_i's if necessary) there exists $\beta_1, \ldots, \beta_r \in L$ such that

$$\beta_1 v_1 + \cdots \beta_r v_r = 0.$$

WLOG, we may assume $\beta_r \neq 0$ here, then replace β_i by $\beta_i' = \beta_i/\beta_r$, one get

$$\beta_1' v_1 + \cdots \beta_{r-1}' v_{r-1} + v_r = 0. \tag{*}$$

for $\beta_1', \ldots, \beta_{r-1}' \in L$. Pick the first row in (*) , we have

$$\beta_1' \sigma_1(\alpha_1) + \cdots + \beta_{r-1}' \sigma_1(\alpha_{r-1}) + \sigma_1(\alpha_r) = 0,$$

which is

$$\beta_1' \alpha_1 + \cdots \beta_{r-1}' \alpha_{r-1} + \alpha_r = 0.$$

Since $\{\alpha_1, \ldots, \alpha_n\}$ forms a basis of L as an L^H vector space, they are L^H-linearly independent. So at least one β_i' is not in L^H. WLOG, assume $\beta_1' \notin L^H$, that means there exists an element $\tau \in H$ such that $\tau(\beta_1') \neq \beta_1'$.

Pick the i-th row in (*)

$$\beta_1' \sigma_i(\alpha_1) + \cdots + \beta_{r-1}' \sigma_i(\alpha_{r-1}) + \sigma_i(\alpha_r) = 0,$$

apply τ to the equation,

$$\tau(\beta_1')\tau\sigma_i(\alpha_1) + \cdots + \tau(\beta_{r-1}')\tau\sigma_i(\alpha_{r-1}) + \tau\sigma_i(\alpha_r) = 0.$$

Note $\{\tau\sigma_1, \ldots, \tau\sigma_d\} = \{\sigma_1, \ldots, \sigma_d\} = H$, so one has

$$\tau(\beta_1')\sigma_j(\alpha_1) + \cdots + \tau(\beta_{r-1}')\sigma_j(\alpha_{r-1}) + \sigma_j(\alpha_r) = 0.$$

for all $1 \leq j \leq d$. Combine all the rows together,

$$\tau(\beta_1')v_1 + \cdots + \tau(\beta_{r-1}')v_{r-1} + v_r = 0.$$

Subtract the above equation by $(*)$, one has:

$$(\tau(\beta_1') - \beta_1')v_1 + \cdots + (\tau(\beta_{r-1}') - \beta_{r-1}')v_{r-1} = 0.$$

with the first term $\tau(\beta_1') - \beta_1' \neq 0$. In other words, we find a smaller nontrivial linear relationship involving only "$r - 1 \leq n$" vectors, which contradict to the minimality of r. □

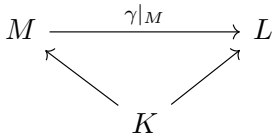
The claim gives $[L : L^H] \leq |H|$. Since $H \leq \mathrm{Aut}_{L^H}(L)$, while Theorem 1.2.3 gives $|\mathrm{Aut}_{L^H}(L)| \leq [L : L^H]$, we have

$$|H| \leq |\mathrm{Aut}_{L^H}(L)| \leq [L : L^H] \leq |H|.$$

Hence $H = \mathrm{Aut}_{L^H}(L)$. Since $L^{\mathrm{Aut}_{L^H}(L)} = L^H$, the extension $L : L^H$ is Galois and $H = \mathrm{Gal}(L/L^H)$. □

Theorem 2.1.5 (Fundamental Theorem of Galois Theory, Part II)**.** *Let $L : K$ be Galois extension, $M : K$ be a subextension of L. Then $M : K$ is Galois if and only if* $\mathrm{Gal}(L/M) \lhd \mathrm{Gal}(L/K)$*. In such case,* $\mathrm{Gal}(M/K) \cong \mathrm{Gal}(L/K)/\mathrm{Gal}(L/M)$*.*

Proof. ($\Rightarrow$): Suppose $M : K$ is Galois extension. Let $\gamma \in \mathrm{Gal}(L/K)$. We consider $\gamma|_M$:



since $M : K$ is splitting field of some separable polynomial, by Theorem 1.3.3 the image $\gamma|_M : M \to L$ are the same for all $\gamma \in \mathrm{Gal}(L/K)$. Pick the identity map, we have the result $\gamma|_M(M) = id_M(M) = M$, so $\gamma|_M \in \mathrm{Gal}(M/K)$.

Define the map

$$\Phi : \mathrm{Gal}(L/K) \to \mathrm{Gal}(M/K)$$

by

$$\gamma \mapsto \gamma|_M.$$

Φ is clearly a group homomorphism. $\Phi(\gamma) = id$ if and only if $\gamma|_M = id_M$, i.e. $\gamma \in \mathrm{Gal}(L/M)$, thus $\ker \Phi = \mathrm{Gal}(L/M)$. Hence $\mathrm{Gal}(L/M) \lhd \mathrm{Gal}(L/K)$.

Applying the first isomorphism theorem of groups to Φ, we get

$$\mathrm{Gal}(L/K)/\mathrm{Gal}(L/M) \cong \mathrm{im}\,\Phi \leq \mathrm{Gal}(M/K).$$

All those Galois groups are finite so we may count the cardinality of groups above:

$$|\mathrm{im}\,\Phi| = \frac{|\mathrm{Gal}(L/K)|}{|\mathrm{Gal}(L/M)|} = \frac{[L : K]}{[L : M]} = [M : K] = |\mathrm{Gal}(M/K)|\,,$$

so $\operatorname{im}\Phi = \operatorname{Gal}(M/K)$ and

$$\operatorname{Gal}(M/K) \cong \operatorname{Gal}(L/K)/\operatorname{Gal}(L/M).$$

($\Leftarrow$): Suppose $\operatorname{Gal}(L/M) \lhd \operatorname{Gal}(L/K)$. Take any $\gamma \in \operatorname{Gal}(L/K)$, then $\gamma(M)$ is a subfield of L. Then

$$\begin{aligned}\operatorname{Gal}(L/\gamma(M)) &= \{\mu \in \operatorname{Gal}(L/K) | \mu(x) = x, \forall x \in \gamma(M)\} \\ &= \{\mu \in \operatorname{Gal}(L/K) | \gamma^{-1}\mu\gamma(x) = x, \forall x \in M\}.\end{aligned}$$

So $\mu \in \operatorname{Gal}(L/\gamma(M))$ if and only if $\gamma^{-1}\mu\gamma \in \operatorname{Gal}(L/M)$ and

$$\operatorname{Gal}(L/\gamma(M)) = \gamma \operatorname{Gal}(L/M)\gamma^{-1}.$$

Since $\operatorname{Gal}(L/M)$ is normal, the equation above implies $\operatorname{Gal}(L/\gamma(M)) = \operatorname{Gal}(L/M)$. Therefore

$$\gamma(M) = L^{\operatorname{Gal}(L/\gamma(M))} = L^{\operatorname{Gal}(L/M)} = M.$$

Since $\gamma(M) = M$ for every $\gamma \in \operatorname{Gal}(L/K)$, we may define the restriction homomorphism

$$\Phi : \operatorname{Gal}(L/K) \longrightarrow \operatorname{Aut}_K(M), \qquad \gamma \longmapsto \gamma|_M.$$

with $\ker\Phi = \{\gamma \in \operatorname{Gal}(L/K) \mid \gamma|_M = \mathrm{id}_M\} = \operatorname{Gal}(L/M)$. Hence, by the First Isomorphism Theorem,

$$|\operatorname{im}\Phi| = \frac{|\operatorname{Gal}(L/K)|}{|\operatorname{Gal}(L/M)|}.$$

Since $L : K$ and $L : M$ are Galois,

$$|\operatorname{Gal}(L/K)| = [L : K], \qquad |\operatorname{Gal}(L/M)| = [L : M].$$

Therefore, by the Tower Law,

$$|\operatorname{im}\Phi| = \frac{[L : K]}{[L : M]} = [M : K].$$

By the equality established in the proof of Theorem 2.1.3,

$$[M : M^{\operatorname{im}\Phi}] = |\operatorname{im}\Phi| = [M : K].$$

where $K \subseteq M^{\operatorname{im}\Phi} \subseteq M$. Thus the Tower Law gives $[M^{\operatorname{im}\Phi} : K] = 1$ and hence

$$M^{\operatorname{im}\Phi} = K.$$

Finally, since $\operatorname{im}\Phi \leq \operatorname{Aut}_K(M)$, we have

$$M^{\operatorname{Aut}_K(M)} \subseteq M^{\operatorname{im}\Phi} = K.$$

On the other hand, every K-automorphism fixes K, so

$$K \subseteq M^{\operatorname{Aut}_K(M)}.$$

Consequently, $M^{\operatorname{Aut}_K(M)} = K$ and the result follows. □

Remark 2.1.6. *In the proof of Theorem 2.1.5, we also proved the following result:*

$$\boxed{M : K \text{ is Galois extension } \iff \forall \gamma \in \mathrm{Gal}(L/K), \gamma(M) = M}$$

Chapter 3

Some Calculations of $\mathrm{Gal}(L/K)$

3.1 First Examples

Proposition 3.1.1. *Let $L : K$ be the splitting field of a separable polynomial $p(x) \in K[x]$. Let $\alpha_1, \ldots \alpha_n \in L$ be the distinct roots of $p(x)$. Then the map $\Phi : \mathrm{Gal}(L/K) \to S_n$ defined by $\gamma \mapsto \Phi(\gamma)$ such that $\gamma(\alpha_i) = \alpha_{\Phi(\gamma)(i)}$ is an injective homomorphism.*

Proof. Φ is well-defined since all α_i are distinct and it is obviously a group homomorphism. Suppose $\Phi(\gamma) = e_{S_n}$, then $\gamma(\alpha_i) = \alpha_i$ for all i. But $\gamma : L = K(\alpha_1, \ldots, \alpha_n) \to L = K(\alpha_1, \ldots, \alpha_n)$ is uniquely determined by $\gamma(\alpha_1) = \alpha_1, \ldots, \gamma(\alpha_n) = \alpha_n$. So $\gamma = id_L$ and $\ker \Phi = \{id_L\}$. Consequently, Φ is injective. □

By abuse of notations, we will treat $\mathrm{Gal}(L/K)$ as a subgroup of S_n whenever L/K is the splitting field of some irreducible polynomial of degree n with no repeated roots.

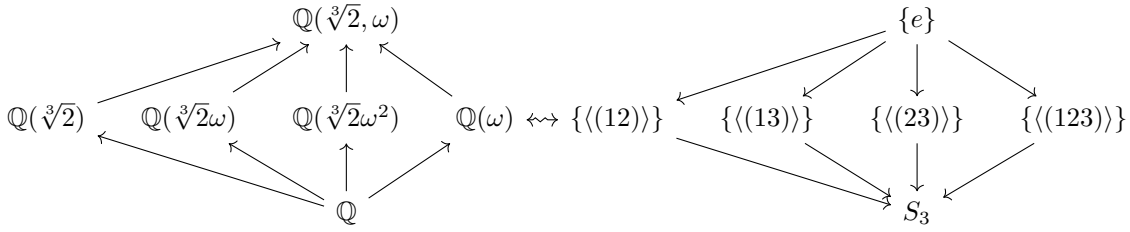
Example 3.1.2. *Let $x^3 - 2 \in \mathbb{Q}[x]$. Then $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2) = \mathbb{Q}(\sqrt[3]{2}, \omega)$ is a splitting field of $x^3 - 2$. So $\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}$ is Galois with $\left|\mathrm{Gal}(\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q})\right| = [\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = 6$. By Proposition 3.1.1, $\mathrm{Gal}(\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}) \subseteq S_3$. So $\mathrm{Gal}(\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}) = S_3$.*

Under Φ defined in Proposition 3.1.1, one can write down $\mathrm{Gal}(\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}) = S_3$ precisely. Namely, let

$$\begin{pmatrix} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \omega \mapsto \omega^2 \end{pmatrix} \leftrightarrow (12), \qquad \begin{pmatrix} \sqrt[3]{2} \mapsto \sqrt[3]{2}\omega \\ \omega \mapsto \omega \end{pmatrix} \leftrightarrow (123)$$

(note that the elements in $\mathrm{Gal}(\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q})$ on the left are of order 2 and 3 respectively).

By Theorem 2.1.3, the relation between subextensions and Galois groups are as follows:



Moreover, for $M = \mathbb{Q}(\sqrt[3]{2})$, $\mathrm{Gal}(L/M) = \{\gamma \in \mathrm{Gal}(L/\mathbb{Q}) | \gamma(\sqrt[3]{2}) = \sqrt[3]{2}\} = \{e, (12)\}$. $|\mathrm{Gal}(L/M)| = [L : M] = 2$.

For $H = \langle(123)\rangle \leq \mathrm{Gal}(L/\mathbb{Q})$, *one has* $L^H = \mathbb{Q}(\omega)$.

To verify Theorem 2.1.5, note that

$$H = \langle(123)\rangle \lhd S_3 \text{ is normal} \iff \mathbb{Q}(\omega)/\mathbb{Q} \text{ is Galois,}$$

$$\langle(12)\rangle \leq S_3 \text{ is not normal} \iff \mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q} \text{ is not Galois.}$$

3.2 Transitivity of $\mathrm{Gal}(L/K)$

Definition 3.2.1. *Let* $H \leq S_n$, *we say* H *is* **transitive** *if for any* $i, j \in \{1, \ldots, n\}$, $\exists \sigma \in H$ *such that* $\sigma(i) = j$.

In other words, if we treat H as an H-action on $S = \{1, \ldots, n\}$, H is transitive means $\mathrm{orb}(i) = S$ for all i.

Example 3.2.2. *Let* S_4 *act on* $\{1, 2, 3, 4\}$, *then* $\langle(1234)\rangle$ *is transitive but* $\langle(12)\rangle$, $\langle(12)(34)\rangle$ *are not.*

Theorem 3.2.3. *Let* $L : K$ *be a splitting extension of* $p(x) \in K[x]$ *with* $p(x)$ *separable and irreducible. Then* $\mathrm{Gal}(L/K) \leq S_n$ *(*$n = \deg p(x)$*) is transitive.*

Lemma 3.2.4. *Let* $L : K$ *be Galois,* $\alpha \in L$. *Then* $m_\alpha(x) \in K[x]$ *satisfies*

$$m_\alpha(x) = \prod_{\beta \in \mathrm{orb}(\alpha)} (x - \beta),$$

where $\mathrm{orb}(\alpha) = \{\gamma(\alpha) | \gamma \in \mathrm{Gal}(L/K)\}$.

Proof. Let $\mu(x) = \prod_{\beta \in \mathrm{orb}(\alpha)}(x - \beta) \in L[x]$. Then $\mu(x) \in L^{\mathrm{Gal}(L/K)}[x] = K[x]$ by the proof of (2) $\Rightarrow$ (3) in the proof of Theorem 1.4.5. Moreover, since $\alpha \in \mathrm{orb}(\alpha)$, one has $\mu(\alpha) = 0$. So

$$m_\alpha(x) \mid \mu(x).$$

We next prove $\mu(x)$ is irreducible. If not, write $\mu(x) = a(x) \cdot b(x)$ with $a(x), b(x) \in K[x]$ and $\deg a, \deg b < \deg \mu$. Since $\mu(\alpha) = 0$, after interchanging a and b if necessary, assume $a(\alpha) = 0$, then for any $\beta = \gamma(\alpha) \in \mathrm{orb}(\alpha)$,

$$a(\beta) = a(\gamma(\alpha)) = \gamma(a(\alpha)) = 0.$$

So β is also a root of $a(x)$, and hence $(x - \beta)|a(x)$ for all $\beta \in \mathrm{orb}(\alpha)$. However $\deg a < \deg \mu = |\mathrm{orb}(\alpha)|$, contradiction! We must have $\mu(x)$ irreducible and thus

$$m_\alpha(x) = \mu(x)$$

is the minimal polynomial. □

Proof of Theorem 3.2.3. Replacing $p(x)$ by its monic associate, we may assume that $p(x)$ is monic. Let $\alpha_1, \ldots, \alpha_n$ be all the roots of $p(x)$, so that $L = K(\alpha_1, \ldots, \alpha_n) : K$ and $p(x) = \prod_{i=1}^n (x - \alpha_i) \in L[x]$. Since $p(x)$ is irreducible in $K[x]$, $p(x)$ must be the minimal polynomial of each α_i, for any $1 \leq i \leq n$. Therefore,

$$\prod_{i=1}^{n} (x - \alpha_i) = p(x) = m_{\alpha_i}(x) = \prod_{\beta \in \mathrm{orb}(\alpha_i)} (x - \beta) \in L[x]$$

for all $1 \leq i \leq n$ by Lemma 3.2.4. By comparing the linear terms on the above equation, one concludes that $\mathrm{orb}(\alpha_1) = \mathrm{orb}(\alpha_2) = \cdots = \mathrm{orb}(\alpha_n) = \{\alpha_1, \ldots, \alpha_n\}$, i.e. Gal($L/K$) is transitive. □

3.3 Parity of Gal(L/K)

Theorem 3.3.1. *Let $p(x) \in K[x]$ and L/K as above. Consider*

$$\Delta = \Delta(\alpha_1, \ldots, \alpha_n) := \prod_{i<j} (\alpha_i - \alpha_j)^2,$$

where $\alpha_1, \ldots, \alpha_n$ are roots of $p(x)$. Then:

- *$\Delta \in K^*$, and it can be expressed in terms of coefficients of $p(x)$*
- *If $\mathrm{char}(K) \neq 2$, then*

$$\mathrm{Gal}(L/K) \leq A_n \iff \Delta \text{ is a square in } K^*, \text{ i.e. } \Delta = a^2 \text{ for some } a \in K^*$$

Example 3.3.2. *For $p(x) = x^3 - 2 \in \mathbb{Q}[x]$, we have $\Delta = -108 \notin \mathbb{Q}^{*2}$, hence*

$$\mathrm{Gal}(\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}) \not\leq A_3.$$

In fact, Gal($\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}$) $= S_3$.

Proof. **(1)** Note that Δ is symmetric in $\alpha_1, \cdots, \alpha_n$. Since $\mathrm{Aut}_K(L) \leq S_n$ by Proposition 3.1.1,

$$\Delta \in L^{\mathrm{Aut}_K(L)} = K.$$

By the Fundamental Theorem of Symmetric Polynomials over a field K,

$$\Delta \in K[s_1(\alpha_1, \alpha_2, ..., \alpha_n), ..., s_n(\alpha_1, \alpha_2, ..., \alpha_n)]$$

On the other hand, if $p(x) = c\prod_{i=1}^{n}(x - \alpha_i)$, then

$$\frac{p(x)}{c} = x^n - s_1(\alpha_1, \ldots, \alpha_n)x^{n-1} + \cdots + (-1)^n s_n(\alpha_1, \ldots, \alpha_n),$$

so every $s_i(\alpha_1, \ldots, \alpha_n)$ is determined by the coefficients of $p(x)$, and hence Δ can be expressed by the coefficients of $p(x)$.

(2) Consider

$$\sqrt{\Delta} := \prod_{i<j}(\alpha_i - \alpha_j) \in L$$

For all $\sigma \in \mathrm{Gal}(L/K)$, σ permutes the α_i's. So one has:

$$\sigma(\sqrt{\Delta}) = \mathrm{sgn}(\sigma)\sqrt{\Delta}$$

where $\mathrm{sgn}(\sigma)$ is the permutation signature (here we treat $\sigma \in S_n$). Therefore, when $\mathrm{char}(K) \neq 2$ (so that $1 \neq -1$ in K), one has:

$$\begin{aligned}
\sqrt{\Delta} \in K^* &\iff \sigma(\sqrt{\Delta}) = \sqrt{\Delta}\ \forall \sigma \in \mathrm{Gal}(L/K) \\
&\iff \mathrm{sgn}(\sigma) = 1\ \forall \sigma \in \mathrm{Gal}(L/K) \\
&\iff \mathrm{Gal}(L/K) \leq A_n
\end{aligned}$$

So the result follows. □

3.4 Cyclotomic Extension

In this section, we'll study $x^n - 1 \in K[x]$. Assume $\mathrm{char}(K) = 0$ throughout (ensuring separability).

Definition 3.4.1. *Let K be any field and $n \in \mathbb{N}$. The nth* **roots of unity in** *K are:*

$$\mu_n(K) := \{\xi \in K \mid \xi^n = 1\}$$

Example 3.4.2.

- $\mu_n(\mathbb{C}) = \{e^{2\pi i a/n}, 0 \leq a \leq n-1\}$
- $\mu_8(\mathbb{Q}(i)) = \{-1, +1, -i, +i\}$

Proposition 3.4.3. *$\mu_n(K)$ is a cyclic group.*

Proof. Note that $\mu_n(K)$ are roots of $x^n - 1$, so there are at most n elements in $\mu_n(K)$, hence $|\mu_n(K)| \leq n$. Obviously, $\mu_n(K)$ is an abelian group.

To see $\mu_n(K)$ is cyclic, we use the fact from MAT3004: For a finite abelian group G:

G is cyclic if and only if for all $d \mid \|G\|$, there exists a unique subgroup $H \leq G$ with $\|H\| = d$.

Suppose $|\mu_n(K)| = k$. For any $\delta \in \mu_n(K)$, $\delta^k = 1$, hence:

$$x^k - 1 = \prod_{\delta \in \mu_n(K)} (x - \delta)$$

For any $d \mid k$, $x^k - 1 \in K[x]$ can be factorized into:

$$x^k - 1 = (x^d - 1)\left(x^{k-d} + x^{k-2d} + \cdots + 1\right) = \left(\prod_{i=1}^{d}(x - \delta_i)\right) (\text{other terms})$$

Thus the roots $H := \{\delta \in K \mid \delta^d = 1\} = \{\delta_1, \delta_2, \ldots, \delta_d\}$ form a subgroup of $\mu_n(K)$ of order d. Moreover, for any $H' \leq \mu_n(K)$ with $|H'| = d$, the elements $\epsilon \in H'$ must satisfy $\epsilon^d = 1$ by Lagrange's Theorem. Hence $\epsilon \in H$ and $H' = H$ is unique. □

Remark 3.4.4. *By the above proposition, one has*

$$x^n - 1 \in K[x] \text{ splits over } K \quad \Leftrightarrow \quad |\mu_n(K)| = n \quad \Leftrightarrow \quad \mu_n(K) \cong \mathbb{Z}/n\mathbb{Z}.$$

For instance, since $\mathbb{Q}(i)$ does not *split $x^8 - 1$, so $\mu_8(\mathbb{Q}(i)) \cong \mathbb{Z}/4\mathbb{Z}$ but not isomorphic to $\mathbb{Z}/8\mathbb{Z}$.*

Definition 3.4.5. *Let K be a field, and $L : K$ be such that $\mu_n(L) \cong \mathbb{Z}/n\mathbb{Z}$ (for instance one can take the splitting extension $L : K$ of $x^n - 1 \in K[x]$). The* **primitive n^{th} roots of unity** *$\omega \in \mu_n(L)$ are the generators of the cyclic group satisfying $\mu_n(L) = \langle\omega\rangle$.*

Example 3.4.6. *The nth primitive roots of $\mu_n(\mathbb{C})$ are $e^{2\pi ia/n}$ for $\gcd(a, n) = 1$. More generally, if $\omega \in \mu_n(L)$ is primitive, then all other primitive roots of $\mu_n(L)$ are of the form*

$$\{\omega^a \mid \gcd(a, n) = 1\} = \{\omega^a \mid a \in \mathbb{Z}_n^*\}.$$

In other words, the nth primitive roots are parametrized by $\mathbb{Z}_n^*$.

Definition 3.4.7. *Let $L : K$ be such that $\mu_n(L) \cong \mathbb{Z}/n\mathbb{Z}$. Define the polynomial*

$$\Phi_{n,K}(x) := \prod_{\beta \text{ primitive root}} (x - \beta) = \prod_{a \in \mathbb{Z}_n^*} (x - \omega^a)$$

for any primitive n^{th} root $\omega \in \mu_n(L)$.

Example 3.4.8. *Let $K = \mathbb{Q}$. When p is prime,*

$$\Phi_p(x) = \prod_{d=1}^{p-1} \left(x - e^{2\pi i d/p}\right) = 1 + x + \cdots + x^{p-1}$$

When $n = 4$:

$$\Phi_4(x) = (x - i)(x + i) = x^2 + 1$$

And when $n = 6$:

$$\Phi_6(x) = (x - \omega)(x - \omega^5) = x^2 - x + 1 \quad (\omega = e^{2\pi i/6})$$

In fact, $\Phi_{n,K}(x)$ is independent of the choice of the splitting field extension $L : K$. This justifies our symbol $\Phi_{n,K}(x)$ (rather than $\Phi_{n,L}(x)$).

Lemma 3.4.9. *$\Phi_{n,K}(x) \in K[x]$, and is independent of the choice of $L : K$ satisfying $\mu_n(L) \cong \mathbb{Z}/n\mathbb{Z}$.*

Proof. For any such $L : K$, let $\omega \in \mu_n(L)$ be a primitive root of unity. Then we have a splitting field $K \subseteq K(\omega) \subseteq L$ where $K(\omega)$ is the splitting field of $x^n - 1 = \prod_{i \in \mathbb{Z}_n}(x - \omega^i)$. Therefore $K(\omega) : K$ is Galois, and

$$\Phi_{n,K}(x) = \prod_{a \in \mathbb{Z}_n^*} (x - \omega^a) \in K(\omega)[x].$$

Note that $\forall \sigma \in \mathrm{Gal}(K(\omega)/K)$, σ maps primitive root to primitive root, that is σ permutes the primitive roots of unity. Since the coefficients of $\Phi_{n,K}(x)$ are symmetric over the primitive roots, they are all fixed by $\sigma \in \mathrm{Gal}(K(\omega)/K)$. In other words, the coefficients of $\Phi_{n,K}$ are in $K(\omega)^{\mathrm{Gal}(K(\omega)/K)} = K$. If a different splitting field is chosen, uniqueness of splitting fields gives a K-isomorphism carrying primitive roots to primitive roots, so the same polynomial is obtained. □

Due to the importance of $K(\omega) : K$, we give the following:

Definition 3.4.10. *The splitting extension $K(\omega) : K$ of $x^n - 1 \in K[x]$ is called the nth* **cyclotomic extension** *of K.*

Theorem 3.4.11. *Let $K(\omega)/K$ be n^{th} cyclotomic extension. Then:*

1. *There exists an injective homomorphism*

$$\phi : \mathrm{Gal}(K(\omega)/K) \hookrightarrow \mathbb{Z}_n^\star$$

2. *ϕ is an isomorphism if and only if the cyclotomic polynomial $\Phi_{n,K}(x)$ is irreducible over K.*

Proof. 1. Since ϕ maps primitive roots to primitive roots, then $\sigma(\omega) = \omega^{a(\sigma)}$, where $a(\sigma) \in \mathbb{Z}_n^*$, we define:

$$\phi : \mathrm{Gal}(K(\omega)/K) \to \mathbb{Z}_n^\star \quad \text{by} \quad \phi(\sigma) := a(\sigma).$$

It is easy to check that this is an injective homomorphism.

2. ($\Rightarrow$) Suppose ϕ is an isomorphism, then $\mathrm{Gal}(K(\omega)/K) \simeq \mathbb{Z}_n^\star$, thus $\mathrm{orb}(\omega) = \{\omega^a \mid a \in \mathbb{Z}_n^\star\}$ consists all primitive roots (Example 3.4.6). So

$$\Phi_{n,K}(x) = \prod_{\beta \text{ primitive roots}} (x - \beta) = \prod_{\omega^a \in \mathrm{orb}(\omega)} (x - \omega^a) = m_\omega(x),$$

where the last equality comes from Lemma 3.2.4. But we know $m_\omega(x)$ is irreducible, so $\Phi_{n,K}(x)$ is also irreducible.
($\Leftarrow$) Suppose $\Phi_{n,K}(x)$ is irreducible, then as above, we have

$$\Phi_{n,K}(x) = m_\omega(x) = \prod_{\beta \text{ primitive roots}} (x - \beta) = \prod_{\omega' \in \mathrm{orb}(\omega)} (x - \omega')$$

then $|\,\mathrm{orb}(\omega)| = |\mathbb{Z}_n^\star|$. Since an automorphism of $K(\omega)$ fixing ω is the identity, the stabilizer of ω is trivial. Hence

$$|\,\mathrm{Gal}(K(\omega)/K)| = |\,\mathrm{orb}(\omega)| = |\mathbb{Z}_n^\star|.$$

Since ϕ is injective, it is an isomorphism.

□

For instance, if $K = \mathbb{C}$, then any $\Phi_{n,\mathbb{C}}(x)$ splits in $\mathbb{C}[x]$, i.e. $\Phi_{n,\mathbb{C}}(x)$ is reducible if its degree is > 1, and hence $\phi : \mathrm{Gal}(\mathbb{C}(\omega)/\mathbb{C}) \to \mathbb{Z}_n^*$ is almost never an isomorphism (which is obviously true since $[\mathbb{C}(\omega) : \mathbb{C}] = 1$. On the other extreme, if $K = \mathbb{Q}$, one has:

Theorem 3.4.12. *$\Phi_{n,\mathbb{Q}}(x) \in \mathbb{Z}[x]$ is always irreducible. Consequently,* $\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \cong \mathbb{Z}_n^\star$.

Proof. We need to show that $m_\omega(x) = \Phi_{n,\mathbb{Q}}(x)$. Note that $m_\omega(x) | x^n - 1$ in $\mathbb{Q}[x]$, so we have

$$x^n - 1 = m_\omega(x) \cdot t(x)$$

for some $t(x) \in \mathbb{Q}[x]$.

- Claim 1: $m_\omega(x), t(x) \in \mathbb{Z}[x]$
- Claim 2: For any prime p such that $\gcd(p, n) = 1$, $m_\omega(\omega^p) = 0$

Suppose the claims hold, then for any a such that $\gcd(a,n) = 1$, $a = p_1 p_2 \cdots p_n$ with $\gcd(p_i, n) = 1$. Then

$$\begin{aligned} m_\omega(\omega) = 0 &\Rightarrow m_\omega(\omega^{p_1}) = 0 \text{ (so } m_\omega = m_{\omega^{p_1}}) \\ &\Rightarrow m_\omega(\omega^{p_1 p_2}) = m_{\omega^{p_1}}((\omega^{p_1})^{p_2}) = 0 \\ &\vdots \\ &\Rightarrow m_\omega(\omega^{p_1 p_2 \cdots p_n}) = m_\omega(\omega^a) = 0 \end{aligned}$$

and then we conclude that $m_\omega(x)$ contains all the primitive roots ω^a for $\gcd(a,n) = 1$, i.e.

$$\Phi_{n,\mathbb{Q}}(x) \mid m_\omega(x)$$

and the equality comes from the fact that $m_\omega(x)$ is irreducible.

We left Claim 1 as an exercise (it uses the proof of Gauss Lemma), now we prove Claim 2 – Suppose on contrary that $m_\omega(\omega^p) \neq 0$. Since

$$m_\omega(\omega^p) t(\omega^p) = (\omega^p)^n - 1 = 0,$$

we have $t(\omega^p) = 0$, and hence $m_\omega(x) \mid t(x^p)$ in $\mathbb{Z}[x]$. Thus there exists $h(x) \in \mathbb{Z}[x]$ such that

$$m_\omega(x) h(x) = t(x^p).$$

Reducing modulo p gives

$$\overline{m_\omega}(x)\, \overline{h}(x) = \overline{t}(x)^p \qquad \text{in } \mathbb{F}_p[x].$$

Hence $\overline{m_\omega}(x)$ and $\overline{t}(x)$ have a nonconstant common factor $g(x)$. Since

$$x^n - 1 = \overline{m_\omega}(x)\, \overline{t}(x) \qquad \text{in } \mathbb{F}_p[x],$$

we get $g(x)^2 \mid x^n - 1$. But $\gcd(p,n) = 1$, so $x^n - 1$ has no repeated roots in $\mathbb{F}_p[x]$, a contradiction. □

3.5 Kummer Extension

In this section, we will study the polynomial $p(x) = x^n - a \in L[x]$, where L satisfies

- $\mathrm{char}(L) \nmid n$ (to ensure $p(x)$ has no repeated roots); and
- $x^n - 1$ splits in $L[x]$ (we will write $\omega \in L$ as an nth primitive root).

Assume $a \in L^\times$. Suppose $M : L$ is the splitting field of $x^n - a$, and let $\beta \in M$ be a root. Then all its other roots are given by $\omega^i \beta \in M$.

Lemma 3.5.1. *Let $M : L$ be the splitting field of $x^n - a \in L[x]$. For any root $\rho \in M$ of $x^n - a$, the map*

$$\psi : \mathrm{Gal}(M/L) \hookrightarrow \mu_n(L) \simeq \mathbb{Z}_n$$
$$\psi(\sigma) := \sigma(\rho)/\rho$$

is a well-defined injective group homomorphism.

Moreover, if $x^n - a$ is irreducible, then ψ is an isomorphism with $\mathrm{Gal}(M/L) \cong \mathbb{Z}_n$.

Proof. Firstly, we need to show $\sigma(\rho)/\rho \in \mu_n(L)$. Since

$$\left(\frac{\sigma(\rho)}{\rho}\right)^n = \frac{\sigma(\rho^n)}{\rho^n} = \frac{\sigma(a)}{a} = \frac{a}{a} = 1,$$

the last two equalities holds since $\sigma \in \mathrm{Gal}(M/L), a \in L$. We observe that $\sigma(\rho)/\rho \in M$ is a root of the polynomial $x^n - 1$. Since $x^n - 1 = \prod_{1 \le i \le n} (x - \omega^i) \in L[x] \subseteq M[x]$. Therefore $\sigma(\rho)/\rho = \omega^i \in \mu_n(L) \subseteq \mu_n(M)$.

Secondly, we need to verify our definition of ψ is independent of the choice of ρ – Suppose that $\chi \in M$ also satisfies $\chi^n = a$, then $\rho/\chi = \omega^i \in \mu_n(L)$. Since $(\rho/\chi)^n = a/a = 1$, we can compute

$$\frac{\sigma(\rho)}{\rho} = \frac{\sigma(\omega^i\chi)}{\omega^i\chi} = \frac{\sigma(\omega^i)\sigma(\chi)}{\omega^i\chi} = \frac{\omega^i\sigma(\chi)}{\omega^i\chi} = \frac{\sigma(\chi)}{\chi},$$

with $\theta' \in L$, the $\sigma(\rho)/\rho$ is well defined.

Thirdly, we need to check that ψ is a group homomorphism: For all $\sigma_1, \sigma_2 \in \mathrm{Gal}(M/L)$,

$$\begin{aligned}
& \psi(\sigma_1\sigma_2) = \psi(\sigma_1)\psi(\sigma_2) \\
\Leftrightarrow \quad & \frac{\sigma_1\sigma_2(\rho)}{\rho} = \frac{\sigma_1(\rho)}{\rho} \cdot \frac{\sigma_2(\rho)}{\rho} \\
\Leftrightarrow \quad & \sigma_1\sigma_2(\rho) = \frac{\sigma_1(\rho)\sigma_2(\rho)}{\rho} \\
\Leftrightarrow \quad & \frac{\sigma_1\sigma_2(\rho)}{\sigma_1(\rho)} = \frac{\sigma_2(\rho)}{\rho} \\
\Leftrightarrow \quad & \sigma_1\left(\frac{\sigma_2(\rho)}{\rho}\right) = \frac{\sigma_2(\rho)}{\rho}.
\end{aligned}$$

Note that $\sigma_2(\rho)/\rho \in \mu_n(L)$, so $\sigma_1 \in \mathrm{Gal}(M/L)$ fixes $\sigma_2(\rho)/\rho$ and hence the last equality holds.

Finally, to show ψ is injective, suppose that $\psi(\sigma) = 1 \in \mu_n(L)$, then $\sigma(\rho)/\rho = 1$ and hence $\sigma(\rho) = \rho$. For all roots $\{\rho\omega^i | 0 \le i \le n-1\}$ of polynomial $x^n - a$, $\sigma(\rho\omega^i) = \sigma(\rho)\sigma(\omega^i) = \rho\omega^i$, since $\omega^i \in L$. Therefore, σ fixes all roots of $x^n - a$, so it is the identity map on $M = L(\rho, \rho\omega, \ldots, \rho\omega^{n-1})$.

If $x^n - a$ is irreducible, then $M = L(\rho)$ and $[M : L] = n$. Also $|\mu_n(L)| = n$. Hence the injective homomorphism ψ is an isomorphism. □

The importance of the Lemma 3.5.1 is that its converse also holds:

Theorem 3.5.2. *Let L be such that $x^n - 1$ splits in L and $\mathrm{char}(L) \nmid n$. Suppose $M : L$ is a Galois extension such that $\mathrm{Gal}(M/L) = \langle\sigma\rangle \cong \mathbb{Z}_n$. Let $\omega \in L$ be a primitive n^{th} root of unity, then*

(a) There exists $\alpha \in M$ such that

$$\beta := \alpha + \omega\sigma(\alpha) + \cdots + \omega^{n-1}\sigma^{n-1}(\alpha) \neq 0.$$

(b) $b := \beta^n \in L$.

(c) $M = L(\beta)$ is a simple extension, with $m_\beta(x) = x^n - b$.

From now on, we call such $M : L$ satisfying the hypotheses of the above theorem a **Kummer extension**.

Proof. (a) is proved in Homework 2 Question 8. For (b), one can easily check that $\sigma(\beta) = \omega^{-1}\beta$ which implies

$$\sigma^i(\beta) = \omega^{-i}\beta$$

Also, $\sigma^i(\beta^n) = \big(\sigma^i(\beta)\big)^n = \omega^{-in}\beta^n = 1^{-i}\beta^n = \beta^n$. Therefore, β^n is fixed by $\langle\sigma\rangle = \mathrm{Gal}(M/L)$, i.e. $\beta^n \in M^{\mathrm{Gal}(M/L)} = L$.

For (c), Obviously, $L(\beta) \subseteq M$. Since $[M : L] = |\mathrm{Gal}(M/L)| = n$, by tower law, we just need to show $[L(\beta) : L] = n$. We study the minimal polynomial $m_\beta(x) \in L[x]$, by lemma 3.2.4

$$m_\beta(x) = \prod_{\gamma \in \mathrm{orb}(\beta)} (x - \gamma).$$

But Equation $(*)$ implies that $\mathrm{orb}(\beta) = \big\{\sigma^i(\beta)\big\} = \{\beta, \omega^{-1}\beta, \ldots, \omega^{-(n-1)}\beta\}$, then

$$m_\beta(x) = \prod_{i=0}^{n-1} \big(x - \sigma^i(\beta)\big) = \prod_{i=0}^{n-1} (x - \omega^{-i}\beta) = x^n - \beta^n = x^n - b$$

Therefore, $[L(\beta) : L] = \deg\left(m_\beta(x)\right) = n$. □

Before Theorem 3.5.2, we compute the Galois group $\mathrm{Gal}(M/L)$ of a given field extension $M : L$. However, Theorem 3.5.2 works in the opposite direction – we obtain the field extension $M : L$ from its Galois group $\mathrm{Gal}(M/L)$. This will play an important role in the study of roots of polynomials in the next section.

Chapter 4

Radical Extensions

4.1 Solvable Groups

Definition 4.1.1. *1. Let G be a finite group. A* **finite filtration** *is a finite sequence*

$$\{e\} \leq G_1 \leq G_2 \leq \cdots \leq G_n = G$$

such that $G_i \lhd G_{i+1}$ for all i.

2. We say a filtration has no redundancies if $G_i \lneqq G_{i+1}$ for all i.

3. We say a filtration has abelian quotients if G_{i+1}/G_i is abelian for all i.

4. We call $\{e\} \leq G$ the trivial filtration of G.

Definition 4.1.2. *A finite group G is called:*

1. **solvable** *if G has a finite filtration with abelian quotients;*

2. **simple** *if G has no nontrivial proper normal subgroups.*

Thus, the only filtration with no redundancies for a simple group G is $\{e\} \lhd G$.

Example 4.1.3. *1. S_3 is solvable: there is a filtration with no redundancies*

$$\overbrace{\{e\} \leq \underbrace{A_3}_{\mathbb{Z}_2} \leq S_3}^{\mathbb{Z}_3}.$$

2. S_4 is solvable: there is a filtration with no redundancies

$$\underbrace{\{e\} \leq \overbrace{\langle (12)(34) \rangle}^{\mathbb{Z}_2}}_{\mathbb{Z}_2} \leq \underbrace{K_4}_{\mathbb{Z}_3} \leq \overbrace{A_4 \leq S_4}^{\mathbb{Z}_2}.$$

3. *All finite abelian groups are solvable.*

4. *A_5 is simple. Since A_5 is nonabelian, A_5 is not solvable.*

Before entering the main content of this chapter, we highlight the idea of solving polynomial equations by radicals:

$$\boxed{\text{Roots of } p(x) \in K[x] \text{ are solvable by radicals } (+, -, \times, \div, \sqrt[n]{})}$$
$$\Updownarrow$$
$$\boxed{\text{The Galois group } \operatorname{Gal}(L/K) \text{ of } p(x) \text{ is solvable}}$$

Proposition 4.1.4. *Let G be a solvable group.*

1. *If $H \leq G$, then H is solvable.*

2. *If $H \lhd G$, then G/H is solvable.*

Proof. Let

$$\{e\} \leq G_1 \leq \cdots \leq G_n = G$$

be a filtration with abelian quotients.

1. Consider

$$\{e\} \leq G_1 \cap H \leq \cdots \leq G_n \cap H = H. \tag{$*$}$$

If $r \in G_{i+1} \cap H$ and $q \in G_i \cap H$, then

$$rqr^{-1} \in G_i \qquad \text{and} \qquad rqr^{-1} \in H.$$

Hence $G_i \cap H \lhd G_{i+1} \cap H$, so $(*)$ is a filtration of H.

Consider the homomorphism

$$\phi : G_{i+1} \cap H \longrightarrow G_{i+1}/G_i, \qquad r \longmapsto rG_i.$$

Its kernel is $G_i \cap H$. Therefore, by the First Isomorphism Theorem,

$$\frac{G_{i+1} \cap H}{G_i \cap H} \cong \operatorname{im}\phi \leq \frac{G_{i+1}}{G_i}.$$

Since G_{i+1}/G_i is abelian, so is $(G_{i+1} \cap H)/(G_i \cap H)$. Thus H is solvable.

2. Let
$$\pi : G \longrightarrow G/H, \qquad g \longmapsto gH.$$
Since $G_i \lhd G_{i+1}$, we have
$$\pi(G_i) \lhd \pi(G_{i+1}).$$
Hence
$$\{e\} \leq \pi(G_1) \leq \cdots \leq \pi(G_n) = G/H \qquad (**)$$
is a filtration of G/H.
The map
$$G_{i+1}/G_i \longrightarrow \pi(G_{i+1})/\pi(G_i), \qquad rG_i \longmapsto \pi(r)\pi(G_i),$$
is a well-defined surjective homomorphism. Therefore $\pi(G_{i+1})/\pi(G_i)$ is a quotient of the abelian group G_{i+1}/G_i, and hence is abelian. Thus G/H is solvable.

□

Proposition 4.1.5. *Let G be a finite group and $H \lhd G$. If H and G/H are solvable, then G is solvable.*

Proof. Let
$$\{e\} = H_0 \leq H_1 \leq \cdots \leq H_m = H$$
and
$$\{e\} = \overline{G}_0 \leq \overline{G}_1 \leq \cdots \leq \overline{G}_n = G/H$$
be filtrations with abelian quotients of H and G/H, respectively.

Let $\pi : G \to G/H$ be the quotient map, and put
$$G_i := \pi^{-1}(\overline{G}_i).$$
Then $G_0 = H$, $G_n = G$, and
$$H = G_0 \leq G_1 \leq \cdots \leq G_n = G$$
is a filtration. Moreover, by the Correspondence Theorem,
$$G_{i+1}/G_i \cong \overline{G}_{i+1}/\overline{G}_i,$$
so these quotients are abelian. Therefore
$$\{e\} = H_0 \leq H_1 \leq \cdots \leq H_m = H = G_0 \leq G_1 \leq \cdots \leq G_n = G$$
is a filtration of G with abelian quotients. Hence G is solvable. □

Example 4.1.6. *5. S_5 is not solvable since $A_5 \leq S_5$ is not solvable.*

6. For any $n \geq 5$, A_n and S_n are not solvable since $A_5 \leq A_n$ and $S_5 \leq S_n$.

Proposition 4.1.7. *Let G be a group such that $|G| = p^n$ for some prime p. Then G is solvable.*

Proof. We prove the proposition by induction on n.

When $n = 1$, $G \cong \mathbb{Z}_p$, so G is solvable.

Suppose the proposition holds for every group of order p^i with $1 \leq i < k$, and let $|G| = p^k$.

Claim. *The center $Z(G)$ is nontrivial.*

Proof of claim. Let G act on itself by conjugation:

$$h \cdot g = hgh^{-1}.$$

Its orbits are the conjugacy classes. By the Orbit–Stabilizer Theorem, every conjugacy class has cardinality a power of p. The conjugacy classes of size 1 are precisely the elements of $Z(G)$.

Write b_i for the number of conjugacy classes of cardinality p^i. Then

$$p^k = |G| = \sum_{i=0}^{k} b_i p^i.$$

Reducing modulo p gives

$$b_0 \equiv 0 \pmod{p}.$$

Since the identity already gives one conjugacy class of size 1, we have $b_0 \geq p$. Thus $|Z(G)| = b_0 > 1$. □

Since $Z(G)$ is abelian, it is solvable. Also,

$$|G/Z(G)| = p^{k'}$$

for some $k' < k$, so $G/Z(G)$ is solvable by induction. By Proposition 4.1.5, G is solvable. □

Definition 4.1.8. *The length of a finite group G is*

$$\text{length}(G) = \sup\{n \mid \{e\} \leq G_1 \leq \cdots \leq G_n = G \text{ is a filtration with no redundancies}\}.$$

Example 4.1.9. *For S_4, there is a filtration*

$$\underbrace{\{e\}}_{2} \leq \overbrace{\underbrace{\langle (12)(34) \rangle}_{2}}^{2} \leq \underbrace{K_4}_{3} \leq \overbrace{A_4}^{2} \leq S_4,$$

so $\text{length}(S_4) \geq 4$.

On the other hand, suppose

$$\{e\} = G_0 \lneqq G_1 \lneqq \cdots \lneqq G_n = S_4$$

is a filtration with no redundancies, and put $a_i = [G_i : G_{i-1}]$. *Then every* $a_i \geq 2$ *and*

$$a_1 a_2 \cdots a_n = |S_4| = 24 = 2^3 \cdot 3.$$

Hence $n \leq 4$. *Therefore* $\text{length}(S_4) = 4$.

Proposition 4.1.10. *Suppose* G *is solvable and*

$$\{e\} = G_0 \lneqq G_1 \lneqq \cdots \lneqq G_{\text{length}(G)} = G$$

is a filtration of maximal length. Then every quotient G_{i+1}/G_i *is cyclic of prime order.*

Proof. Fix i. By maximality of the filtration, the quotient G_{i+1}/G_i is simple. Indeed, if

$$\{e\} \lneqq H/G_i \lneqq G_{i+1}/G_i$$

were a proper nontrivial normal subgroup, then

$$G_i \lneqq H \lneqq G_{i+1}, \qquad G_i \lhd H \lhd G_{i+1},$$

and we could insert H into the filtration, contradicting maximality.

Since G is solvable, Proposition 4.1.4 implies that G_{i+1} is solvable, and hence G_{i+1}/G_i is solvable. Thus G_{i+1}/G_i is both simple and solvable.

A nontrivial finite simple solvable group must be abelian: otherwise a filtration with abelian quotients would contain a proper nontrivial normal subgroup. A finite simple abelian group is cyclic of prime order. Therefore

$$G_{i+1}/G_i \cong \mathbb{Z}_p$$

for some prime p. □

4.2 Solvability by Radicals

In this section, we assume $\text{char}(K) = 0$ for simplicity.

Definition 4.2.1. *An extension* $M : K$ *is called* **radical** *if*

1. $M = K(\alpha_1, \ldots, \alpha_k)$;

2. *there exist* $n_1, \ldots, n_k \in \mathbb{N}$ *such that*

$$\alpha_i^{n_i} \in K(\alpha_1, \ldots, \alpha_{i-1}) \qquad (1 \leq i \leq k),$$

where $K(\alpha_1, \ldots, \alpha_0) := K$.

Example 4.2.2. $\mathbb{Q}(\sqrt{2+\sqrt[3]{2}}) : \mathbb{Q}$ *is radical since*

$$\mathbb{Q}(\sqrt{2+\sqrt[3]{2}}) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{2+\sqrt[3]{2}}),$$

and

$$(\sqrt[3]{2})^3 \in \mathbb{Q}, \qquad (\sqrt{2+\sqrt[3]{2}})^2 \in \mathbb{Q}(\sqrt[3]{2}).$$

Definition 4.2.3. *Let* $p(x) \in K[x]$ *be separable, and let* $L : K$ *be its splitting field. We say* $L : K$ *is a* **solvable extension** *if there exists an extension* $M : L : K$ *such that* $M : K$ *is radical.*

Remark 4.2.4.
1. *If* $L : K$ *is solvable, then every root* $\alpha \in L$ *of* $p(x)$ *lies in* $M = K(\alpha_1, \ldots, \alpha_k)$*, and can therefore be expressed using the operations* $+, -, \times, \div, \sqrt[n]{}$ *starting from elements of* K.
2. *Cyclotomic and Kummer extensions are radical extensions.*
3. *If* $M' : M$ *and* $M : K$ *are radical, then* $M' : K$ *is also radical.*

Here is our **MAIN GOAL**: if $L : K$ is Galois, then

$$\boxed{\text{Gal}(L/K) \text{ is solvable} \iff L : K \text{ is a solvable extension.}}$$

We begin with the $\Rightarrow$ direction.

Lemma 4.2.5. *Let* $P : K$ *be a finite extension. Then there exists an extension* $J : P$ *such that* $J : K$ *is Galois.*

Proof. Since $\text{char}(K) = 0$, the Primitive Element Theorem gives

$$P = K(\delta)$$

for some $\delta \in P$. Let $m_\delta(x) \in K[x]$ be the minimal polynomial of δ, and let J be its splitting field in an algebraic closure containing P. Then $J : K$ is Galois and $P = K(\delta) \subseteq J$. $\square$

Theorem 4.2.6. *Let* $L : K$ *be Galois with*

$$|\,\text{Gal}(L/K)| = m.$$

Suppose $\text{Gal}(L/K)$ *is solvable. Let* ω *be a primitive* m*-th root of unity. Then there exists an extension* $P : L$ *such that:*

1. *$P : K$ is Galois;*
2. *P contains the cyclotomic extension $K(\omega)$;*
3. *$P : K(\omega)$ is a composition of Kummer extensions.*

Proof. Set

$$P := L(\omega).$$

Since $L : K$ and $K(\omega) : K$ are Galois, their compositum $P : K$ is Galois. This proves (1), while (2) is immediate from the definition of P.

Consider the restriction map

$$\rho : \mathrm{Gal}(P/K(\omega)) \longrightarrow \mathrm{Gal}(L/K), \qquad \sigma \longmapsto \sigma|_L.$$

It is well-defined because $L : K$ is Galois, so every K-automorphism of P preserves L. It is injective: if $\sigma|_L = \mathrm{id}_L$, then σ fixes both L and ω, hence fixes $P = L(\omega)$.

Thus $\mathrm{Gal}(P/K(\omega))$ is a subgroup of the solvable group $\mathrm{Gal}(L/K)$, and is therefore solvable by Proposition 4.1.4. By Proposition 4.1.10, there is a filtration

$$\{e\} = H_0 \lhd H_1 \lhd \cdots \lhd H_r = \mathrm{Gal}(P/K(\omega))$$

such that

$$H_{i+1}/H_i \cong \mathbb{Z}_{q_i}$$

for some prime q_i.

Let

$$P_i := P^{H_i}.$$

By the Fundamental Theorem of Galois Theory,

$$P = P_0 \supseteq P_1 \supseteq \cdots \supseteq P_r = K(\omega),$$

and, since $H_i \lhd H_{i+1}$,

$$\mathrm{Gal}(P_i/P_{i+1}) \cong H_{i+1}/H_i \cong \mathbb{Z}_{q_i}.$$

Since every q_i divides

$$|\mathrm{Gal}(P/K(\omega))|,$$

and this group injects into $\mathrm{Gal}(L/K)$, we have $q_i \mid m$. Therefore $K(\omega)$, and hence every P_{i+1}, contains all q_i-th roots of unity. Since

$$\mathrm{Gal}(P_i/P_{i+1}) \cong \mathbb{Z}_{q_i},$$

Theorem 3.5.2 implies that each

$$P_i : P_{i+1}$$

is a Kummer extension. Hence $P : K(\omega)$ is a composition of Kummer extensions. □

Corollary 4.2.7. *Let $L : K$ be Galois. If $\mathrm{Gal}(L/K)$ is solvable, then $L : K$ is a solvable extension.*

Proof. By Theorem 4.2.6, there is an extension $P : L$ such that $P : K(\omega)$ is a composition of Kummer extensions and $K(\omega) : K$ is cyclotomic. By Remark 4.2.4(2), both cyclotomic and Kummer extensions are radical, and by Remark 4.2.4(3), a composition of radical extensions is radical. Hence $P : K$ is radical, so $L : K$ is solvable. □

We now prove the $\Leftarrow$ direction. We first need a slightly stronger version of Lemma 4.2.5.

Lemma 4.2.8. *Let $P : K$ be a radical extension. Then there exists an extension $P' : P$ such that $P' : K$ is both radical and Galois.*

Proof. Write

$$P = K(\alpha_1, \ldots, \alpha_k), \qquad \alpha_i^{n_i} \in K(\alpha_1, \ldots, \alpha_{i-1}).$$

Let $J : P : K$ be a Galois closure as in Lemma 4.2.5, and write

$$\mathrm{Gal}(J/K) = \{\sigma_1 = e, \sigma_2, \ldots, \sigma_r\}.$$

Define

$$P' := K\big(\sigma_j(\alpha_i) \mid 1 \le j \le r,\ 1 \le i \le k\big).$$

Since $\sigma_1 = e$, we have $P \subseteq P'$.

We claim first that $P' : K$ is radical. Order the generators block by block:

$$\sigma_1(\alpha_1), \ldots, \sigma_1(\alpha_k), \sigma_2(\alpha_1), \ldots, \sigma_2(\alpha_k), \ldots.$$

For every j and i,

$$\big(\sigma_j(\alpha_i)\big)^{n_i} = \sigma_j(\alpha_i^{n_i}) \in K\big(\sigma_j(\alpha_1), \ldots, \sigma_j(\alpha_{i-1})\big),$$

which is contained in the field generated by the preceding elements in this ordering. Thus $P' : K$ is radical.

Finally, for every $\tau \in \mathrm{Gal}(J/K)$, left multiplication by τ permutes the elements σ_j of $\mathrm{Gal}(J/K)$. Hence

$$\tau(P') = P'.$$

By Remark 2.1.6, $P' : K$ is Galois. □

Theorem 4.2.9. *Let $L : K$ be Galois. If $L : K$ is a solvable extension, then $\mathrm{Gal}(L/K)$ is a solvable group.*

Proof. By hypothesis, there is an extension $P : L$ such that $P : K$ is radical. By Lemma 4.2.8, there exists $P' : P$ such that $P' : K$ is both radical and Galois.

Write

$$P' = K(\alpha_1, \ldots, \alpha_k), \qquad \alpha_i^{n_i} \in K(\alpha_1, \ldots, \alpha_{i-1}) \quad (1 \leq i \leq k).$$

Put

$$t := \prod_{i=1}^{k} n_i,$$

let ω be a primitive t-th root of unity, and set

$$E := P'(\omega).$$

Since $P' : K$ and $K(\omega) : K$ are Galois, their compositum $E : K$ is Galois.

For $0 \leq i \leq k$, define

$$E_i := K(\omega)(\alpha_1, \ldots, \alpha_i), \qquad E_0 = K(\omega).$$

Then

$$E_0 \subseteq E_1 \subseteq \cdots \subseteq E_k = E.$$

Since $n_i \mid t$, the polynomial $x^{n_i} - 1$ splits in E_{i-1}. Moreover,

$$\alpha_i^{n_i} \in K(\alpha_1, \ldots, \alpha_{i-1}) \subseteq E_{i-1},$$

so α_i is a root of

$$x^{n_i} - \alpha_i^{n_i} \in E_{i-1}[x].$$

Because E_{i-1} contains all n_i-th roots of unity, adjoining α_i adjoins every root of $x^{n_i} - \alpha_i^{n_i}$. Hence $E_i : E_{i-1}$ is Galois. By Lemma 3.5.1,

$$\mathrm{Gal}(E_i/E_{i-1}) \hookrightarrow \mu_{n_i}(E_{i-1}),$$

so $\mathrm{Gal}(E_i/E_{i-1})$ is cyclic, and in particular abelian. Thus $E_i : E_{i-1}$ is a Kummer extension (of degree dividing n_i). Put

$$G_i := \mathrm{Gal}(E/E_i), \qquad 0 \leq i \leq k.$$

Since $E_i : E_{i-1}$ is Galois, the Fundamental Theorem of Galois Theory gives

$$G_i \lhd G_{i-1} \qquad \text{and} \qquad G_{i-1}/G_i \cong \mathrm{Gal}(E_i/E_{i-1}),$$

so every quotient G_{i-1}/G_i is abelian. Also,

$$G_k = \{e\},$$

and, because $K(\omega) : K$ is cyclotomic,

$$\frac{\mathrm{Gal}(E/K)}{G_0} \cong \mathrm{Gal}(K(\omega)/K)$$

is abelian. Thus

$$\{e\} = G_k \lhd G_{k-1} \lhd \cdots \lhd G_0 \lhd \mathrm{Gal}(E/K)$$

is a filtration with abelian quotients. Hence $\mathrm{Gal}(E/K)$ is solvable.

Finally, $L \subseteq E$ and $L : K$ is Galois, so

$$\mathrm{Gal}(L/K) \cong \mathrm{Gal}(E/K)/\mathrm{Gal}(E/L).$$

Therefore $\mathrm{Gal}(L/K)$ is solvable by Proposition 4.1.4(2). □

4.3 Cubic Polynomials

We now give a general solution to any irreducible cubic equation

$$x^3 + ax^2 + bx + c = 0.$$

After the change of variable $x \mapsto x - a/3$, it is enough to consider a depressed cubic

$$f(x) = x^3 + px + q \in \mathbb{Q}[x].$$

Let $L : \mathbb{Q}$ be its splitting field. Since f is irreducible, transitivity implies

$$\mathrm{Gal}(L/\mathbb{Q}) = A_3 \qquad \text{or} \qquad S_3.$$

Let $\alpha_1, \alpha_2, \alpha_3 \in L$ be the three roots of $f(x)$, so

$$\alpha_1 + \alpha_2 + \alpha_3 = 0,$$

and let ω be a primitive cube root of unity. Put

$$P = L(\omega),$$

and define

$$\beta := \alpha_1 + \omega\alpha_2 + \omega^2\alpha_3, \qquad \gamma := \alpha_1 + \omega^2\alpha_2 + \omega\alpha_3.$$

Since both $L : \mathbb{Q}$ and $\mathbb{Q}(\omega) : \mathbb{Q}$ are Galois, their compositum $P : \mathbb{Q}$ is Galois.

We now compute $\beta^3 + \gamma^3$ and $\beta^3\gamma^3$. First,

$$\beta + \gamma = 2\alpha_1 + (\omega + \omega^2)(\alpha_2 + \alpha_3) = 3\alpha_1,$$

since $\omega + \omega^2 = -1$ and $\alpha_2 + \alpha_3 = -\alpha_1$. Also,

$$\begin{aligned}\beta\gamma &= \alpha_1^2 + \alpha_2^2 + \alpha_3^2 - (\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3) \\ &= -3p,\end{aligned}$$

where we used

$$\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3 = p$$

and

$$\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = -2p.$$

Therefore

$$\begin{aligned}\beta^3 + \gamma^3 &= (\beta + \gamma)^3 - 3\beta\gamma(\beta + \gamma) \\ &= 27(\alpha_1^3 + p\alpha_1) \\ &= -27q,\end{aligned}$$

and

$$\beta^3\gamma^3 = (\beta\gamma)^3 = -27p^3.$$

Thus β^3 and γ^3 are the roots of

$$T^2 + 27qT - 27p^3 = 0.$$

Hence

$$\boxed{\beta^3, \gamma^3 = \frac{-27q \pm \sqrt{729q^2 + 108p^3}}{2}.}$$

Choose cube roots β, γ of these two numbers compatibly so that

$$\beta\gamma = -3p.$$

Then

$$\alpha_1 = \frac{\beta + \gamma}{3}, \qquad \alpha_2 = \frac{\beta\omega^2 + \gamma\omega}{3}, \qquad \alpha_3 = \frac{\beta\omega + \gamma\omega^2}{3}.$$

This gives the roots of the cubic by radicals.

4.4 Insolvability of Quintic Polynomials

Let $f(x) \in K[x]$ be irreducible of degree 5, and let $L : K$ be its splitting field. Then

$$\mathrm{Gal}(L/K) \leq S_5.$$

To show that $f(x)$ is not solvable by radicals, it is enough to find an example for which the Galois group is not solvable; in particular, it is enough to find an example with Galois group S_5.

Theorem 4.4.1. *Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of prime degree p. If $f(x)$ has exactly $p-2$ real roots, then the splitting field L of $f(x)$ satisfies*

$$\mathrm{Gal}(L/\mathbb{Q}) \cong S_p.$$

Proof. Let

$$G := \mathrm{Gal}(L/\mathbb{Q}) \leq S_p.$$

Since $f(x)$ is irreducible of degree p, for any root α of f we have

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = p.$$

As

$$\mathbb{Q} \subseteq \mathbb{Q}(\alpha) \subseteq L,$$

the Tower Law gives

$$p \mid [L : \mathbb{Q}] = |G|.$$

By Cauchy's Theorem, G contains an element σ of order p. Since $G \leq S_p$, such an element must be a p-cycle.

By hypothesis, $f(x)$ has exactly two nonreal roots. Since f has real coefficients, these roots are complex conjugates. Complex conjugation preserves the splitting field L, fixes $\mathbb{Q}$, fixes the $p-2$ real roots, and interchanges the two nonreal roots. Hence its restriction to L gives a transposition

$$\tau \in G.$$

It remains to show that a p-cycle together with a transposition generates S_p. Relabel the roots so that

$$\sigma = (0\ 1\ \cdots\ p-1), \qquad \tau = (0\ d)$$

for some $d \not\equiv 0 \pmod{p}$. Then

$$\sigma^k \tau \sigma^{-k} = (k\ k+d) \qquad (k \in \mathbb{Z}/p\mathbb{Z}).$$

Since p is prime, d generates the additive group $\mathbb{Z}/p\mathbb{Z}$. Thus the graph with edges

$$\{k, k+d\}$$

is connected. The transpositions corresponding to the edges of a connected graph generate the full symmetric group. Hence

$$\langle \sigma, \tau \rangle = S_p.$$

Since $\sigma, \tau \in G \leq S_p$, we conclude that $G = S_p$. □

Example 4.4.2. *Consider*

$$f(x) = x^5 - 6x + 3.$$

By Eisenstein's criterion at 3, $f(x)$ *is irreducible over* $\mathbb{Q}$.

We claim that f *has exactly three real roots. First,*

$$f(-2) < 0 < f(-1), \qquad f(0) > 0 > f(1), \qquad f(1) < 0 < f(2),$$

so the Intermediate Value Theorem gives at least three real roots. On the other hand,

$$f'(x) = 5x^4 - 6$$

has exactly two real zeros. By Rolle's Theorem, f *can therefore have at most three real roots. Hence* f *has exactly three real roots.*

The theorem therefore gives

$$\mathrm{Gal}(L/\mathbb{Q}) \cong S_5.$$

Since S_5 *is not solvable,* $f(x)$ *is not solvable by radicals.*

Chapter 5

Algebraic Integers

5.1 Preliminaries

From this Chapter on, we will study some basics in Algebraic Number Theory. To begin with, we will focus on **number fields**:

Definition 5.1.1. *A number field K is a finite extension of $\mathbb{Q}$.*

Note that by Primitive Element Theorem (Theorem 0.1.5), there exists an element $\theta \in K$ such that $K = \mathbb{Q}(\theta)$ is a simple extension. Moreover, if $[K : \mathbb{Q}] = n$, then $K = \mathbb{Q}(\theta) = \mathrm{Span}_{\mathbb{Q}}\{1, \theta, \ldots, \theta^{n-1}\}$.

In Algebraic Number Theory, we would like to study the 'integers' of K defined as follows:

Definition 5.1.2. *Let $K : \mathbb{Q}$ be a number field. An element $\alpha \in K$ is an algebraic integer if α is a root of a monic polynomial $p(x) \in \mathbb{Z}[x]$.*

The collection of all algebraic integers of K is called the **ring of integers** $\mathcal{O}_K$.

It is easy to check (see Example 5.3.3 below) that the algebraic integers of $\mathbb{Q}$ are the usual integers $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$, and the algebraic integers of $\mathbb{Q}(i)$ are the Gaussian integers $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$.

We will prove in Theorem 5.3.5 below that $\mathcal{O}_K$ is always a ring (and hence an integral domain). The main question is the following:

Is $\mathcal{O}_K$ a PID? If not, how far is it away from being a PID?

By knowing whether $\mathcal{O}_K$ is PID or not, one can solve some Diophantine equations such as the **Pythagorean Equation**:

Find all integer solutions of $x^2 + y^2 = z^2$.

Instead of working in $\mathbb{Z}$, one can work on Gaussian integers $\mathbb{Z}[i]$, which is known to be a ED (and hence PID):

$$(x+yi)(x-yi) = z^2$$

By using unique factorization of $\mathbb{Z}[i]$, one can get a general solution of $x, y, z \in \mathbb{Z}$ (MAT3004).

Before we move on, we need some preliminary definitions and theorems:

Definition 5.1.3. *A* **free abelian group** *G of rank n with basis $e_1, \ldots, e_n \in G$ is defined by*

$$G = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \cdots \oplus \mathbb{Z}e_n = \left\{ \sum_{i=1}^{n} a_i e_i \mid a_i \in \mathbb{Z} \right\}.$$

Proposition 5.1.4. *Let G be a free abelian group of rank n with basis $\{e_1, \ldots, e_n\}$, and $M = (m_{ij})$ be an $n \times n$ matrix with $m_{ij} \in \mathbb{Z}$ such that $\det(M) \neq 0$. Suppose $f_1, \ldots, f_n \in G$ defined by $f_i := \sum_{j=1}^{n} m_{ij} e_j$. and*

$$H := \langle f_1, \ldots, f_n \rangle_{\mathbb{Z}} = \left\{ \sum_{i=1}^{n} \alpha_i f_i \mid \alpha_i \in \mathbb{Z} \right\}.$$

Then H is a subgroup of G, with

$$[G : H] = |\det(M)|$$

In particular, $f_1, \ldots, f_n$ is a basis of G if and only if $\det(M) = \pm 1$ (such $M \in M_{n\times n}(\mathbb{Z})$ are called **unimodular** *matrices).*

5.2 Discriminants, Norms and Traces

Theorem 5.2.1. *Let $K : \mathbb{Q}$ be a number field of degree n. Then there are n distinct injective homomorphisms $\sigma_i : K \hookrightarrow \mathbb{C}$.*

Proof. Let $K = \mathbb{Q}(\theta)$ for some $\theta \in K$ with minimal polynomial $m_\theta(x) \in \mathbb{Q}[x]$ of θ. Suppose $\theta_1, \theta_2, \ldots, \theta_n \in \mathbb{C}$ are the (distinct) roots of $m_\theta(x)$ in $\mathbb{C}$, then Proposition 1.2.1 implies that there are precisely n $\mathbb{Q}$-injections $\sigma_i : K = \mathbb{Q}(\theta) \hookrightarrow \mathbb{C}$ given by $\sigma_i(\theta) := \theta_i \in \mathbb{C}$. □

Example 5.2.2. *Let $K = \mathbb{Q}(\theta)$, where $\theta^3 = 2$. Then $[K : \mathbb{Q}] = 3$, and we have 3 injections $\sigma_i : K \to \mathbb{C}$ given by*

$$\sigma_1(\theta) = \sqrt[3]{2}, \quad \sigma_2(\theta) = \sqrt[3]{2}\omega, \quad \sigma_3(\theta) = \sqrt[3]{2}\omega^2,$$

where $\omega = e^{\frac{2\pi i}{3}}$ is the primitive cube root of unity in $\mathbb{C}$.

Definition 5.2.3. *Let $K = \mathbb{Q}(\theta)$, and $\sigma_i : K \to \mathbb{C}$ are the embeddings given in Theorem 5.2.1. The* **field polynomial** *of $\alpha \in K$ is defined as:*

$$F_\alpha(t) = \prod_{i=1}^{n}(t - \sigma_i(\alpha))$$

Theorem 5.2.4. *The coefficients of $F_\alpha(t)$ are in $\mathbb{Q}$, i.e. $F_\alpha(t) \in \mathbb{Q}[t]$.*

Proof. Let $K = \mathbb{Q}(\theta) = \mathrm{Span}_{\mathbb{Q}}(1, \theta, \ldots, \theta^{n-1})$. For any $\alpha \in K$, α can be expressed as $\alpha = a_0 + a_1\theta + \cdots + a_{n-1}\theta^{n-1}$ for some $a_i \in \mathbb{Q}$. Then

$$\sigma_i(\alpha) = \sigma_i(a_0 + a_1\theta + \cdots + a_{n-1}\theta^{n-1}) = a_0 + a_1\theta_i + \cdots + a_{n-1}\theta_i^{n-1}$$

where θ_i are roots of $m_\theta(x)$ in $\mathbb{C}$. Therefore,

$$F_\alpha(t) = \prod_{i=1}^{n}(t - \sigma_i(\alpha)) = t^n - s_1(\sigma_1(\alpha), \ldots, \sigma_n(\alpha))t^{n-1} + \cdots + (-1)^n s_n(\sigma_1(\alpha), \ldots, \sigma_n(\alpha))$$

where each coefficient $s_1(\sigma_1(\alpha), \ldots, \sigma_n(\alpha))$ of $F_\alpha(t)$ is symmetric over the roots $\theta_i \in \mathbb{C}$ of $m_\theta(x)$. By Theorem 0.3.2, these coefficients can be written as combinations of the symmetric polynomials $s_i(\theta_1, \theta_2, \ldots, \theta_n)$, i.e.

$$s_i(\sigma_1(\alpha), \ldots, \sigma_n(\alpha)) \in \mathbb{Q}[\theta_1, \theta_2, \ldots, \theta_n]^{S_n} = \mathbb{Q}[s_1(\theta_1, \theta_2, \ldots, \theta_n), \ldots, s_n(\theta_1, \theta_2, \ldots, \theta_n)].$$

On the other hand, each $s_i(\theta_1, \theta_2, \ldots, \theta_n)$ is the coefficient of x^{n-i} in $m_\theta(x) \in \mathbb{Q}[x]$, hence it is in $\mathbb{Q}$ and the result follows. □

Definition 5.2.5. *Let $[K : \mathbb{Q}] = n$ with $\sigma_i : K \to \mathbb{C}$ be the embeddings of K into $\mathbb{C}$.*

For any $\alpha \in K$:

(a) The **norm** *of $\alpha \in K$ is defined as $Norm_{K/\mathbb{Q}}(\alpha) := \prod_{i=1}^{n} \sigma_i(\alpha)$*

(b) The **trace** *of $\alpha \in K$ is defined as $Tr_{K/\mathbb{Q}}(\alpha) := \sum_{i=1}^{n} \sigma_i(\alpha)$.*

*(c) The K-**conjugates** of α are given by $\{\sigma_1(\alpha), \ldots, \sigma_n(\alpha)\}$.*

(note that in $Norm_{K/\mathbb{Q}}(\alpha)$, $Tr_{K/\mathbb{Q}}(\alpha) \in \mathbb{Q}$ by the theorem above).

For any $\omega := \{\omega_1, \ldots, \omega_n\} \subseteq K$:

(d) The **discriminant** *of ω is defined by $\Delta(\omega) := \det(\sigma_i(\omega_j))$.*

Lemma 5.2.6. $\Delta(\omega)^2 = \det(Tr_{K/\mathbb{Q}}(\omega_i\omega_j)) \in \mathbb{Q}$.

Proof. Let $A := (\sigma_i(\omega_j))$. Then

$$\begin{aligned}\Delta(\omega)^2 = \det(A^t A) &= \det\left(\sum_k (\sigma_i(\omega_j))^t_{ik}(\sigma_i(\omega_j))_{kj}\right)\\ &= \det\left(\sum_k \sigma_k(\omega_i)\sigma_k(\omega_j)\right)\\ &= \det\left(\sum_k \sigma_k(\omega_i\omega_j)\right)\end{aligned}$$

where the summand in the last term is equal to $\mathrm{Tr}_{K/\mathbb{Q}}(\omega_i\omega_j)$. □

Example 5.2.7. *Let $K = \mathbb{Q}(\theta), \omega = \{1, \theta, \ldots, \theta^{n-1}\}$ a basis of K over $\mathbb{Q}$. Let $\theta_1, \ldots, \theta_n \in \mathbb{C}$ be the roots of $m_\theta(x) \in \mathbb{Q}[x]$ as before, then*

$$\Delta(\omega)^2 = \det(\sigma_i(\theta^{j-1}))^2 = \det((\theta_i^{j-1})_{i,j})^2 = \det\begin{pmatrix}1 & \theta_1 & \ldots & \theta_1^{n-1}\\ 1 & \theta_2 & \ldots & \theta_2^{n-1}\\ \vdots & \vdots & \ddots & \vdots\\ 1 & \theta_n & \ldots & \theta_n^{n-1}\end{pmatrix}^2 = \prod_{1\le i<j\le n}(\theta_i - \theta_j)^2$$

Indeed, this is just the determinant of Vandermonde matrix. So we get back our definition of $\Delta(\theta_1, \ldots, \theta_n)^2 \in \mathbb{Q}$ in Homework 1.

Obviously, since $m_\theta(x)$ is separable, $\theta_i - \theta_j \neq 0$ for all $i \neq j$ and hence $\Delta(1, \theta, \ldots, \theta^{n-1})^2 \neq 0$. In Homework 4, you will prove that for any basis $\beta = \{\beta_1, \ldots, \beta_n\}$ of K (as a $\mathbb{Q}$-vector space), $\Delta(\beta_1, \ldots, \beta_n)^2 \neq 0$

5.3 Ring of Integers $\mathcal{O}_K$

Let's recall the definition of algebraic integers at the beginning of this section:

Definition 5.3.1. *Let $K : \mathbb{Q}$ be a number field.*

(a) An element $\alpha \in K$ is an **algebraic integer** *if there exists a monic polynomial $g(x) \in \mathbb{Z}[x]$ such that $g(\alpha) = 0$.*

(b) The **ring of integers** *of K is the collection of all algebraic integers, i.e.,*

$$\mathcal{O}_K := \{\alpha \in K \mid g(\alpha) = 0 \text{ for some monic } g \in \mathbb{Z}[x]\}.$$

Note that we have not justified the term *ring* of integers above yet. Later on, we will prove that $\mathcal{O}_K$ is indeed a ring (and hence an integral domain). Before doing so, here is an easier criterion to check whether an element $\alpha \in K$ is an algebraic integer or not:

Lemma 5.3.2. *Let $\alpha \in K$. Then $\alpha \in \mathcal{O}_K$ if and only if the minimal polynomial $m_\alpha(x) \in \mathbb{Q}[x]$ is in $\mathbb{Z}[x]$.*

Proof. ($\Leftarrow$) is automatic by taking $g(x) = m_\alpha(x)$.
($\Rightarrow$) Suppose $g(x) \in \mathbb{Z}[x]$ is monic such that $g(\alpha) = 0$. If $g(x)$ is irreducible over $\mathbb{Q}$, then $g = m_\alpha$, and we are done. Otherwise, by Gauss's lemma, $g(x) = h(x) \cdot r(x)$ is reducible in $\mathbb{Z}[x]$. By multiplying by -1 if necessary, both $h(x)$ and $r(x)$ are monic, with either $h(\alpha)$ or $r(\alpha) = 0$. So one can repeat the process on $h(x)$ or $r(x) \in \mathbb{Z}[x]$ until you hit the irreducible polynomial, which is the minimal polynomial $m_\alpha(x)$. □

Example 5.3.3. *Let $K = \mathbb{Q}$. Then by Lemma 5.3.2,*

$$\alpha \in \mathcal{O}_{\mathbb{Q}} \iff m_\alpha(x) = x - \alpha \in \mathbb{Z}[x] \iff \alpha \in \mathbb{Z}.$$

In fact, for any $K \supseteq \mathbb{Q}$, the same argument implies that $\mathbb{Z} \subseteq \mathcal{O}_K$.

Example 5.3.4. *We now study the ring of integers for quadratic extensions $K = \mathbb{Q}(\sqrt{d})$, where $d \in \mathbb{Z}$ is* **square-free**, *i.e. d doesn't have a power ≥ 2 in its prime factorization (e.g., $d = 12 = 2^2 \cdot 3$ is NOT square-free). For $\alpha = a + b\sqrt{d} \in K$ with $b \neq 0$, the minimal polynomial is*

$$m_\alpha(x) = x^2 - 2ax + (a^2 - db^2).$$

When $b = 0$, the conclusion below is immediate from $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$. Thus we may assume $b \neq 0$. Therefore, $\alpha \in \mathcal{O}_K$ iff $2a$, $a^2 - db^2 \in \mathbb{Z}$., and one has

$$(2a)^2 - d(2b)^2 \in \mathbb{Z} \implies d \cdot (2b)^2 \in \mathbb{Z}.$$

Since d is square-free, we must have $2b \in \mathbb{Z}$.

Let $A := 2a$ and $B := 2b \in \mathbb{Z}$. Since $a^2 - db^2 \in \mathbb{Z}$, multiplying by 4 gives $A^2 - dB^2 \in 4\mathbb{Z}$. Thus,

$$A^2 \equiv dB^2 \equiv 0, 1 \pmod 4 \quad (*)$$

We divide two cases for d, noting that $d \notin 4\mathbb{Z}$ since it is square-free.

Case (I): *If $d \equiv 2, 3 \pmod 4$, then $B^2 \equiv 0 \pmod 4$ (by (*)), and hence $B \in 2\mathbb{Z}$, which immediately implies b (as well as a) are both in $\mathbb{Z}$.*

Case (II): *If $d \equiv 1 \pmod 4$, then (*) implies that $A^2 \equiv B^2 \pmod 4 \implies A \equiv B \pmod 2$. Therefore, a and b are both in $\mathbb{Z}$ or both in $\mathbb{Z} + \frac{1}{2}$.*

Conclusion: *The above arguments gives a necessary condition for $\alpha = a + b\sqrt{d}$ to be in $\mathcal{O}_K$. On the other hand, if α satisfies the above conditions, then one can easily check that $m_\alpha(x) \in \mathbb{Z}[x]$. Consequently, one has:*

$$\mathcal{O}_K = \begin{cases} \langle 1, \sqrt{d} \rangle_{\mathbb{Z}} & \text{if } d \equiv 2, 3 \pmod 4, \\ \langle 1, \frac{1+\sqrt{d}}{2} \rangle_{\mathbb{Z}} & \text{if } d \equiv 1 \pmod 4, \end{cases}$$

where $\langle \alpha_1, \ldots, \alpha_k \rangle_{\mathbb{Z}} := \{\sum_{i=1}^k m_i \alpha_i \mid m_i \in \mathbb{Z}\}$.

Theorem 5.3.5. *$\mathcal{O}_K$ is a ring.*

Proof. Suppose $\alpha, \beta \in \mathcal{O}_K$ with minimal polynomials $m_\alpha(x) = x^n + \cdots + a_1 x + a_0$, $m_\beta(x) = x^m + \cdots + b_1 x + b_0$ in $\mathbb{Z}[x]$. We want to show that $\alpha \pm \beta,\ \alpha\beta \in \mathcal{O}_K$.

Consider the $\mathbb{Z}$-module $\mathbb{Z}[\alpha, \beta]$. Since α^n and β^m can be expressed as $\mathbb{Z}$-linear combination of terms of lower powers, then so does α^N and β^M for all $N > n$ and $M > m$. This implies the $\mathbb{Z}$-module $\mathbb{Z}[\alpha, \beta]$ is mn-generated:

$$\mathbb{Z}[\alpha, \beta] := \langle \alpha^i \beta^j \mid 0 \leq i \leq n-1, 0 \leq j \leq m-1 \rangle_{\mathbb{Z}}$$

(they may be linearly dependent over $\mathbb{Z}$). Now since $\mathbb{Z}$ is a PID, recall from MAT3042 on modules over PID, we know

$$\boxed{M \text{ is } k\text{-generated} \Rightarrow \text{all submodules } N \leq M \text{ are } k\text{-generated.}}$$

Now consider the submodule $\mathbb{Z}[\alpha + \beta] \leq \mathbb{Z}[\alpha, \beta]$. We can conclude that $\mathbb{Z}[\alpha + \beta]$ is mn-generated. Hence we have

$$\mathbb{Z}[\alpha + \beta] = \langle p_1(\alpha + \beta), p_2(\alpha + \beta), \cdots p_{mn}(\alpha + \beta) \rangle_{\mathbb{Z}}$$

for some polynomials $p_i(\alpha+\beta) \in \mathbb{Z}[\alpha+\beta]$. Consider an integer $t > \max\{\deg p_1, \cdots \deg p_{mn}\}$, then there exists $a_i \in \mathbb{Z}$ such that

$$(\alpha + \beta)^t = a_1 p_1(\alpha + \beta) + \cdots + a_{mn} p_{mn}(\alpha + \beta) \quad \in \mathbb{Z}[\alpha + \beta]$$

So the monic polynomial

$$g(x) = x^t - a_1 p_1(x) - \cdots - a_{mn} p_{mn}(x) \in \mathbb{Z}[x]$$

satisfies $g(\alpha + \beta) = 0$, i.e. $\alpha + \beta \in \mathcal{O}_K$. Similarly one can consider the $\mathbb{Z}$-modules $\mathbb{Z}[\alpha - \beta],\ \mathbb{Z}[\alpha\beta] \leq \mathbb{Z}[\alpha, \beta]$, to conclude that $\alpha - \beta, \alpha\beta \in \mathcal{O}_K$ □

Corollary 5.3.6. *Let $K : \mathbb{Q}$ be a number field, and $\sigma_i : K \hookrightarrow \mathbb{C}$ be the embeddings. If $\alpha \in \mathcal{O}_K$, then the field polynomial (Definition 5.2.3) $F_\alpha(t) := \prod_{i=1}^n (t - \sigma_i(\alpha))$ is in $\mathbb{Z}[t]$ (Recall in Theorem 5.2.4 that $F_\alpha(t) \in \mathbb{Q}[t]$ for every $\alpha \in K$). In particular,*

$$\mathrm{Tr}_{K/\mathbb{Q}}(\alpha), \mathrm{Norm}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}.$$

Proof. Consider the minimal polynomial $m_\alpha(x) \in \mathbb{Z}[x]$ of $\alpha \in \mathcal{O}_K$. Then for any $\sigma_i(\alpha) \in \mathbb{C}$, $\sigma_i(\alpha)$ is a root of $m_\alpha(x)$. Let $L : \mathbb{Q}$ be any algebraic extension containing $\sigma_1(\alpha), \cdots \sigma_n(\alpha) \in \mathbb{C}$, then $\sigma_i(\alpha) \in \mathcal{O}_L$. By the fact that $\mathcal{O}_L$ is a ring, we have

$$s_j(\sigma_1(\alpha), \cdots, \sigma_n(\alpha)) \in \mathcal{O}_L.$$

Meanwhile, we know $F_\alpha(t) = t^n - s_1(\sigma_1(\alpha), \cdots, \sigma_n(\alpha)) t^{n-1} + \cdots + (-1)^n s_n(\sigma_1(\alpha), \cdots, \sigma_n(\alpha)) \in \mathbb{Q}[t]$ and hence

$$s_j(\sigma_1(\alpha), \cdots, \sigma_n(\alpha)) \in \mathcal{O}_L \cap \mathbb{Q},$$

and hence it is in $\mathbb{Z}$. □

Corollary 5.3.7. *Let $\alpha, \beta \in \mathcal{O}_K$. Then the following holds:*

(a) α is a unit in $\mathcal{O}_K \Leftrightarrow \mathrm{Norm}_{K/\mathbb{Q}}(\alpha) = \pm 1$.

(b) If $\alpha \sim \beta$ are associates (i.e. $\alpha = u\beta$ for a unit $u \in \mathcal{O}_K$), then $\mathrm{Norm}_{K/\mathbb{Q}}(\alpha) = \pm \mathrm{Norm}_{K/\mathbb{Q}}(\beta)$.

(c) If $\mathrm{Norm}_{K/\mathbb{Q}}(\alpha) = p$ is a prime number, then α is irreducible (i.e. if $\alpha = \gamma\delta$ for some $\gamma, \delta \in \mathcal{O}_K$, then either γ or δ are units).

Proof. (a) ($\Rightarrow$) If α is a unit, then there exists $\alpha^{-1} \in \mathcal{O}_K$, and we have:

$$\mathrm{Norm}(\alpha) \cdot \mathrm{Norm}(\alpha^{-1}) = \mathrm{Norm}(\alpha\alpha^{-1}) = \mathrm{Norm}(1) = 1.$$

($\Leftarrow$) If $\mathrm{Norm}(\alpha) = \pm 1$, then

$$F_\alpha(t) = t^n + \cdots + a_1 t + (-1)^n \mathrm{Norm}(\alpha) = t^n + \cdots + a_1 t + (-1)^n \pm 1.$$

Note that for $\sigma_1 : K \hookrightarrow \mathbb{C}$, $\sigma_1(F_\alpha(\alpha)) = F_\alpha(\sigma_1(\alpha)) = F_\alpha(\alpha_1) = \prod_{i=1}^n (\alpha_1 - \alpha_i) = 0$. Since σ_1 is injective, one has $F_\alpha(\alpha) = 0$, and:

$$\alpha^n + \cdots + a_1\alpha = \mp 1, \quad \Rightarrow \quad \alpha\left(\mp(\alpha^{n-1} + \cdots + a_1)\right) = 1$$

and hence $\alpha^{-1} = \mp(\alpha^{n-1} + \cdots + a_1) \in \mathcal{O}_K$.

(b) This follows immediately from (a) and the fact that $\mathrm{Norm}_{K/\mathbb{Q}}(\gamma\beta) = \mathrm{Norm}_{K/\mathbb{Q}}(\gamma)\mathrm{Norm}_{K/\mathbb{Q}}(\beta)$ is multiplicative.

(c) Again, this follows from the multiplicativity of $\mathrm{Norm}_{K/\mathbb{Q}}$, and by Corollary 5.3.6 that $\mathrm{Norm}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$.

□

Note that the converse of (b) and (c) in the above Corollary are not true.

5.4 Integral Basis Theorem

Theorem 5.4.1 (Integral Basis)**.** *Let $K : \mathbb{Q}$ be a number field with $[K : \mathbb{Q}] = n$, and $\mathcal{O}_K$ its ring of integers. Then there exists $\omega_1, \ldots, \omega_n \in \mathcal{O}_K$ such that:*

1. *$Span_{\mathbb{Q}}\{\omega_1, \ldots, \omega_n\} = K$ (i.e. $\omega_1, \ldots, \omega_n$ is a basis of K as a vector space over $\mathbb{Q}$); and*
2. *$\langle \omega_1, \ldots, \omega_n \rangle_{\mathbb{Z}} = \mathcal{O}_K$ (i.e., $\omega_1, \ldots, \omega_n$ is a basis of $\mathcal{O}_K$ as a free module over $\mathbb{Z}$).*

Remark 5.4.2. *If $\theta_1, \ldots, \theta_n \in \mathcal{O}_K$ satisfies $\langle \theta_1, \ldots, \theta_n \rangle_{\mathbb{Z}} = \mathcal{O}_K$, then $\theta_1, \ldots, \theta_n$ is automatically a basis of K over $\mathbb{Q}$ – Suppose on contrary that $\{\theta_1, \ldots, \theta_n\}$ is not a basis. Then $\{\theta_1, \ldots, \theta_n\}$ must be linearly dependent and does not span K. In particular, there exists $\beta \in K \setminus \mathrm{Span}_{\mathbb{Q}}\{\theta_1, \ldots, \theta_n\}$.*

Claim: *$\exists M \in \mathbb{N}$ such that $M\beta \in \mathcal{O}_K$.*

Proof of Claim. Let $m_\beta(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 x + b_0 \in \mathbb{Q}[x]$ be the minimal polynomial. Then:

$$m_\beta(\beta) = \beta^k + b_{k-1}\beta^{k-1} + \cdots + b_1\beta + b_0 = 0.$$

Multiply by M^k:

$$(M\beta)^k + b_{k-1}M(M\beta)^{k-1} + \cdots + b_1 M^{k-1}(M\beta) + M^k b_0 = 0.$$

Choose M such that $b_{k-i}M^i \in \mathbb{Z}$ for all i. Then $g(x) = x^k + b_{k-1}Mx^{k-1} + \cdots + M^k b_0 \in \mathbb{Z}[x]$, so $M\beta \in \mathcal{O}_K = \langle \theta_1, \ldots, \theta_n \rangle_{\mathbb{Z}}$. □

Therefore, there exists integers $c_1, \ldots, c_n \in \mathbb{Z}$ such that

$$M\beta = c_1\theta_1 + \cdots + c_n\theta_n \quad \Rightarrow \quad \beta = \frac{c_1}{M}\theta_1 + \cdots + \frac{c_n}{M}\theta_n$$

In other words, $\beta \in Span_{\mathbb{Q}}\{\theta_1, \ldots, \theta_n\}$, a contradiction.

In view of the above remark, one only needs to prove statement (2) of Theorem 5.4.1.

Proof of Integral Basis Theorem: Let $\tilde{\omega}_1, \ldots, \tilde{\omega}_n$ be any basis of K over $\mathbb{Q}$. By multiplying $\tilde{\omega}_i$ by a large integer M_i (as in the Claim above), we get $\omega_i = M_i\tilde{\omega}_i \in \mathcal{O}_K$, and $\{\omega_1, \ldots, \omega_n\} \subseteq \mathcal{O}_K$ remains a basis of K over $\mathbb{Q}$. Consider:

$$\Delta^2(\omega) = \det\left(\mathrm{Tr}_{K/\mathbb{Q}}(\omega_i\omega_j)\right) \in \mathbb{Z}.$$

Among all bases $\underline{\omega} = \{\omega_1, \ldots, \omega_n\} \subseteq \mathcal{O}_K$ of K (as a vector space over $\mathbb{Q}$) -

Claim: The ones with minimal $\lvert\Delta^2(\omega)\rvert$ are integral bases of $\mathcal{O}_K$.

Suppose not. Then $\exists \beta \in \mathcal{O}_K \setminus \langle \omega_1, \ldots, \omega_n \rangle_{\mathbb{Z}}$. Then $\beta \in K = Span_{\mathbb{Q}}\{\omega_1, \ldots, \omega_n\}$, i.e.

$$\beta = \delta_1\omega_1 + \cdots + \delta_n\omega_n, \quad \delta_i \in \mathbb{Q}$$

with some $\delta_i \notin \mathbb{Z}$ (we assume $\delta_1 \notin \mathbb{Z}$ without loss of generality). By subtracting β by some element in $\langle \omega_1, \ldots, \omega_n \rangle_{\mathbb{Z}}$, i.e. some integer linear combination of $\omega_1, \ldots, \omega_n$, one obtains $\beta' \in \mathcal{O}_K \setminus \langle \omega_1, \ldots, \omega_n \rangle_{\mathbb{Z}}$ such that

$$\beta' = \delta_1'\omega_1 + \cdots + \delta_n'\omega_n, \quad 0 < |\delta_1'| \leq 1/2.$$

Replace ω_1 with β' to form a new basis $\underline{\omega}' = \{\beta', \omega_2, \ldots, \omega_n\}$, the change of basis matrix looks like $A := \begin{pmatrix} \delta_1' & & & \\ \delta_2' & 1 & & \\ \vdots & & \ddots & \\ \delta_n' & & & 1 \end{pmatrix}$, and hence discriminant becomes:

$$|\Delta^2(\omega')| = |\det(A)|^2 |\Delta^2(\omega)| = |\delta_1'|^2 |\Delta^2(\omega)| \leq \frac{1}{4} |\Delta^2(\omega)|,$$

(the first equality comes from Homework 4 Question 3 - $\Delta(\omega') = \det(A)\Delta(\omega)$). So it contradicts the minimality of $\Delta(\omega)$. □

Definition 5.4.3. *The discriminant of a number field $K : \mathbb{Q}$ is*

$$\Delta^2(K) := \Delta^2(\omega)$$

for any integral basis ω of $\mathcal{O}_K$.

Remark 5.4.4. *Let $\underline{\omega}$ be an integral basis of $\mathcal{O}_K$. Then for any $\underline{\theta} = \{\theta_1, \ldots, \theta_n\} \subseteq \mathcal{O}_K$ such that $\underline{\theta}$ spans K as a vector space over $\mathbb{Q}$, one has $\langle \underline{\theta} \rangle_{\mathbb{Z}} \leq \mathcal{O}_K = \langle \underline{\omega} \rangle_{\mathbb{Z}}$ as a $\mathbb{Z}$-module (or as abelian groups). Proposition 5.1.4 implies that the index of the subgroup is given by*

$$|\det(A)| = [\langle \underline{\omega} \rangle_{\mathbb{Z}} : \langle \underline{\theta} \rangle_{\mathbb{Z}}] = [\mathcal{O}_K : \langle \underline{\theta} \rangle_{\mathbb{Z}}]$$

where A is the change of basis matrix $\underline{\omega} \mapsto \underline{\theta}$. On the other hand, by Homework 4 Question 3, $\Delta^2(\underline{\theta}) = \det(A)^2 \Delta^2(\underline{\omega})$. So one has

$$\Delta^2(\underline{\theta}) = [\mathcal{O}_K : \langle \underline{\theta} \rangle_{\mathbb{Z}}]^2 \Delta^2(K).$$

In other words, one always has

$$\Delta^2(K) \mid \Delta^2(\underline{\theta})$$

and they are equal if and only if $\underline{\theta}$ is an integral basis of $\mathcal{O}_K$.

Procedure of finding an Integral Basis:

(a) Take any $\{\tilde{\alpha}_1, \cdots, \tilde{\alpha}_n\}$ basis of K over $\mathbb{Q}$.

(b) We can multiply large integers to each term in the above basis, and obtain a basis of K over $\mathbb{Q}$ such that $\underline{\alpha} := \{\alpha_1, \cdots, \alpha_n\}$ are all in $\mathcal{O}_K$. Find all primes $p_1 < p_2 < \cdots < p_k$ satisfying $p_i^2 \mid |\Delta^2(\alpha)|$.

(c) Apply the following algorithm on $\underline{\alpha}$:

(1) Start with $\ell = 1$.

(2) For all $1 \leq i \leq n$, compute the minimal polynomial $m_\gamma(x) \in \mathbb{Q}[x]$ for all

$$\gamma := \frac{a_1}{p_\ell}\alpha_1 + \cdots + \frac{a_{i-1}}{p_\ell}\alpha_{i-1} + \frac{1}{p_\ell}\alpha_i + \frac{a_{i+1}}{p_\ell}\alpha_{i+1} + \cdots + \frac{a_n}{p_\ell}\alpha_n$$

satisfying $0 \leq a_j < p_\ell$ for all $j \neq i$.

(3) If there exists a γ in Step (2) such that $m_\gamma(x) \in \mathbb{Z}[x]$ (i.e. $\gamma \in \mathcal{O}_K$), replace

$$\underline{\alpha} = \{\alpha_1, \alpha_2, \cdots, \alpha_n\} \quad \longrightarrow \quad \underline{\alpha'} = \{\alpha_1, \cdots, \alpha_{i-1}, \gamma, \alpha_{i+1}, \cdots, \alpha_n\},$$

so that $\Delta^2(\underline{\alpha'}) = (\frac{1}{p_\ell})^2\Delta^2(\underline{\alpha})$. There are two possibilities for $\underline{\alpha}$:

(i) If $p_\ell^2 \mid \Delta^2(\underline{\alpha'})$, repeat Step (1) for $\underline{\alpha'}$ with the same ℓ.

(ii) If $p_\ell^2 \nmid \Delta^2(\underline{\alpha'})$, repeat Step (1) for $\underline{\alpha'}$ with ℓ replaced by $\ell + 1$.

(4) If $m_\gamma(x) \notin \mathbb{Z}[x]$ for all γ in Step (2), repeat Step (1) for $\underline{\alpha}$ with ℓ replaced by $\ell + 1$.

(d) After running through all $1 \leq \ell \leq k$ in the above algorithm, we obtain $\underline{\omega}$. **This is an integral basis of $\mathcal{O}_K$.**

To see why the above algorithm works, suppose on contrary $[\mathcal{O}_K : \langle\underline{\omega}\rangle_\mathbb{Z}] \neq 1$. Then there exists a prime r such that

$$r \mid [\mathcal{O}_K : \langle\underline{\omega}\rangle_\mathbb{Z}].$$

Note that by Remark 5.4.4 (and the obvious fact that $\Delta^2(\underline{\omega}) \mid \Delta^2(\underline{\alpha})$), $r = p_\ell$ for some $1 \leq \ell \leq k$. Then the quotient group $\mathcal{O}_K/\langle\underline{\omega}\rangle_\mathbb{Z}$ has order equal to a multiple of p_ℓ. By Cauchy's theorem, there must be some $\beta \in \mathcal{O}_K$ such that the element

$$\beta + \langle\underline{\omega}\rangle_\mathbb{Z} \in \mathcal{O}_K/\langle\underline{\omega}\rangle_\mathbb{Z}$$

of order p_ℓ, i.e.

$$p_\ell \cdot (\beta + \langle\underline{\omega}\rangle_\mathbb{Z}) = 0_{\mathcal{O}_K/\langle\underline{\omega}\rangle_\mathbb{Z}} \quad \Leftrightarrow \quad p_\ell \cdot \beta = A_1\omega_1 + \cdots + A_n\omega_n \in \langle\underline{\omega}\rangle_\mathbb{Z}$$

for some $A_1, \ldots, A_n \in \mathbb{Z}$ not all multiples of p_ℓ. Choose i such that $p_\ell \nmid A_i$, and let $B_i \in \mathbb{Z}$ be such that $A_iB_i \equiv 1 \pmod{p_\ell}$, then

$$B_i\beta = \frac{B_iA_1}{p_\ell}\omega_1 + \cdots + \frac{B_iA_i}{p_\ell}\omega_i + \cdots + \frac{B_iA_n}{p_\ell}\omega_n \in \mathcal{O}_K.$$

by subtracting each $\frac{B_iA_j}{p_\ell}\omega_j$ in the above expression by a suitable integer multiple of $\omega_j \in \mathcal{O}_K$, one can conclude that there exists

$$\gamma = \frac{a_1}{p_\ell}\omega_1 + \cdots + \frac{a_{i-1}}{p_\ell}\omega_{i-1} + \frac{1}{p_\ell}\omega_i + \frac{a_{i+1}}{p_\ell}\omega_{i+1} + \cdots + \frac{a_n}{p_\ell}\omega_n \in \mathcal{O}_K$$

satisfying $0 \leq a_j < p_\ell$ for all $j \neq i$. However, by our construction of $\underline{\omega}$, all such elements cannot be in $\mathcal{O}_K$ (or else we can replace $\underline{\omega}$ with $\underline{\omega'}$ in Step (3), so that $\underline{\omega}$ is not the final outcome). So we have a contradiction.

With the recipe above, we can re-obtain $\mathcal{O}_K$ for the quadratic extension $K = \mathbb{Q}(\sqrt{d})$:

Example 5.4.5. *Let $K = \mathbb{Q}(\sqrt{d}) : \mathbb{Q}$, with d square-free, then one can take $\underline{\alpha} = \{1, \sqrt{d}\}$ in the above algorithm, which has*

$$\Delta^2(\underline{\alpha}) = \det \begin{pmatrix} 1 & \sqrt{d} \\ 1 & -\sqrt{d} \end{pmatrix}^2 = (-2\sqrt{d})^2 = 4d$$

Since d is square free, the only p such that $p^2 \mid \Delta^2(\underline{\alpha})$ is $p_1 = 2$. By Step (2), we consider

$$\beta = \frac{1}{2}, \quad \frac{1}{2}\sqrt{d}, \quad \frac{1}{2} + \frac{1}{2}\sqrt{d}$$

with minimal polynomials:

$$m_{1/2}(x) = x - \frac{1}{2}, \quad m_{\sqrt{d}/2}(x) = x^2 - \frac{d}{4}, \quad m_{(1+\sqrt{d})/2}(x) = x^2 - x - \frac{d-1}{4}$$

respectively.

If $d \equiv 2, 3 \pmod 4$, then none of the above are in $\mathbb{Z}[x]$, so immediately hit Step (4) and continue our algorithm with another prime $p_2 > p_1 = 2$. But there is no such p_2, so we stop here and conclude that $\underline{\alpha} = \{1, \sqrt{d}\}$ is an integral basis.

On the other hand, if $d \equiv 1 \pmod 4$, then we have $\beta = \frac{1}{2} + \frac{1}{2}\sqrt{d} \in \mathcal{O}_K$, so we replace $\underline{\alpha} = \{1, \sqrt{d}\}$ by $\underline{\alpha'} = \{1, \frac{1+\sqrt{d}}{2}\}$, with

$$\Delta^2(\underline{\alpha'}) = \frac{1}{2^2}\Delta^2(\underline{\alpha}) = \frac{1}{2^2}(4d) = d$$

So we are in the situation of Step (3)(ii), and consider $\underline{\alpha'}$ with $p_2 > p_1 = 2$. Once again, since we do not have such p_2, $\underline{\alpha'} = \{1, \frac{1+\sqrt{d}}{2}\}$ is an integral basis.

In conclusion, we have:

- *$\mathcal{O}_K = \langle 1, \sqrt{d} \rangle_{\mathbb{Z}}$ if $d \equiv 2, 3 \pmod 4$, with $\Delta^2(K) = 4d$.*
- *$\mathcal{O}_K = \langle 1, \frac{1+\sqrt{d}}{2} \rangle_{\mathbb{Z}}$,if $d \equiv 1 \pmod 4$, with $\Delta^2(K) = d$.*

Also, we can find the integral basis to the cyclotomic number field $K = \mathbb{Q}(\omega)$, where ω is a primitive p^{th} root of unity for prime p:

Theorem 5.4.6. *Let p be a rational prime, and ω is a primitive p^{th} root of unity. Consider $K = \mathbb{Q}(\omega)$, then $\mathcal{O}_K = \mathbb{Z}[\omega] = \langle 1, \omega, \ldots, \omega^{p-2} \rangle_{\mathbb{Z}}$, i.e. $\{1, \omega, \omega^2, \cdots, \omega^{p-2}\}$ is an integral basis.*

Proof. For $p = 2$ the result is immediate, so assume p is odd. The minimal polynomial of ω is $m_\omega(x) = \prod_{i=1}^{p-1}(x - \omega^i) = x^{p-1} + x^{p-2} + \cdots + 1$. Hence $[K : \mathbb{Q}] = \deg m_\omega(x) = p - 1$, and $\{1, \omega, \omega^2, \ldots, \omega^{p-2}\}$ forms a basis for $K/\mathbb{Q}$. We need to show $\mathbb{Z}[\omega] = \mathcal{O}_K$.

Note that $\omega \in \mathcal{O}_K$ by above, and hence $\mathbb{Z}[\omega] \subseteq \mathcal{O}_K$. It suffices to prove the reverse inclusion. By Homework 4, the discriminant calculation gives:

$$\begin{aligned}
\Delta^2(1, \omega, \ldots, \omega^{p-2}) &= (-1)^{\frac{(p-1)(p-2)}{2}} \operatorname{Norm}_{K/\mathbb{Q}}(m'_\omega(\omega)) \\
&= (-1)^{\frac{(p-1)(p-2)}{2}} \operatorname{Norm}_{K/\mathbb{Q}}\left(\frac{p\omega^{p-1}}{\omega - 1}\right) \\
&= (-1)^{\frac{(p-1)(p-2)}{2}} \operatorname{Norm}_{K/\mathbb{Q}}\left(\frac{p}{\omega - 1}\right) \\
&= (-1)^{\frac{(p-1)(p-2)}{2}} \prod_{i=1}^{p-1}\left(\frac{p}{\omega^i - 1}\right) \\
&= (-1)^{\frac{(p-1)(p-2)}{2}} \frac{(-1)^{p-1}p^{p-1}}{\prod_{i=1}^{p-1}(1 - \omega^i)} \\
&= (-1)^{\frac{(p-1)p}{2}} \frac{p^{p-1}}{m_\omega(1)} \\
&= (-1)^{\frac{p-1}{2}} p^{p-2}
\end{aligned}$$

Therefore, by our algorithm of finding integral basis, one only needs to check whether $p \nmid [\mathcal{O}_K : \mathbb{Z}[\omega]]$.

Suppose on contrary that $p \mid [\mathcal{O}_K : \mathbb{Z}[\omega]]$. Consider $\theta = \omega - 1$, so that the discriminant $\Delta^2(1, \theta, \ldots, \theta^{p-2}) = \Delta^2(1, \omega, \ldots, \omega^{p-2})$. Therefore, $\mathbb{Z}[\theta] = \mathbb{Z}[\omega]$ and hence

$$p \mid [\mathcal{O}_K : \mathbb{Z}[\theta]].$$

By our discussions in the previous section, this implies there exists $\varepsilon \in \mathcal{O}_K$ of the form:

$$\varepsilon = \frac{1}{p}(a_0 + a_1\theta + \cdots + a_{p-2}\theta^{p-2})$$

with $0 \le a_i \le p - 1$ not all zeros. Let r be minimal with $a_r \neq 0$. Then:

$$\theta^{p-2-r}\varepsilon = \frac{a_r}{p}\theta^{p-2} + \frac{a_{r+1}}{p}\theta^{p-1} + \cdots + \frac{a_{p-2}}{p}\theta^{2p-4-r} \in \mathcal{O}_K$$

Using $\omega^p = 1 \Rightarrow (\theta + 1)^p = 1$, we derive $\theta^{p-1} = -\left(\binom{p}{1}\theta^{p-2} + \cdots + \binom{p}{p-1}\right)$. Since $\binom{p}{\ell}$ is a multiple of p for all $0 < \ell < p$, hence $\frac{\theta^{p-1}}{p} \in \mathbb{Z}[\theta]$ $(\subseteq \mathcal{O}_K)$. More generally, $\frac{\theta^k}{p} \in \mathbb{Z}[\theta]$ $(\subseteq \mathcal{O}_K)$ for all $k \ge p - 1$, and hence

$$\frac{a_r}{p}\theta^{p-2} = (-a_{r+1})\frac{\theta^{p-1}}{p} + \cdots + (-a_{p-2})\frac{\theta^{2p-4-r}}{p} + \theta^{p-2-r}\varepsilon \in \mathcal{O}_K.$$

Now study the norm of $\frac{a_r}{p}\theta^{p-2} \in \mathcal{O}_K$:

$$\begin{aligned}
\mathrm{Norm}_{K/\mathbb{Q}}\left(\frac{a_r}{p}\theta^{p-2}\right) &= \left(\frac{a_r}{p}\right)^{p-1} \prod_{i=1}^{p-1} \sigma_i(\theta)^{p-2} \\
&= \left(\frac{a_r}{p}\right)^{p-1} \left(\prod_{i=1}^{p-1} (\omega^i - 1)\right)^{p-2} \\
&= (-1)^{p-1} \left(\frac{a_r}{p}\right)^{p-1} m_\omega(1)^{p-2} \\
&= \left(\frac{a_r}{p}\right)^{p-1} \cdot p^{p-2} \\
&= \frac{a_r^{p-1}}{p}
\end{aligned}$$

which is not in $\mathbb{Z}$. However, this contradicts Corollary 5.3.6. Hence $[\mathcal{O}_K : \mathbb{Z}[\theta]] = 1$ and the result follows. □

Chapter 6

Unique Factorization

6.1 Unique Factorization Domain

In this section, we will focus on studying whether the ring of integers $\mathcal{O}_K$ is a unique factorization domain (UFD) or not. Indeed, one can solve a lot of Diophantine equations by checking whether $\mathcal{O}_K$ is a UFD or not, as we see in the following example:

Example 6.1.1. *The only integer solutions of*

$$y^2 + 2 = x^3$$

are $(x, y) = (3, \pm 5)$.

Proof. To begin with, it is obvious that y must be odd. Now consider $K = \mathbb{Q}(\sqrt{-2})$ with $\mathcal{O}_K = \mathbb{Z}[\sqrt{-2}]$. Recall that $\mathbb{Z}[\sqrt{-2}]$ is Euclidean with respect to the norm, hence a UFD. Thus

$$(y + \sqrt{-2})(y - \sqrt{-2}) = x^3 \qquad (*)$$

Suppose there exists $\alpha \in \mathcal{O}_K$ irreducible such that:

$$\alpha \mid (y + \sqrt{-2}) \quad \text{and} \quad \alpha \mid (y - \sqrt{-2}),$$

then

$$\alpha \mid [(y + \sqrt{-2}) - (y - \sqrt{-2})] = 2\sqrt{-2} \quad \Rightarrow \quad \alpha \mid -(\sqrt{-2})^3 \quad \Rightarrow \quad \alpha \sim \sqrt{-2}.$$

Therefore,

$$\sqrt{-2} \mid (y + \sqrt{-2}) \quad \Rightarrow \quad \sqrt{-2} \mid y$$

and since $\mathrm{Norm}(ab) = \mathrm{Norm}(a)\mathrm{Norm}(b)$ is multiplicative,

$$\mathrm{Norm}(\sqrt{-2}) \mid \mathrm{Norm}(y) \quad \Rightarrow \quad 2 \mid y^2$$

which implies y is even, contradicting our observation in the beginning of the proof. Therefore, $y + \sqrt{-2}$ and $y - \sqrt{-2}$ has no common factor.

By to Equation $(*)$, suppose $x^3 = (\gamma_1 \dots \gamma_k)^3 = \gamma_1^3 \dots \gamma_k^3$ is factorized into irreducibles. Then by unique factorization, $\gamma_i \mid y + \sqrt{-2}$ implies $\gamma_i^3 \mid y + \sqrt{-2}$, otherwise γ_i is also a factor of $y - \sqrt{-2}$, violating the fact that they $y \pm \sqrt{-2}$ has no common factors. In other words,

$$y + \sqrt{-2} = \beta^3$$

for some $\beta \in \mathcal{O}_K$.

Let $\beta = a + b\sqrt{-2}$ for $a, b \in \mathbb{Z}$, then:

$$y + \sqrt{-2} = a(a^2 - 6b^2) + b(3a^2 - 2b^2)\sqrt{-2}$$

which implies that $1 = b(3a^2 - 2b^2)$. Therefore $b = 1$ and $a = \pm 1$. Substituting back, one gets $y = \pm 5$, which forces $x = 3$. And they are the only solutions of the equation. □

However, this is **NOT** true in general that $\mathcal{O}_K$ is a UFD. Indeed, let $K = \mathbb{Q}(\sqrt{-5})$, then $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ and

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

in $\mathcal{O}_K$. One can see that the above terms are not associates with each other. Indeed, note that the norms of these numbers are:

$$4,\ 9,\ 6,\ 6$$

Therefore $2, 3 \nsim (1 \pm \sqrt{-5})$ by Corollary 5.3.7(b).

Moreover, the terms in the factorization are all irreducible. For instance, if $\alpha = u + v\sqrt{-5} \in \mathcal{O}_K$ is a nonunit divisor of 2, then by multiplicativity of Norm,

$$\text{Norm}(\alpha) = u^2 + 5v^2 \mid \text{Norm}(2) = 4.$$

Since there are no elements of norm 2, one must have $\text{Norm}(\alpha) = 4$, hence $\alpha = \pm 2$. The same norm argument shows that 3 and $1 \pm \sqrt{-5}$ are irreducible. Consequently, we factorize $6 \in \mathcal{O}_K$ into irreducibles into **two** different ways, and hence it is not a UFD.

Instead of having unique factorization of **elements** in $\mathcal{O}_K$ into primes, we will prove unique factorization of **ideals** of $\mathcal{O}_K$ into prime ideals. To begin with, we make the following:

Definition 6.1.2. *Let $\mathcal{O}_K$ be the ring of integers for a number field $K : \mathbb{Q}$. For any ideals $I, J \lhd \mathcal{O}_K$, the product $IJ \lhd \mathcal{O}_K$ is defined as*

$$IJ := \{i_1 j_1 + \dots + i_n j_n \mid i_p \in I,\ j_q \in J, n \in \mathbb{N}\}.$$

Before moving on, let's make some some simple observations:

- Obviously $IJ = JI$ and $(IJ)P = I(JP)$ for all $I, J, P \lhd \mathcal{O}_K$. In other words, the product of ideals is commutative and associative.
- If $(\alpha) \lhd \mathcal{O}_K$ is a principal ideal, the product $(\alpha)I$ can be expressed as

$$(\alpha)I = \{\alpha i \mid i \in I\}.$$

  In particular, one can easily check that $(\alpha)(\beta) = (\alpha\beta)$.

We can now state the theorem of unique factorization of ideals in $\mathcal{O}_K$:

Theorem 6.1.3 (Dedekind's Theorem)**.** *Let $I \lhd \mathcal{O}_K$ be any proper, nonzero ideal. Then one can factorize $I = P_1 \dots P_k$ into product of prime ideals $P_i \lhd \mathcal{O}_K$. Moreover, if there are two factorizations of prime ideals of I:*

$$I = P_1 \dots P_k = Q_1 \dots Q_l.$$

Then one has $k = l$, and there exists a permutation $\sigma \in S_k = S_l$ such that

$$P_i = Q_{\sigma(i)}$$

for all $1 \leq i \leq k$.

6.2 Ideal Class

We first give the definition of ideal class using the equivalence of ideals.

Definition 6.2.1. *Let $I, J \lhd \mathcal{O}_K$ be nonzero ideals. We say $I \sim J$ if $\exists \alpha, \beta \in \mathcal{O}_K \setminus \{0\}$ such that $(\alpha)I = (\beta)J$.*

The collection of all equivalence classes of $I \lhd \mathcal{O}_K$ is called the **ideal classes** *of K:*

$$C_K := \{[I] \mid I \lhd \mathcal{O}_K \text{ nonzero}\},$$

where $[I] := \{J \lhd \mathcal{O}_K \mid J \sim I\}$. is the equivalence class of I.

The **class number** *of K is defined by $h_K = |C_K|$.*

One can easily check that $\sim$ defines an equivalence relationship. For instance, if $I \sim J$ and $J \sim K$, then $(\alpha)I = (\beta)J$ and $(\gamma)J = (\delta)K$. Therefore, $(\alpha\gamma)I = (\gamma)(\alpha)I = (\gamma)(\beta)J = (\beta)(\gamma)J = (\beta)(\delta)K = (\beta\delta)K$ and hence $I \sim K$.

Example 6.2.2. *Consider the equivalence classes of $\mathcal{O}_K = (1)$. We claim that:*

$$[\mathcal{O}_K] = \{\textit{All nonzero principal ideals } (a) \lhd \mathcal{O}_K\}.$$

Proof of the claim. If $I = (\gamma)$ is principal, it is clear that

$$(1)I = (1)(\gamma) = (\gamma)(1) = (\gamma)\mathcal{O}_K,$$

hence $I \sim \mathcal{O}_K$.

Conversely, if $I \sim \mathcal{O}_K$, then there exists $a, b \in \mathcal{O}_K$ such that $(a)I = (b)\mathcal{O}_K = (b)$. Thus we may write $b = ai$ for some $i \in I$ and $\frac{b}{a} = i \in I \subseteq \mathcal{O}_K$. Therefore $(\frac{b}{a}) \subseteq I$ On the other hand for any $i' \in I$, we have $ai' \in (b)$. We may write $ai' = bx$ for some $x \in \mathcal{O}_K$, then $i' = \frac{b}{a} \cdot x \in (\frac{b}{a})$. Thus $I \subseteq (\frac{b}{a})$ and hence

$$I = (\frac{b}{a})$$

is a principal ideal. □

As a direct consequence of the example above, one has:

Theorem 6.2.3. *$\mathcal{O}_K$ is a PID if and only if $h_K = 1$.*

Therefore C_K (or h_K) gives a measure on how "far" $\mathcal{O}_K$ from being PID. We will first study the structure of C_K in this section. The ultimate goal of this section is to prove:

For number field $K : \mathbb{Q}$, the ideal class C_K has a finite abelian group structure

Goal I: $h_K = |C_K| < \infty$**:**

Lemma 6.2.4 (Hurwitz)**.** *Let $K : \mathbb{Q}$ be a number field and $[K : \mathbb{Q}] = n$. Then there exists a large integer M such that $\forall \gamma \in K$, there exist an integer $1 \leq t \leq M$ and $\omega \in \mathcal{O}_K$ with $\left|\mathrm{Norm}_{K/\mathbb{Q}}(t\gamma - \omega)\right| < 1$.*

Proof. Let $\{\omega_1, \ldots, \omega_n\}$ be an integral basis of $\mathcal{O}_K$, i.e. $\mathcal{O}_K = \langle \omega_1, \ldots, \omega_n \rangle_{\mathbb{Z}}$. For any $\gamma \in K = \mathrm{Span}_{\mathbb{Q}}\{\omega_1, \ldots, \omega_n\}$, assume $\gamma = a_1\omega_1 + \cdots + a_n\omega_n$. Let

$$\lfloor \gamma \rfloor = \lfloor a_1 \rfloor \omega_1 + \cdots + \lfloor a_n \rfloor \omega_n, \quad \{\gamma\} = b_1\omega_1 + \cdots + b_n\omega_n,$$

where $b_i = a_i - \lfloor a_i \rfloor$ satisfies $0 \leq b_i < 1$, i.e. $\lfloor \gamma \rfloor \in \mathcal{O}_K$ is the integer part of γ and $\{\gamma\}$ is the decimal part of γ, so that

$$\gamma = \lfloor \gamma \rfloor + \{\gamma\}.$$

Consider

$$\begin{aligned}\left|\mathrm{Norm}_{K/\mathbb{Q}}(\gamma)\right| &= \left|\prod_{j=1}^{n}(\sum_{i=1}^{n} a_i\sigma_j(\omega_i))\right| \\ &\leq \prod_{j=1}^{n}(\sum_{i=1}^{n}|a_i|\cdot|\sigma_j(\omega_i)|) \\ &\leq (\max_i\{|a_i|\})^n\prod_{j=1}^{n}(\sum_{i=1}^{n}|\sigma_j(\omega_i)|) \\ &\leq C\cdot(\max_i\{|a_i|\})^n \qquad \text{where } C := \prod_{j=1}^{n}(\sum_{i=1}^{n}|\sigma_j(\omega_i)|)\end{aligned}$$

Take any integer $m > C^{1/n}$ and let $M = m^n$. Consider the map

$$\phi : K \to \mathbb{R}^n, \quad a_1\omega_1 + \cdots + a_n\omega_n \mapsto (a_1, \ldots, a_n),$$

then $\phi(\{\gamma\}) = (b_1, \ldots, b_n) \in \mathscr{A} = \{(x_1, \ldots, x_n) \mid 0 \leq x_i < 1\}$. If we divide $\mathscr{A}$ into $\frac{1}{m}\times\cdots\frac{1}{m}$ cubes, there are m^n such cubes in $\mathscr{A}$. By Pigeonhole principle, $\exists 1 \leq k < l \leq M+1$ such that $\phi(\{k\gamma\})$ and $\phi(\{l\gamma\})$ lies in the same $\frac{1}{m}\times\cdots\times\frac{1}{m}$ cube.

Take $t := l - k$, then

$$t\gamma = l\gamma - k\gamma = (\lfloor l\gamma\rfloor - \lfloor k\gamma\rfloor) + (\{l\gamma\} - \{k\gamma\}) = \omega + (\{l\gamma\} - \{k\gamma\}),$$

where $\omega := \lfloor l\gamma\rfloor - \lfloor k\gamma\rfloor \in \mathcal{O}_K$. Since $t\gamma - \omega = \{l\gamma\} - \{k\gamma\}$, write

$$\{l\gamma\} - \{k\gamma\} = c_1\omega_1 + \cdots + c_n\omega_n,$$

where $|c_i| < 1/m$ because the two points lie in the same cube. Applying the inequality above gives

$$\left|\mathrm{Norm}_{K/\mathbb{Q}}(t\gamma - \omega)\right| < C\left(\frac{1}{m}\right)^n < 1,$$

and the result follows. □

Theorem 6.2.5. *Let $K : \mathbb{Q}$ be a number field, then $h_K < \infty$.*

Proof. Let $I \lhd \mathcal{O}_K$ be a nonzero ideal. Take any nonzero $\beta \in I$ such that $\left|\mathrm{Norm}_{K/\mathbb{Q}}(\beta)\right|$ is minimal among all elements in I. By Lemma 6.2.4, there exists M such that for any $\alpha \in I$, $\exists 1 \leq t \leq M$ and $\omega \in \mathcal{O}_K$ with $\left|\mathrm{Norm}_{K/\mathbb{Q}}(t\frac{\alpha}{\beta} - \omega)\right| < 1$. Therefore

$$\left|\mathrm{Norm}_{K/\mathbb{Q}}(t\alpha - \omega\beta)\right| = \left|\mathrm{Norm}_{K/\mathbb{Q}}(t\frac{\alpha}{\beta} - \omega)\right| \cdot \left|\mathrm{Norm}_{K/\mathbb{Q}}(\beta)\right| < \left|\mathrm{Norm}_{K/\mathbb{Q}}(\beta)\right|.$$

Since $t\alpha - \omega\beta \in I$, by the minimality of $\mathrm{Norm}_{K/\mathbb{Q}}(\beta)$, we must have $t\alpha - \omega\beta = 0$. Consequently, for any $i \in I$, $M!i \in (\beta)$.

Consider

$$J = \left\{ M!\frac{i}{\beta} \mid i \in I \right\} \subseteq \mathcal{O}_K.$$

It can be easily checked that $J \lhd \mathcal{O}_K$ is an ideal and $(\beta)J = (M!)I$, i.e. $I \sim J$.

We claim that there are only finitely such choices of J. To see so, note that

$$\mathcal{O}_K \supseteq J \supseteq (M!).$$

By the correspondence theorem,

$$\{J \mid \mathcal{O}_K \supseteq J \supseteq (M!)\} \xleftrightarrow{1 \text{ to } 1} \{H \mid \mathcal{O}_K/(M!) \geq H \geq \{0\}\}.$$

$\mathcal{O}_K = \langle \omega_1, \ldots, \omega_n \rangle_{\mathbb{Z}} = \mathbb{Z}\omega_1 \oplus \cdots \oplus \mathbb{Z}\omega_n$ and $(M!) = (M!\mathbb{Z})\omega_1 \oplus \cdots \oplus (M!\mathbb{Z})\omega_n$, thus,

$$\mathcal{O}_K/(M!) \cong \mathbb{Z}/M!\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/M!\mathbb{Z}.$$

It is a finite abelian group and hence there are only finitely many $\{0\} \leq H \leq \mathcal{O}_K/(M!)$. So the claim is proved.

Suppose there are N choices of such $\{0\} \leq H \leq \mathcal{O}_K/(M!)$, then they correspond to $J_1, \ldots, J_N \lhd \mathcal{O}_K$. In other words:

$$\text{For all } I \lhd \mathcal{O}_K,\ I \sim J_i \text{ for some } 1 \leq i \leq N.$$

Consequently, $C_K = \{[J_1], \ldots, [J_N]\}$ (there may be repetitions), and $h_K \leq N < \infty$ is finite. □

Goal II: C_K has a group structure:

We define a binary operator $* : C_K \times C_K \to C_K$ by

$$[I] * [J] := [IJ].$$

To make C_K a group, we have several things to check:

1. Well-definedness: if $I_1 \sim I_2$ and $J_1 \sim J_2$, then $I_1J_1 \sim I_2J_2$ by Homework 5.
2. Associativity: $([I] * [J]) * [K] = [I] * ([J] * [K])$ is obvious true by the associativity of $\mathcal{O}_K$. Note it is also clear that $[I] * [J] = [J] * [I]$.
3. Identity element: take $e = [\mathcal{O}_K] = [(1)]$. Then clearly $[I] * [\mathcal{O}_K] = [\mathcal{O}_K] * [I] = [I]$.

We also have to show that $[I]$ has an inverse.

Lemma 6.2.6. *Let $I, J \lhd \mathcal{O}_K$ be ideals with $I \neq (0)$ such that $JI = I$. Then $J = \mathcal{O}_K$.*

Proof. We know $\mathcal{O}_K = \langle \omega_1, \dots, \omega_n \rangle_{\mathbb{Z}}$ is n-generated $\mathbb{Z}$-module, so by MAT3042, $I = \langle i_1, \dots, i_n \rangle_{\mathbb{Z}}$ is also an n-generated $\mathbb{Z}$-module.

Since $JI = I$, there are $b_{ij} \in J$ such that

$$\begin{cases} b_{11}i_1 + \cdots + b_{1n}i_n = i_1 \\ b_{21}i_1 + \cdots + b_{2n}i_n = i_2 \\ \qquad \vdots \\ b_{n1}i_1 + \cdots + b_{nn}i_n = i_n \end{cases}$$

Therefore,

$$A = \begin{pmatrix} b_{11} - 1 & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} - 1 & \cdots & b_{2n} \\ \vdots & & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} - 1 \end{pmatrix}$$

has determinant $\det(A) = 0$. Expand this determinant, we have $(-1)^n + \beta = 0$, where β is combination of multiples of b_{ij}. Therefore $\beta \in J$ and therefore $1 \in J$. Thus $J = \mathcal{O}_K$. □

Lemma 6.2.7. *Let $I, J \lhd \mathcal{O}_K$ with $I \neq (0)$, and let $\omega \neq 0$. Suppose $IJ = (\omega)I$, then $J = (\omega)$.*

Proof. For any $j \in J$ and any $i \in I$, $ji = \omega i'$ for some $i' \in I$, so $\frac{j}{\omega} i = i' \in I$ and hence $\frac{j}{\omega} I = \{\frac{j}{\omega} i \mid i \in I\} \subseteq I$ (here $\frac{j}{\omega} \in K$).

Let $I = \langle i_1, \dots, i_n \rangle_{\mathbb{Z}}$ as in the lemma above, then there exists $x_{ij} \in \mathbb{Z}$ such that

$$\begin{cases} x_{11}i_1 + \cdots + x_{1n}i_n = \dfrac{j}{\omega} i_1 \\ \qquad \vdots \\ x_{n1}i_1 + \cdots + x_{nn}i_n = \dfrac{j}{\omega} i_n \end{cases}$$

Thus $B \begin{pmatrix} i_1 \\ \vdots \\ i_n \end{pmatrix} = \frac{j}{\omega} \begin{pmatrix} i_1 \\ \vdots \\ i_n \end{pmatrix}$ for $B := \begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix}$, i.e. $\frac{j}{\omega} \in K$ is an eigenvalue of B, and hence the (monic) characteristic polynomial $\chi_B(t) \in \mathbb{Z}[t]$ has $\frac{j}{\omega}$ as a root. So $\frac{j}{\omega} \in \mathcal{O}_K$.

It is easy to see $\frac{1}{\omega} J = \{\frac{j}{\omega} | j \in J\} \lhd \mathcal{O}_K$ defines an ideal, and $JI = (\omega)I$ implies $(\frac{1}{\omega} J)I = I$. By Lemma 6.2.6, $\frac{1}{\omega} J = \mathcal{O}_K$ and $J = (\omega)$. □

Theorem 6.2.8. *Let $I \lhd \mathcal{O}_K$ be a nontrivial ideal. Then $\exists 1 \leq l \leq h_K < \infty$ such that $I^l = (\gamma)$ is principal. Consequently, $[I]^l = [(\gamma)] = [(1)] = e$ and $[I]^{l-1} = [I]^{-1}$.*

Proof. Consider the set of ideals $\{I^k | 1 \leq k \leq h_K + 1\}$, there are $1 \leq x < y \leq h_K + 1$ such that $[I^x] = [I^y]$. So $I^x \sim I^y$ and $(a)I^x = (b)I^{y-x}I^x$ for some $a, b \in \mathcal{O}_K$. Let $l = y - x$, by Lemma 6.2.7, $(a) = (b)I^l$. Thus $I^l \sim \mathcal{O}_K$ and it is principal. □

So we have finished proving **Goal II**.

6.3 Proof of Dedekind Theorem

Before going into the proof of Dedekind theorem (Theorem 6.1.3), we need some tools to study $I \lhd \mathcal{O}_K$.

Lemma 6.3.1. *Let $I \lhd \mathcal{O}_K$ be a nonzero ideal. Then $I \cap \mathbb{Z}$ contains a nonzero integer.*

Proof. Let any $\alpha \in I$ with minimal polynomial $m_\alpha(x) = x^k + \cdots + a_1 x + a_0$. Since $m_\alpha(x)$ is irreducible, $a_0 \neq 0$. So $m_\alpha(\alpha) - a_0 = -a_0 \in (\alpha) \subseteq I$. □

Proposition 6.3.2. *Let $I \lhd \mathcal{O}_K$ be nonzero ideal. Then $[\mathcal{O}_K : I] < \infty$.*

Proof. Pick $a \in I \cap \mathbb{Z}$ in lemma above. Then $(a) \subseteq I \subseteq \mathcal{O}_K$. Take an integral basis $\{\omega_1, \ldots, \omega_n\}$ of $\mathcal{O}_K$. Note that

$$(a) \cong (a\mathbb{Z})\omega_1 \oplus \cdots \oplus (a\mathbb{Z})\omega_n,$$

we have

$$\mathcal{O}_K/(a) \cong \mathbb{Z}/a\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a\mathbb{Z}$$

being a finite group. By the second isomorphism theorem, $\mathcal{O}_K/I \cong \frac{\mathcal{O}_K/(a)}{I/(a)}$ and hence $|\mathcal{O}_K/I| \leq |a|^n$ is finite. □

Definition 6.3.3. *Let $I \lhd \mathcal{O}_K$ be a nonzero ideal. The* **norm of ideal** *I is defined as $N(I) = [\mathcal{O}_K : I] = |\mathcal{O}_K/I|$.*

Remark 6.3.4.

1. *If $I \lhd \mathcal{O}_K$ is "small" then $N(I)$ is "big"; if $I \lhd \mathcal{O}_K$ is "big" then $N(I)$ is small.*
2. *If $N(I) = p$ is a rational prime, then $I \lhd \mathcal{O}_K$ is maximal and hence prime.*
3. *In the special case when $K = \mathbb{Q}$ and $\mathcal{O}_K = \mathbb{Z}$, the norm of any nonzero $I = (m)$ is $N((m)) = |\mathbb{Z}/m\mathbb{Z}| = |m|$.*

Proposition 6.3.5. *Let $(\alpha) \lhd \mathcal{O}_K$ be a principal ideal. Then $N((\alpha)) = \left|\mathrm{Norm}_{K/\mathbb{Q}}(\alpha)\right|$.*

Proof. Let $\mathcal{O}_K = \langle \omega_1, \dots, \omega_n \rangle_{\mathbb{Z}}$ be a integral basis, then $(\alpha) = \langle \alpha\omega_1, \dots, \alpha\omega_n \rangle_{\mathbb{Z}}$ is a $\mathbb{Z}$-basis of (α). By Proposition 5.1.4, $[\mathcal{O}_K : (\alpha)] = |\det(A)|$, where A is the change of basis matrix $\{\omega_1, \dots, \omega_n\} \to \{\alpha\omega_1, \dots, \alpha\omega_n\}$.

On the other hand, Homework 4 Question 3 gives $\Delta^2(\alpha\omega_1, \dots, \alpha\omega_n) = \det(A)^2 \Delta^2(\omega_1, \dots, \omega_n)$, where

$$\begin{aligned}
\Delta^2(\alpha\omega_1, \dots, \alpha\omega_n) &= \det \begin{pmatrix} \sigma_1(\alpha\omega_1) & \dots & \sigma_n(\alpha\omega_1) \\ \vdots & & \vdots \\ \sigma_1(\alpha\omega_n) & \dots & \sigma_n(\alpha\omega_n) \end{pmatrix}^2 \\
&= (\sigma_1(\alpha) \cdots \sigma_n(\alpha))^2 \cdot \det \begin{pmatrix} \sigma_1(\omega_1) & \cdots & \sigma_n(\omega_1) \\ \vdots & & \vdots \\ \sigma_1(\omega_n) & \cdots & \sigma_n(\omega_n) \end{pmatrix}^2 \\
&= \Delta^2(\omega_1, \dots, \omega_n) \mathrm{Norm}_{K/\mathbb{Q}}(\alpha)^2.
\end{aligned}$$

So $|\det(A)| = \left|\mathrm{Norm}_{K/\mathbb{Q}}(\alpha)\right|$ and $N((\alpha)) = |\mathrm{Norm}_{K/\mathbb{Q}}(\alpha)|$. □

Lemma 6.3.6. *Suppose $A, B, C \lhd \mathcal{O}_K$ be ideals such that $AB = AC$, then $B = C$.*

Proof. By Theorem 6.2.8, there exists $k \in \mathbb{N}$ such that $A^k = (\alpha)$ is a principal ideal. Therefore $AB = AC$ implies $A^{k-1}AB = A^{k-1}AC$, so $(\alpha)B = (\alpha)C$. Write it explicitly, we have $\{\alpha b | b \in B\} = \{\alpha c | c \in C\}$, so $B = C$. □

Definition 6.3.7. *Let $A, B \lhd \mathcal{O}_K$ be ideals. We say B* **divides** *A (denoted by $B \mid A$) if $\exists C \lhd \mathcal{O}_K$ such that $BC = A$.*

Obviously, if $B \mid A$, then $B \supseteq A$. The converse also holds:

Proposition 6.3.8. *Let $A, B \lhd \mathcal{O}_K$ be nonzero ideals, then $B \mid A$ if and only if $B \supseteq A$.*

Proof. ($\Rightarrow$) is clear. For ($\Leftarrow$), let $B^l = (\beta)$ for some $l \in \mathbb{N}$ and $\beta \in \mathcal{O}_K$. Then $A \subseteq B$ implies $AB^{l-1} \subseteq B^l = (\beta)$. Consider $J = \frac{1}{\beta} AB^{l-1}$, then J is an ideal of $\mathcal{O}_K$. $BJ = \frac{1}{\beta} BAB^{l-1} = \frac{1}{\beta} A(\beta) = A$. Thus $BJ = A$. □

Remark 6.3.9. *If $B \mid A$, then $N(B) \leq N(A)$.*

Theorem 6.3.10 (Dedekind Theorem, Part I)**.** *Let $A \lhd \mathcal{O}_K$ be a nonzero proper ideal. Then there exists $P_1, \dots, P_k$ such that $A = P_1 \cdots P_k$.*

Proof. Suppose on the contrary that there are $I \lhd \mathcal{O}_K$ cannot be divided into product of prime ideals. Then take $B \lhd \mathcal{O}_K$ to be on such ideals with minimal norm. Then there exists a maximal ideal P (which is also prime!) such that $B \subseteq P$.

Proposition 6.3.8 implies $P \mid B$, i.e. $PC = B$ for some $C \lhd \mathcal{O}_K$. Note that $B \neq C$, otherwise Lemma 6.2.6 applied to $PB = B$ would imply $P = \mathcal{O}_K$. Since $B = PC \subseteq C$, we

have $B \subsetneq C$ and hence $N(C) < N(B)$. By the minimality of $N(B)$, C can be written as product of prime ideals $C = Q_1 \cdots Q_l$. Thus $B = PC = PQ_1 \cdots Q_l$ can be factorized into prime ideals. Contradiction! □

In order to show the factorization is unique, we need

Lemma 6.3.11. *Let $I \lhd \mathcal{O}_K$ be a nonzero proper ideal. Then I is maximal if and only if I is prime.*

Proof. The only if direction is clear. For the if direction, we need several facts from MAT3004:

1. $I \lhd R$ is maximal ideal $\iff$ R/I is a field;
2. $I \lhd R$ is prime ideal $\iff$ R/I is an integral domain.
3. Any integral domain S with $|S| < \infty$ is a field.

For ideal $I \lhd \mathcal{O}_K$, $N(I) = [\mathcal{O}_K/I]$ is finite. Since I is prime, $\mathcal{O}_K/I$ is integral domain thus a field. So I is maximal. □

Theorem 6.3.12 (Dedekind Theorem, Part II)**.** *Suppose $A \lhd \mathcal{O}_K$ has two factorization $A = P_1 \cdots P_k = Q_1 \cdots Q_l$, where P_i, Q_j are all prime ideals. Then $k = l$ and $\exists \sigma \in S_k$ such that $P_i = Q_{\sigma(i)}$ for all i.*

Proof. $P_1 \cdots P_k = Q_1 \cdots Q_l$ so $P_1 \mid Q_1 \cdots Q_l$. We leave an exercise to show that if P is prime ideal and $P \mid AB$, then $P \mid A$ or $P \mid B$. So $P_1 \mid Q_1$ or $P_1 \mid Q_2 \cdots Q_l$. If $P_1 \mid Q_1$, we are done, otherwise $P_1 \mid Q_2 \cdots Q_l$ so $P_1 \mid Q_2$ or $P \mid Q_3 \cdots Q_l$. Continuing these steps we can find j such that $P_1 \mid Q_j$, and $Q_j \subseteq P_1$. Since Q_j and P_1 are prime ideals, they are all maximal, we conclude that $P_1 = Q_j$.

Therefore, we have $P_1 P_2 \cdots P_k = P_1 Q_1 \cdots \hat{Q}_j \cdots Q_l$. By the cancellation of ideals, $P_2 \cdots P_k = Q_1 \cdots \hat{Q}_j \cdots Q_l$. The result follows by induction on the number of prime factors k. □

Example 6.3.13. *Let $K = \mathbb{Q}(\sqrt{-5})$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ is not a UFD, but we have unique factorization of ideals of $\mathcal{O}_K$. For example,*

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

are two prime factorizations of elements of $\mathcal{O}_K$. However,

$$(6) = (3)(2) = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

is not a factorization of prime ideals. To be more precise, $(2) = (2, 1 + \sqrt{-5})(2, 1 - \sqrt{-5})$ and $(3) = (3, 1+\sqrt{-5})(3, 1-\sqrt{-5})$, $(1+\sqrt{-5}) = (2, 1+\sqrt{-5})(3, 1+\sqrt{-5})$ and $(1-\sqrt{-5}) = (2, 1 - \sqrt{-5})(3, 1 - \sqrt{-5})$. Therefore the actual prime factorization of (6) is

$$(6) = (2, 1 + \sqrt{-5})(2, 1 - \sqrt{-5})(3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5}),$$

which is unique.

6.4 Multiplicativity of Norm

Definition 6.4.1. *Let $A, B \lhd R$ be ideals. We say A, B are coprime if*

$$A + B := \{a + b | a \in A, b \in B\} = R.$$

Remark 6.4.2. *If $R = \mathcal{O}_K$, then $A, B \lhd \mathcal{O}_K$ coprime if and only if there are no common prime factors $P \mid A$ and $P \mid B$.*

To see why, suppose $A + B = I \neq \mathcal{O}_K$, then we have a maximal ideal P such that $A + B \subseteq P$, so $P \mid A$ and $P \mid B$. The converse holds since $P \mid A$, $P \mid B$ implies $A + B \subseteq P$.

Lemma 6.4.3. *If $A, B \lhd \mathcal{O}_K$ are coprime ideals, then $AB = A \cap B$.*

Proof. We obviously have $AB \subseteq A \cap B$. $A \cap B \subseteq A$ and $A \cap B \subseteq B$ implies $A \mid A \cap B$ and $B \mid A \cap B$. Since A, B are coprime, they have no common factor, so $AB \mid A \cap B$ and $A \cap B \subseteq AB$. □

Theorem 6.4.4. *If $A, B \lhd \mathcal{O}_K$ are coprime ideals, then $N(AB) = N(A)N(B)$.*

Proof. This directly follows from above lemma and Chinese Remainder Theorem:

$$N(AB) = |\mathcal{O}_K/AB| = |\mathcal{O}_K/(A \cap B)| = |\mathcal{O}_K/A \times \mathcal{O}_K/B| = |\mathcal{O}_K/A| \times |\mathcal{O}_K/B| = N(A)N(B).$$

□

Lemma 6.4.5. *Let $P \lhd \mathcal{O}_K$ be a prime ideal. For all $l \in \mathbb{N}$, $N(P) = |\mathcal{O}_K/P| = \left|P^l/P^{l+1}\right|$.*

Proof. Obviously $P^l \neq P^{l+1}$. Take an element $\pi \in P^l \backslash P^{l+1}$. $(\pi) \subseteq P^l$ so $P^l \mid (\pi)$, so we can write $P^l B = (\pi)$ for some $B \lhd \mathcal{O}_K$. Note that P, B are coprime, otherwise $P^{l+1} \mid (\pi)$. Define

$$\Phi : \mathcal{O}_K \to P^l/P^{l+1}, \quad \alpha \mapsto \alpha\pi + P^{l+1}.$$

We show that Φ induces an isomorphism between $\mathcal{O}_K/P$ and P^l/P^{l+1}.

First, $\ker \Phi = P$: For $\alpha \in P$, then $\alpha \in \ker \Phi$ so $P \subseteq \ker \Phi$. On the other hand, if $\Phi(\beta) = 0$, then $\pi\beta \in P^{l+1}$, $(\beta\pi) \subseteq P^{l+1}$. We have $P^{l+1} \mid (\beta)(\pi)$, $(\beta)(\pi) = (\beta)BP^l$. Since B and P are coprime, $P \mid (\beta)B$ implies $P \mid (\beta)$. So $\beta \in P$ and $\ker \Phi = P$.

Then we show Φ is surjective: Note that

$$(\pi) + P^{l+1} = P^l B + P^{l+1} = P^l(B + P) = P^l \cdot \mathcal{O}_K = P^l$$

. For all $\beta \in P^l$, we may write $\beta = \alpha\pi + \gamma$ for some $\alpha \in \mathcal{O}_K$ and $\gamma \in P^{l+1}$. So $\Phi(\alpha) = \alpha\pi + P^{l+1} = \beta + P^{l+1}$.

By the first isomorphism theorem, we have $\mathcal{O}_K/P \cong P^l/P^{l+1}$. Hence $|\mathcal{O}_K/P| = \left|P^l/P^{l+1}\right|$. □

Theorem 6.4.6. *Let $P \lhd \mathcal{O}_K$ be a prime ideal, then $N(P^m) = (N(P))^m$.*

Proof.

$$N(P^m) = |\mathcal{O}_K/P^m| = |\mathcal{O}_K/P| \times \left|P/P^2\right| \times \cdots \times \left|P^{m-1}/P^m\right| = |\mathcal{O}_K/P|^m = N(P)^m.$$

□

Corollary 6.4.7. *For all $I, J \lhd \mathcal{O}_K$ ideals, the product of norms satisfies*

$$N(IJ) = N(I)N(J).$$

Proof. By Dedekind Theorem 6.1.3, $I = P_1^{e_1} \cdots P_k^{e_k}$, $J = Q_1^{f_1} \cdots Q_l^{f_l}$. By relabeling, we may assume $P_1 = Q_1, \ldots P_m = Q_m$ and $\{P_{m+1}, \ldots, P_k\} \cap \{Q_{m+1}, \ldots, Q_l\}$ is empty. Then the prime factorization of IJ is

$$IJ = P_1^{e_1+f_1} \cdots P_m^{e_m+f_m} P_{m+1}^{e_{m+1}} \cdots P_k^{e_k} Q_{m+1}^{f_{m+1}} \cdots Q_l^{f_l}.$$

So

$$\begin{aligned}
N(IJ) &= N(P_1^{e_1+f_1}) \cdots N(P_m^{e_m+f_m}) N(P_{m+1}^{e_{m+1}}) \cdots N(P_k^{e_k}) N(Q_{m+1}^{f_{m+1}}) \cdots N(Q_l^{f_l}) \\
&= N(P_1)^{e_1+f_1} \cdots N(P_m)^{e_m+f_m} N(P_{m+1})^{e_{m+1}} \cdots N(P_k)^{e_k} N(Q_{m+1})^{f_{m+1}} \cdots N(Q_l)^{f_l} \\
&= (\prod_{i=1}^{k} N(P_i)^{e_i})(\prod_{j=1}^{l} N(Q_j)^{f_j}) \\
&= N(\prod_{i=1}^{k} P_i^{e_i}) N(\prod_{j=1}^{l} Q_j^{f_j}) \\
&= N(I)N(J).
\end{aligned}$$

□

Example 6.4.8. *Recall that $(6) = (2)(3) = (1+\sqrt{-5})(1-\sqrt{-5})$. We claim that*

$$(1+\sqrt{-5}) = (2, 1+\sqrt{-5})(3, 1+\sqrt{-5}).$$

First

$$(2, 1+\sqrt{-5}) \supsetneq (1+\sqrt{-5}) \text{ and } (3, 1+\sqrt{-5}) \supsetneq (1+\sqrt{-5}).$$

So $(2, 1+\sqrt{-5}) \mid (1+\sqrt{-5})$, $(3, 1+\sqrt{-5}) \mid (1+\sqrt{-5})$. By Corollary 6.4.7,

$$N((2, 1+\sqrt{-5})) \mid N(1+\sqrt{-5}) = 6 \text{ and } N((2, 1+\sqrt{-5})) \mid N(2) = 4,$$

so $N(2, 1+\sqrt{-5}) = 2$. Similarly, $N(3, 1+\sqrt{-5}) = 3$. So both $(2, 1+\sqrt{-5})$ and $(3, 1+\sqrt{-5})$ are prime (also maximal) ideals.

Since $(2, 1+\sqrt{-5})(3, 1+\sqrt{-5}) \mid (1+\sqrt{-5})$ and $N((2, 1+\sqrt{-5}))N((3, 1+\sqrt{-5})) = N(1+\sqrt{-5})$, we conclude that $(2, 1+\sqrt{-5})(3, 1+\sqrt{-5}) = (1+\sqrt{-5})$.

Chapter 7

Prime Ideals in $\mathcal{O}_K$

7.1 Ideals Lying Above a Rational Prime

We now study the structure of nonzero prime ideals $P \lhd \mathcal{O}_K$. To begin with, note that $P \cap \mathbb{Z} = (p)_{\mathbb{Z}} \lhd \mathbb{Z}$ for some rational prime p. So $(p) \subseteq P$ as ideals in $\mathcal{O}_K$. Therefore, all prime ideals $P \lhd \mathcal{O}_K$ must be a factor of (p) for some rational prime p.

Definition 7.1.1. *Let $p \in \mathbb{N}$ be a rational prime and $(p) = P_1^{e_1} \cdots P_k^{e_k}$ be the prime factorization of (p).*

1. *P_i are called the* **prime ideals lying above** *$p \in \mathbb{N}$;*
2. *e_i is called the* **ramification index** *of P_i;*
3. *P_i is* **ramified** *if $e_i > 1$;*
4. *p is* **ramified** *in K if there exists some ramification index $e_i > 1$.*

As discussed above, in order to get all prime ideals of $\mathcal{O}_K$, one studies the prime ideals P_i lying over all rational primes p. The theorem below gives you the (unique) factorization of (p) into prime ideals in $\mathcal{O}_K$:

Theorem 7.1.2. *Let $K : \mathbb{Q}$ be a number field, and $\alpha \in \mathcal{O}_K$ be such that*

1. *$K = \mathbb{Q}(\alpha)$.*
2. *$p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$.*

Suppose the minimal polynomial $m_\alpha(x) \in \mathbb{Z}[x]$ factorizes as

$$\overline{m_\alpha}(x) = \prod_{i=1}^{k} \overline{g_i}(x)^{e_i} \in \mathbb{Z}_p[x].$$

then

$$(p) = \prod_{i=1}^{k} (p, h_i(\alpha))^{e_i},$$

where $h_i \in \mathbb{Z}[x]$ is any polynomial whose image in $\mathbb{Z}_p[x]$ satisfies $\overline{h_i}(x) = \overline{g_i}(x) \in \mathbb{Z}_p[x]$. Moreover, each factor $(p, h_i(\alpha))$ is a prime ideal for all i.

Example 7.1.3. *Let $K = \mathbb{Q}(\sqrt{-5})$ and $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$. We may choose $\alpha = \sqrt{-5}$ and then $K = \mathbb{Q}(\alpha)$ and $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 1$. So the assumption holds for all rational prime $p \in \mathbb{N}$.*

For $p = 2$, $\overline{m_\alpha}(x) = x^2 + 1 = (x+1)^2$, so $(2) = (2, 1 + \sqrt{-5})^2$ is ramified.

For $p = 3$, $\overline{m_\alpha}(x) = x^2 + 2 = (x+1)(x+2)$, so $(3) = (3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5})$ is unramified.

For $p = 5$, $\overline{m_\alpha}(x) = x^2$, so $(5) = (\sqrt{-5})^2$ is ramified.

Example 7.1.4. *Let $K = \mathbb{Q}(\sqrt{-3})$, then $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$. Then for $\alpha = \sqrt{-3}$, $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$. The theorem works for any rational prime $p \neq 2$. $m_\alpha(x) = x^2 + 3$.*

For $p = 5$, $\overline{m_\alpha}(x) = x^2 + 3$ is irreducible in $\mathbb{Z}_5[x]$, so $(5) = (5, \alpha^2 + 3) = (5, 0) = (5)$ is itself a prime ideal in $\mathcal{O}_K$.

If we want to deal with $p = 2$, we may choose $\alpha' = \frac{1+\sqrt{-3}}{2}$ instead, so that $[\mathcal{O}_K : \mathbb{Z}[\alpha']] = 1$. Then $\overline{m_{\alpha'}}(x) = x^2 + x + 1$ is irreducible in $\mathbb{Z}_2[x]$. So (2) is prime ideal in $\mathcal{O}_K$.

Proof of Theorem 7.1.2. Suppose $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then consider the homomorphism

$$\Phi : \mathbb{Z}[\alpha] \longrightarrow \frac{\mathcal{O}_K}{(p)} \qquad \Phi(\beta) := \beta + (p).$$

An element $x \in \mathbb{Z}[\alpha]$ lies in $\ker \Phi$ exactly when $x = p\beta$ for some $\beta \in \mathcal{O}_K$.

We *claim* that $\beta \in \mathbb{Z}[\alpha]$ - if not, then $\beta + \mathbb{Z}[\alpha]$ is an order p element in $\mathcal{O}_K/\mathbb{Z}[\alpha]$. Consequently, the cyclic subgroup

$$\langle \beta + \mathbb{Z}[\alpha] \rangle \leq \mathcal{O}_K/\mathbb{Z}[\alpha]$$

has order p. and hence $p \mid |\mathcal{O}_K/\mathbb{Z}[\alpha]| = [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ by Lagrange's theorem, which contradicts the hypothesis of the theorem.

Consequently, $\ker(\Phi) = p\mathbb{Z}[\alpha]$. By first isomorphism theorem, one therefore has

$$\mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha] \cong \mathrm{im}(\Phi) \leq \mathcal{O}_K/(p) \qquad (*)$$

Let's count the number of element on both sides of equation: Suppose $\deg(m_\alpha(x)) = n$, then $\mathbb{Z}[\alpha] = \mathrm{Span}_{\mathbb{Z}}\{1, \alpha, \ldots, \alpha^{n-1}\}$ and $\mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha] = \mathrm{Span}_{\mathbb{Z}_p}\{1, \alpha, \ldots, \alpha^{n-1}\} \cong \mathbb{Z}_p[x]/\langle \overline{m}_\alpha(x) \rangle$ (**WARNING:** This is **NOT** the same as $\mathbb{Z}_p[x]/\langle \mu_\alpha(x) \rangle$, where $\mu_\alpha(x) \in \mathbb{Z}_p[x]$ is the minimal polynomial of α in $\mathbb{Z}_p[x]$). Therefore,

$$|\mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha]| = |\mathbb{Z}_p[x]/\langle \overline{m}_\alpha(x) \rangle| = p^n.$$

On the other hand,

$$|\mathcal{O}_K/(p)| = N((p)) = \mathrm{Norm}_{K/\mathbb{Q}}(p) = p^{[K:\mathbb{Q}]} = p^{[\mathbb{Q}(\alpha):\mathbb{Q}]} = p^{\deg(m_\alpha(x))} = p^n$$

So the $\leq$ in $(*)$ above is an equality, i.e. one has

$$\mathcal{O}_K/(p) \cong \mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha] \cong \mathbb{Z}_p[x]/\langle \overline{m}_\alpha(x)\rangle.$$

Now we study all prime ideals $P \lhd \mathcal{O}_K$ lying above p: Note that

$$\mathcal{O}_K \supseteq P \supseteq (p) \;\leftrightarrow\; \mathcal{O}_K/(p) \supseteq P/(p) \supseteq \{0\} \;\leftrightarrow\; \mathbb{Z}_p[x]/\langle \overline{m}_\alpha(x)\rangle \supseteq Q' \supseteq \{0\} \;\leftrightarrow\; \mathbb{Z}_p[x] \supseteq Q \supseteq \langle \overline{m}_\alpha(x)\rangle$$

where the first and last $\leftrightarrow$ are given by the correspondence theorem, and the middle $\leftrightarrow$ is the isomorphism given above. Moreover, one has

- $P \lhd \mathcal{O}_K$ is a prime ideal $\leftrightarrow$ $Q \lhd \mathbb{Z}_p[x]$ is a prime ideal.
- $[\mathcal{O}_K : P] = [\mathbb{Z}_p[x] : Q]$.

under the correspondence.

Now the prime ideals $Q \lhd \mathbb{Z}_p[x]$ containing $\langle \overline{m}_\alpha(x)\rangle$ are precisely $Q_i := \langle \bar{g}_i(x)\rangle$, where $\overline{m}_\alpha(x) = \prod_{i=1}^k \bar{g}_i(x)^{e_i}$ with $\bar{g}_i(x)$ irreducible in $\mathbb{F}_p[x]$. Tracing the correspondences, one can check that Q_i correspond to the prime ideals

$$P_i := (p, g_i(\alpha))$$

as given in the theorem.

So we are left to prove $(p) = \prod_{i=1}^k P_i^{e_i}$. To see this, Since $\prod_i g_i(x)^{e_i} \equiv m_\alpha(x) \pmod p$, we have $\prod_i g_i(\alpha)^{e_i} \in (p)$. Hence

$$\prod_{i=1}^k P_i^{e_i} = \prod_{i=1}^k (p, g_i(\alpha))^{e_i} \subseteq \left(p, \prod_{i=1}^k g_i(\alpha)^{e_i}\right) = (p).$$

(exercise: check the above $\subseteq$ holds). Comparing norms on both sides:

$$\begin{aligned} N\left(\prod_{i=1}^k P_i^{e_i}\right) = \prod_{i=1}^k [\mathcal{O}_K : P_i]^{e_i} = \prod_{i=1}^k [\mathbb{Z}_p[x] : Q_i]^{e_i} &= \prod_{i=1}^k (p^{\deg(\bar{g}_i(x))})^{e_i} \\ &= p^{\deg(\prod \bar{g}_i(x)^{e_i})} \\ &= p^{\deg(\bar{m}_\alpha(x))} \\ &= p^n = N((p)). \end{aligned}$$

As a consequence, the inclusions are indeed equalities, and hence $(p) = \prod_{i=1}^k P_i^{e_i}$. $\square$

Corollary 7.1.5. *Let $[K:\mathbb{Q}] = n$ be a number field. $\alpha \in \mathcal{O}_K$ be such that $K = \mathbb{Q}(\alpha)$. If $p \in \mathbb{N}$ ramifies in $\mathcal{O}_K$, then $p \mid \Delta^2(1, \alpha, \ldots, \alpha^{n-1})$.*

Example 7.1.6. *Let $K = \mathbb{Q}(\sqrt{-5})$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$. Take $\alpha = \sqrt{-5}$, then $\Delta^2(1, \alpha) = -20$. So the only possible $p \in \mathbb{N}$ that ramifies in $\mathcal{O}_K$ are 2 and 5. Indeed, from Theorem 7.1.2, $(2) = (2, 1+\sqrt{-5})^2$ and $(5) = (\sqrt{-5})^2$ both ramify in $\mathcal{O}_K$.*

Proof of Corollary 7.1.5. The proof is divided into two cases -

Case I: $p \mid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ Since

$$\Delta^2(1, \alpha, \ldots, \alpha^{n-1}) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \Delta^2(K),$$

therefore $p \mid \Delta^2(1, \alpha, \ldots, \alpha^{n-1})$ regardless of whether p ramifies in $\mathcal{O}_K$ or not.

Case II: $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ Let $L : K$ be a splitting field of $m_\alpha(x) \in \mathbb{Z}[x]$, whose roots are $\alpha = \alpha_1, \ldots, \alpha_n \in \mathcal{O}_L$. By Example 5.2.7,

$$\Delta^2((1, \alpha, \ldots, \alpha^{n-1})) = \prod_{i<j} (\alpha_i - \alpha_j)^2 = \Delta_{m_\alpha(x)} \in \mathbb{Z}.$$

where $\Delta_{p(x)}$ is the discriminant of a polynomial $p(x) \in F[x]$ as defined in Theorem 3.3.1. In particular, $\Delta \neq 0$ if and only if $p(x)$ has no repeated roots (in its splitting extension).

By Theorem 3.3.1, the discriminant $\Delta_{p(x)}$ for any $p(x)$ can be expressed in terms of its coefficients (see Homework 1 Question 7 for the formula of Δ for cubic polynomials). Therefore, the discriminant of the reduced polynomial $\overline{m}_\alpha(x) \in \mathbb{Z}_p[x]$ satisfies:

$$\Delta_{\overline{m}_\alpha(x)} = \Delta_{m_\alpha(x)} \mod p \qquad (*)$$

On the other hand, the hypothesis of Theorem 7.1.2 holds in this Case. Since p ramifies in $\mathcal{O}_K$, then by recipe in the factorization of (p) given by Theorem 7.1.2 implies that

$$\overline{m}_\alpha(x) = \overline{g}(x)^e \cdots$$

has a repeated irreducible factor. Consequently, $\overline{m}_\alpha(x)$ has a repeated root γ, and hence

$$\Delta_{\overline{m}_\alpha(x)} \equiv 0 \mod p$$

By $(*)$, this implies $\Delta^2(1, \alpha, \ldots, \alpha^{n-1}) = \Delta_{m_\alpha(x)}$ is a multiple of p, and hence the result follows. □

7.2 Prime Ideals in Galois Extension

Let $K : \mathbb{Q}$ be Galois extension with $\mathrm{Gal}(K/\mathbb{Q}) = \{\sigma_1 = id, \sigma_2, \ldots, \sigma_n\}$. Let $P \lhd \mathcal{O}_K$ be a prime ideal over p with $N(P) = p^f$. Then

- $\sigma_i(P)$ is also a prime ideal.
- $N(\sigma_i(P)) = N(P) = p^f$.

Let $orb(P) = \{\sigma_i(P) | \sigma_i \in \mathrm{Gal}(K/\mathbb{Q})\} = \{P_1 = P, P_2, \ldots, P_g\}$. By Theorem 0.2.5, $g \mid n = |\mathrm{Gal}(K/\mathbb{Q})|$.

Theorem 7.2.1. *Let $K : \mathbb{Q}$ be Galois extension. Let $P \lhd \mathcal{O}_K$ be a prime ideal with ramification index e and lying above p. Then $(p) = P_1^e \cdots P_g^e$. In particular, all prime ideals lying above p are Galois conjugates. Therefore,*

$$p^{[K:\mathbb{Q}]} = N((p)) = (N(P)^e)^g = p^{efg}$$

and hence $[K : \mathbb{Q}] = efg$.

Example 7.2.2. *Let $K = \mathbb{Q}(\omega)$, where ω is the primitive p-th root of unity. Then $K : \mathbb{Q}$ is Galois and $\mathbb{Z}[\omega] = \langle 1, \omega, \ldots, \omega^{p-2} \rangle_{\mathbb{Z}}$ gives an integral basis. Denote $\underline{\omega} = \{1, \omega, \ldots, \omega^{p-2}\}$, then $\left|\Delta^2(\underline{\omega})\right| = p^{p-2}$.*

Now we study the prime ideals in $\mathcal{O}_K$. We will apply Corollary 7.1.5. Let q be a rational prime.

- *If $q \neq p$, then (q) does not ramify. So $(q) = Q_1 \cdots Q_k$ for some prime ideals $Q_i \lhd \mathcal{O}_K$. $N(Q_i) = q^{f_i}$ for some f_i. Q_i are all Galois conjugates, so all f_i are equal and $kf = p - 1$.*
- *If $q = p$. Then (p) may ramify: Take $\alpha = \omega$, $m_\omega = x^{p-1} + \cdots + x + 1 = \frac{x^p - 1}{x - 1} \in \mathbb{Z}[x]$. Note*

  $$\overline{m_\omega}(x) = \frac{x^p - 1}{x - 1} = \frac{(x-1)^p}{x-1} = (x-1)^{p-1} \in \mathbb{Z}_p[x],$$

  by Theorem 7.1.2, $(p) = (p, \omega - 1)^{p-1}$.

Chapter 8

Determining the Class Group C_K

Recall $C_K = \{[I] | I \lhd \mathcal{O}_K \text{ nonzero ideal}\}$ is a finite abelian group. $|C_K| = 1$ if and only if $\mathcal{O}_K$ is a PID. We will study the structure of C_K and use it to solve Diophantine equations.

8.1 Minkowski's Convex Body Theorem

Let $\mathscr{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be an $\mathbb{R}$-basis of $\mathbb{R}^n$ and let

$$L := \{\sum_{i=1}^{n} a_i \mathbf{v}_i | a_i \in \mathbb{Z}\}$$

be a **lattice** generated by $\mathscr{B}$. The **fundamental domain** of L is

$$D = \{\sum_{i=1}^{n} \alpha_i \mathbf{v}_i | 0 \leq \alpha_i < 1\}.$$

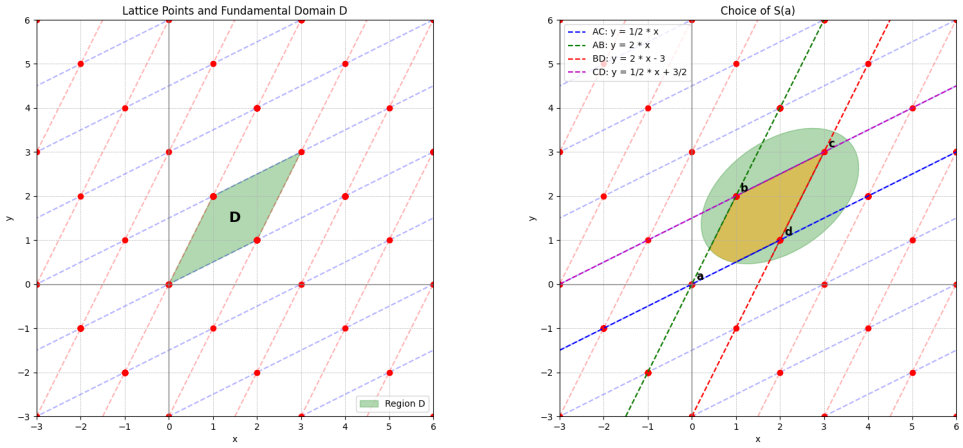
For all $\mathbf{v} \in \mathbb{R}^n$, $\mathbf{v}$ is uniquely expressed as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ where $\mathbf{u} \in L$ and $\mathbf{w} \in D$. Define the volume of L by $vol(L) := vol(D) = |\det(\mathbf{v}_1, \ldots, \mathbf{v}_n)|$.

Proposition 8.1.1. *Let L be a lattice in $\mathbb{R}^n$, $S \subseteq \mathbb{R}^n$ be a bounded and measurable set such that $vol(S) > vol(L)$. Then $\exists \mathbf{x}, \mathbf{y} \in S$ such that $\mathbf{x} \neq \mathbf{y}$ and $\mathbf{x} - \mathbf{y} \in L$.*

Proof. For all $\mathbf{a} \in L$, let $S(\mathbf{a}) = (S - \mathbf{a}) \cap D$. Then $S = \coprod_{\mathbf{a} \in L} S(\mathbf{a}) + \mathbf{a}$. Since

$$vol(S) = \sum_{\mathbf{a} \in L} vol(S(\mathbf{a})) > vol(L),$$

there must be some $\mathbf{a} \neq \mathbf{b} \in L$ such that $S(\mathbf{a}) \cap S(\mathbf{b}) \neq \varnothing$. We may find $\mathbf{x} = \mathbf{v} + \mathbf{a}$, $\mathbf{y} = \mathbf{v} + \mathbf{b} \in S$, so $\mathbf{x} - \mathbf{y} = \mathbf{a} - \mathbf{b} \in L$. □



(a) Fundamental Domain D (b) Example of $S(\mathbf{a})$: Yellow Area

Figure 8.1: $\mathbb{R}^2$ Example

Theorem 8.1.2 (Minkowski's Convex Body)**.** *Let L be a lattice in $\mathbb{R}^n$. $S \subseteq \mathbb{R}^n$ be a bounded, measurable, convex ($\forall \mathbf{x}, \mathbf{y} \in S$ and $\forall 0 \leq t \leq 1$, the linear combination $t\mathbf{x} + (1-t)\mathbf{y} \in S$) and symmetric ($\mathbf{x} \in S \Rightarrow -\mathbf{x} \in S$). Suppose $vol(S) > 2^n vol(L)$, then there exists a nonzero $\mathbf{v} \in S \cap L$.*

Proof. Consider $vol(\frac{1}{2}S) = \frac{1}{2^n} vol(S)$. Then by above proposition, there exists $\mathbf{x}, \mathbf{y} \in \frac{1}{2}S$ such that $\mathbf{x} - \mathbf{y} \in L$. Since S is symmetric, $2\mathbf{x}, -2\mathbf{y} \in S$. By the convexity of S, $\mathbf{x} - \mathbf{y} = \frac{1}{2}(2\mathbf{x}) + \frac{1}{2}(-2\mathbf{y}) \in S$. Since $\mathbf{x} \neq \mathbf{y}$, this is a nonzero element of $S \cap L$. □

Remark 8.1.3. *If S is closed, then one can replace $vol(S) > 2^n vol(L)$ by $vol(S) \geq 2^n vol(L)$. To see so, apply the theorem to $(1 + \frac{1}{n})S$. Then there exist $\mathbf{v}_n \in (1 + \frac{1}{n})S \cap L$. Since*

$$\bigcap_{n=1}^{\infty} (1 + \frac{1}{n})S = S$$

and

$$S \subseteq \cdots \subseteq (1 + \frac{1}{n})S \subseteq 2S,$$

Since $2S \cap L$ is finite, some nonzero lattice point occurs infinitely often among the $\mathbf{v}_n$. Hence there is a nonzero $\mathbf{v} \in S \cap L$.

Example 8.1.4. *Let $p \equiv 1 \pmod 4$ be a rational prime, then $\exists x, y \in \mathbb{Z}$ s.t. $x^2 + y^2 = p$.*

Proof. Let $s \in \mathbb{Z}_p^*$ be an element of order 4, such element exists since $\mathbb{Z}_p^*$ is an cyclic group of order $p - 1$. Then $s^2 = -1 \mod p$.

For $x, y \in \mathbb{Z}$ such that $x^2 + y^2 = 0 \mod p$, then $(x + sy)(x - sy) = 0 \mod p$, i.e. $x = \pm sy \mod p$. By switching $s \to -s$ if necessary, we may assume $x = sy \mod p$. So $x = sy + zp$ for some $z \in \mathbb{Z}$. We may write

$$(x, y) = (sy + zp, y) = (s, 1)y + (p, 0)z.$$

We only need to find such (x, y). Consider the lattice $L = \langle (s, 1), (p, 0) \rangle_{\mathbb{Z}}$, then $vol(L) = p$. Let $S = \{(x, y) | x^2 + y^2 < 2p\}$, then $vol(S) = 2\pi p > 2^2 vol(L)$. By Theorem 8.1.2, there exists $(x, y) \in S \cap L$, i.e. $x^2 + y^2 < 2p$ and $x = sy \mod p$. The only possible case is $x^2 + y^2 = p$. □

8.2 Minkowski's Bound

In this section, we will find a lattice and a bounded, measurable, convex and symmetric set and we will use Theorem 8.1.2 to prove Theorem 8.2.3.

Let $K : \mathbb{Q}$ be a number field with $[K : \mathbb{Q}] = n = r + 2s$, where

$$\sigma_1, \ldots, \sigma_r : K \hookrightarrow \mathbb{R} \quad \text{real embeddings}$$

$$\left.\begin{matrix} \sigma_{r+1}, \ldots, \sigma_{r+s} \\ \overline{\sigma_{r+1}}, \ldots, \overline{\sigma_{r+s}} \end{matrix}\right\} : K \hookrightarrow \mathbb{C} \quad \text{complex embeddings}$$

Example 8.2.1. *1. Let $K = \mathbb{Q}(\sqrt{-5})$, then*

$$\left.\begin{matrix} \sigma_1 : a + b\sqrt{-5} \mapsto a + b\sqrt{-5} \\ \sigma_2 : a + b\sqrt{-5} \mapsto a - b\sqrt{-5} \end{matrix}\right\} \quad \textit{are all complex embeddings}$$

2. Let $K = \mathbb{Q}(\sqrt{3})$, then

$$\left.\begin{matrix} \sigma_1 : a + b\sqrt{3} \mapsto a + b\sqrt{3} \\ \sigma_2 : a + b\sqrt{3} \mapsto a - b\sqrt{3} \end{matrix}\right\} \quad \textit{are all real embeddings}$$

3. Let $K = \mathbb{Q}(\sqrt[3]{2})$, then

$$\sigma_1 : \sqrt[3]{2} \mapsto \sqrt[3]{2} \quad \textit{is real embedding}$$

$$\left.\begin{matrix} \sigma_2 : \sqrt[3]{2} \mapsto \sqrt[3]{2}\omega \\ \overline{\sigma_2} : \sqrt[3]{2} \mapsto \sqrt[3]{2}\omega^2 \end{matrix}\right\} \quad \textit{are complex embeddings}$$

Define $\sigma : K \to \mathbb{R}^r \times \mathbb{C}^s \cong \mathbb{R}^n$ given by

$$\sigma : k \mapsto \begin{pmatrix} \sigma_1(k) \\ \vdots \\ \sigma_r(k) \\ \mathrm{Re}(\sigma_{r+1}(k)) \\ \mathrm{Im}(\sigma_{r+1}(k)) \\ \vdots \\ \mathrm{Re}(\sigma_{r+s}(k)) \\ \mathrm{Im}(\sigma_{r+s}(k)) \end{pmatrix}$$

Remark 8.2.2. *Let $\{k_1, \ldots, k_n\} \subseteq K$ be a $\mathbb{Q}$-basis of K (e.g. integral basis $\{\omega_1, \ldots, \omega_n\}$ of $\mathcal{O}_K$). Then*

$$\det \begin{pmatrix} \uparrow & & \uparrow \\ \sigma(k_1) & \ldots & \sigma(k_n) \\ \downarrow & & \downarrow \end{pmatrix} = \det \begin{pmatrix} \sigma_1(k_1) & & \sigma_1(k_n) \\ \vdots & & \vdots \\ \sigma_r(k_1) & & \sigma_r(k_n) \\ \mathrm{Re}(\sigma_{r+1}(k_1)) & & \mathrm{Re}(\sigma_{r+1}(k_n)) \\ \mathrm{Im}(\sigma_{r+1}(k_1)) & \cdots & \mathrm{Im}(\sigma_{r+1}(k_n)) \\ \vdots & & \vdots \\ \mathrm{Re}(\sigma_{r+s}(k_1)) & & \mathrm{Re}(\sigma_{r+s}(k_n)) \\ \mathrm{Im}(\sigma_{r+s}(k_1)) & & \mathrm{Im}(\sigma_{r+s}(k_n)) \end{pmatrix}$$

$$= \frac{1}{(2i)^s} \det \begin{pmatrix} \sigma_1(k_1) & & \sigma_1(k_n) \\ \vdots & & \vdots \\ \sigma_r(k_1) & & \sigma_r(k_n) \\ \sigma_{r+1}(k_1) & \cdots & \sigma_{r+1}(k_n) \\ \overline{\sigma_{r+1}}(k_1) & & \overline{\sigma_{r+1}}(k_n) \\ \vdots & & \vdots \\ \overline{\sigma_{r+s}}(k_1) & & \overline{\sigma_{r+s}}(k_n) \end{pmatrix}$$

(the last equality comes from simple elementary row operation $\begin{pmatrix} x \\ y \end{pmatrix} \to \begin{pmatrix} x+yi \\ x-yi \end{pmatrix}$). In particular,

$$\det \begin{pmatrix} \uparrow & & \uparrow \\ \sigma(\omega_1) & \ldots & \sigma(\omega_n) \\ \downarrow & & \downarrow \end{pmatrix} = \frac{1}{(2i)^s} \det(\sigma_i(\omega_j)) = \frac{1}{(2i)^s} \Delta(\underline{\omega}) = \frac{1}{(2i)^s} \sqrt{\Delta^2(K)}.$$

Now we consider $I \lhd \mathcal{O}_K$ a nonzero ideal. Note that $[\mathcal{O}_K : I] = N(I) < \infty$. $\mathcal{O}_K = \langle \omega_1, \ldots, \omega_n \rangle_{\mathbb{Z}}$ where $\omega_1, \ldots, \omega_n$ is the integral basis (also a $\mathbb{Q}$-basis of K). Then $\mathcal{O}_K$ forms a lattice generated by $\{\sigma(\omega_1), \ldots, \sigma(\omega_n)\}$.

I is a sublattice of finite index so $I = \langle k_1, \ldots, k_n \rangle_{\mathbb{Z}}$, where $\{k_1, \ldots, k_n\}$ is also a $\mathbb{Q}$-basis of K. Denote the sublattice correspond to I by L_I, which is generated by $\{\sigma(k_1), \ldots, \sigma(k_n)\}$. By the previous remark,

$$\begin{aligned} vol(L_I) &= |\det(\sigma(k_1), \ldots, \sigma(k_n))| = \frac{1}{2^s}\sqrt{|\Delta^2(k_1, \ldots, k_n)|} \\ &= \frac{1}{2^s}\sqrt{[\mathcal{O}_K : I]^2 \, |\Delta^2(K)|} = \frac{N(I)}{2^s}\sqrt{|\Delta^2(K)|}. \end{aligned}$$

This L_I will serve as the lattice.

For $t > 0$, define set

$$R_t = \{ \begin{pmatrix} x_1 \\ \vdots \\ x_r \\ z_{r+1} \\ \vdots \\ z_{r+s} \end{pmatrix} \in \mathbb{R}^r \times \mathbb{C}^s \cong \mathbb{R}^n | \sum_{i=1}^{r} |x_i| + \sum_{j=1}^{s} 2\,|z_{r+j}| \leq t \}.$$

Then clearly R_t is compact, convex and symmetric, whose volume is $vol(R_t) = \frac{2^r t^n (\frac{\pi}{2})^s}{n!}$.

Theorem 8.2.3. *Let $I \lhd \mathcal{O}_K$ be a nonzero ideal, then $\exists \alpha \in I \backslash \{0\}$ such that*

$$\left| Norm_{K/\mathbb{Q}}(\alpha) \right| \leq m_K N(I),$$

where $m_K = (\frac{4}{\pi})^s \frac{n!}{n^n}\sqrt{|\Delta^2(K)|}$.

Proof. Take $t > 0$ such that $\frac{\pi^s t^n}{n!} = 4^s \sqrt{|\Delta^2(K)|} N(I)$ and consider $R_t \subseteq \mathbb{R}^n$.

$$vol(R_t) = \frac{2^r t^n (\frac{\pi}{2})^s}{n!} = 2^{r+2s} vol(L_I) = 2^n vol(L_I).$$

Since R_t is compact, by the closed-set version of Theorem 8.1.2 in the preceding remark, there exists $\alpha \in I \backslash \{0\}$ s.t. $\sigma(\alpha) \in R_t$, i.e.

$$\sum_{i=1}^{r} |\sigma_i(\alpha)| + \sum_{j=1}^{s} 2\,|\sigma_{r+j}(\alpha)| \leq t.$$

So

$$\frac{1}{n}\left(\sum_{i=1}^{r} |\sigma_i(\alpha)| + \sum_{j=1}^{s} (|\sigma_{r+j}(\alpha)| + |\overline{\sigma_{r+j}}(\alpha)|) \right) \leq \frac{t}{n},$$

by AM-GM inequality,

$$\sqrt[n]{\prod_{\sigma\in \mathrm{Hom}(K,\mathbb{C})} |\sigma(\alpha)|} \leq \frac{t}{n}.$$

Hence

$$\left|Norm_{K/\mathbb{Q}}(\alpha)\right| \leq (\frac{t}{n})^n = (\frac{4}{\pi})^s \frac{n!}{n^n}\sqrt{|\Delta^2(K)|}N(I).$$

□

Corollary 8.2.4. *Let $[K : \mathbb{Q}] = n$ be a number field. For any $c \in C_K = \{[I]|I \lhd \mathcal{O}_K$ nonzero$\}$, there exists $J \lhd \mathcal{O}_K$ such that $c = [J]$ and $N(J) \leq m_K$.*

Proof. Fix $c \in C_K$ and consider $I \lhd \mathcal{O}_K$ such that $[I] = c^{-1}$. By Theorem 8.2.3, there exists $\alpha \in I\backslash\{0\}$ such that $\left|Norm_{K/\mathbb{Q}}(\alpha)\right| \leq m_K N(I)$. Note that

$$(\alpha) \subseteq I \quad \Rightarrow \quad I \mid (\alpha) \quad \Rightarrow \quad \text{there exists } J \lhd \mathcal{O}_K \text{ such that } IJ = (\alpha).$$

So

$$N(I)N(J) = N(IJ) = N((\alpha)) = \left|Norm_{K/\mathbb{Q}}(\alpha)\right| \leq m_K N(I)$$

which gives $N(J) \leq m_K$. Moreover, since $IJ = (\alpha)$, $[I] * [J] = [(\alpha)] = e$ and hence $[J] = [I]^{-1} = c$. So the result follows. □

As an immediate consequence of the above corollary, one has $C_K = \{[J] \mid N(J) \leq m_K\}$. In other words, we can list all elements in C_K by finding all ideals J of norm less than or equal to m_K.

8.3 Calculating Class Groups

Example 8.3.1. *Let $K = \mathbb{Q}(\sqrt{-5})$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$. Then $r = 0$ and $s = 1$, $\left|\Delta^2(K)\right| = 20$, so $m_K = \frac{4\sqrt{5}}{\pi} \approx 2.\cdots$. So*

$$C_K = \{[J] \mid N(J) \leq 2\}.$$

So we only have to study ideals with $N(J) = 1$ or 2. In the case $N(J) = 1$, $J = \mathcal{O}_K$ and hence $[J] = e$ is the identity element in C_K. As for $N(J) = 2$, $[\mathcal{O}_K : J] = 2$, so $J \lhd \mathcal{O}_K$ must be a prime ideal lying above 2. By Theorem 7.1.2, the only possibility is the non-principal ideal $J = (2, 1 + \sqrt{-5})$. Therefore,

$$C_K = \{e := [\mathcal{O}_K], \quad \tau := [(2, 1 + \sqrt{-5})]\}$$

has only two elements, i.e. $C_K \cong \mathbb{Z}_2$. Alternatively, one can check directly that $\tau^2 = [(2, 1 + \sqrt{-5})^2] = [(2)] = e$.

Example 8.3.2. *Let $K = \mathbb{Q}(\sqrt{-6}), \mathcal{O}_K = \mathbb{Z}[\sqrt{-6}]$. One can compute $m_K = \frac{2}{\pi}\sqrt{24} \approx 3.\cdots$. Therefore*

$$C_K = \{[J] \mid N(J) = 1, 2, 3\}$$

Note that

$$\begin{aligned} N(J_1) = 1 &\Leftrightarrow J_1 = \mathcal{O}_K \\ N(J_2) = 2 &\Leftrightarrow J_2 \triangleleft \mathcal{O}_K \text{ is a maximal ideal above } 2 \\ &\Leftrightarrow J_2 = (2, \sqrt{-6}), \qquad \text{since } (2) = (2, \sqrt{-6})^2, \\ N(J_3) = 3 &\Leftrightarrow J_3 = (3, \sqrt{-6}). \end{aligned}$$

Therefore,

$$C_K = \{e = [\mathcal{O}_K], [J_2], [J_3]\}.$$

We can easily check that $[J_2] \neq e$ is not principal, otherwise suppose $J_2 = (a + b\sqrt{-6})$ for some $a, b \in \mathbb{Z}$, $N(J_2) = a^2 + 6b^2 = 2$. The only choice for b is $b = 0$, implies $a^2 = 2$, $a \notin \mathbb{Z}$ contradiction. Similarly one has $[J_3] \neq e$. Also, note that

$$[J_2]^2 = [(2)], \quad [J_3]^2 = [(3)] \qquad \Rightarrow \qquad [J_2]^2 = [J_3]^2 = e.$$

which implies C_K has no elements of order 3. Thus $C_K \ncong \mathbb{Z}_3$, implying that $[J_2] = [J_3]$ and

$$C_K = \{e, [J_2] = [J_3]\} \cong \mathbb{Z}_2.$$

Alternatively, one can also check directly that $[J_2] = [J_3]$: since $\sqrt{-6} = 3\sqrt{-6} + \sqrt{-6}(-2)$ we have

$$\sqrt{-6} \in J_3 J_2 \quad \Rightarrow \quad (\sqrt{-6}) \subseteq J_3 J_2 \quad \Rightarrow \quad J_3 J_2 | (\sqrt{-6})$$

By computing norm $N(J_2)N(J_3) = 2 \times 3 = 6 = N\left((\sqrt{-6})\right)$, we can conclude that

$$J_3 J_2 = (\sqrt{-6}) \quad \Rightarrow \quad [J_3] * [J_2] = [(\sqrt{-6})] = e \quad \Rightarrow \quad [J_3] = [J_2]^{-1} = [J_2]$$

since $[J_2]^2 = e$.

Remark 8.3.3. *By Minkowski Bound, we can easily have*

$$C_K = \{[J] | N(J) \leq m_K\}.$$

If we only care about generators of C_K, it is enough to study prime ideals P with $N(P) \leq m_K$. Since the norm of a prime ideal is a prime power p^f, it is enough to inspect prime ideals lying above rational primes $p \leq m_K$.

For instance, suppose $N(J) = 6$. Then for all $\alpha \in \mathcal{O}_K$, $6\alpha \in J$ (since $N(J) = |\mathcal{O}_K/J| = 6$). We have

$$(6) \subseteq J \quad \Rightarrow \quad J|(6) \quad \Rightarrow \quad J|(2)(3).$$

By Dedekind's theorem, J can be factorized into prime ideals lying above 2 and 3.

Example 8.3.4. *Let $K = \mathbb{Q}(\sqrt{-163})$ and $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{-163}}{2}]$. One can easily compute $m_K = \frac{4}{\pi}\frac{1}{2}\sqrt{163} = 8.\cdots$.*

By the Remark above,

$$C_K = \langle [J] \mid N(J) = 2, 3, 5, 7 \rangle$$

So we can apply theorem 7.1.2, with $\alpha = \frac{1+\sqrt{-163}}{2}$ to find the ideals J lying above $2, 3, 5, 7$. Since the minimal polynomial of α is

$$m_\alpha(x) = x^2 - x + 41,$$

we can check that

(2): $\bar{m}_\alpha(x) = x^2 + x + 1$ irreducible in $\mathbb{Z}_2[x] \Rightarrow (2)$ is prime;

(3): $\bar{m}_\alpha(x) = x^2 + 2x + 2$ irreducible in $\mathbb{Z}_3[x] \Rightarrow (3)$ is prime;

(5): $\bar{m}_\alpha(x) = x^2 - x + 1$ irreducible in $\mathbb{Z}_5[x] \Rightarrow (5)$ is prime;

(7): $\bar{m}_\alpha(x) = x^2 - x - 1$ irreducible in $\mathbb{Z}_7[x] \Rightarrow (7)$ is prime.

Thus the prime ideals above $2, 3, 5, 7$ are the principal ideals $(2), (3), (5), (7)$. In particular, the only one of these with norm at most m_K is (2), with norm 4, and its class is trivial. Hence

$$C_K = \{e\}.$$

Therefore $\mathcal{O}_K$ is a PID.

Example 8.3.5. *Let $K = \mathbb{Q}(\sqrt{-29})$ and $\mathcal{O}_K = \mathbb{Z}[\sqrt{-29}]$. One can easily compute $m_K = \frac{2}{\pi}\sqrt{116} = 6.\cdots$. So we need to study $J \triangleleft \mathcal{O}_K$ with $N(J) = 2, 3, 5$, where*

$$\begin{aligned}(2) &= P^2 = (2, 1 + \sqrt{-29})^2;\\ (3) &= QQ' = (3, 1 + \sqrt{-29})(3, 1 - \sqrt{-29});\\ (5) &= RR' = (5, 1 + \sqrt{-29})(5, 1 - \sqrt{-29}).\end{aligned}$$

we can get $C_K = \langle [P], [Q], [Q'], [R], [R'] \rangle$. We want to study the relation between ideal classes $[P], [Q], [R], [Q'], [R']$. Obviously

$$[P]^2 = [Q][Q'] = [R][R'] = e.$$

Hence $[Q'] = [Q]^{-1}$ and $[R'] = [R]^{-1}$, so

$$C_K = \langle [P], [Q], [R] \rangle.$$

Moreover, P is not principal since $a^2 + 29b^2 = 2$ has no integer solutions. Thus $[P]$ has order 2.

To look at the orders of $[Q]$ and $[R]$, one needs to do some trial and error. By observation, $\mathrm{Norm}_{K/\mathbb{Q}}(3+2\sqrt{-29}) = 3^2 + 4\cdot 29 = 125 = 5^3$*, so we study the principal ideal* $I = (3+2\sqrt{-29})$.

By the argument given in Remark 8.3.3,

$$I \mid (125) = (5)^3 = R^3(R')^3,$$

thus there are 4 possible ways in prime factorization as follows

$$I = R^3, R^2R', R(R')^2, (R')^3.$$

Suppose $R|I$, implying that $I \subseteq R$, we have

$$3+2\sqrt{-29} \in I \subseteq R, \qquad 2(1+\sqrt{-29}) \in R,$$

Therefore,

$$\left\{(3+2\sqrt{-29}) - 2(1+\sqrt{-29})\right\} \in R \quad \Rightarrow \quad 1 \in R \quad \Rightarrow \quad R = \mathcal{O}_K,$$

which is a contradiction. Therefore the only choice for prime factorization is $I = (R')^3$, and we can compute

$$[R']^3 = [I] = e \quad \Rightarrow \quad \left([R]^{-1}\right)^3 = e \quad \Rightarrow \quad [R]^3 = e.$$

Since R is not principal (the equation $a^2 + 29b^2 = 5$ has no integer solutions), we conclude that the order of $[R]$ is 3.

Now we study $I' = (1+\sqrt{-29})$ with $N(I') = 30$, one can easily check that $I'|(2)(3)(5)$, which implies that

$$I'|P^2 \cdot QQ' \cdot RR'.$$

Since $N(I') = 30 = 2\times 3\times 5$, therefore we have

$$I' = \begin{cases} PQR \\ PQ'R \\ PQR' \\ PQ'R' \end{cases} \quad \Rightarrow \quad [I'] = \begin{cases} [P][Q][R] \\ [P][Q]^{-1}[R] \\ [P][Q][R]^{-1} \\ [P][Q]^{-1}[R]^{-1} \end{cases}$$

hence $[Q]$ can be expressed as

$$[Q] = [P][R] \quad or \quad [P][R]^{-1} \in \langle [P], [R] \rangle.$$

As a conclusion,

$$C_K = \langle [P], [R] \mid [P]^2 = e,\ [R]^3 = e \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6.$$

8.4 Applications to Diophantine Equations

We will end by presenting two examples on how the knowledge on C_K helps solve certain Diophantine equations:

Example 8.4.1. *Show that there are no integer solutions to* $Y^2 + 37 = X^3$.

Proof. Let $K = \mathbb{Q}(\sqrt{-37})$, with $\mathcal{O}_K = \mathbb{Z}[\sqrt{-37}]$. One can check that $C_K \cong \mathbb{Z}_2$. Then the equation can be factorized as:

$$\{Y + \sqrt{-37}\}\{Y - \sqrt{-37}\} = X^3$$

Claim 1. $2, 37 \nmid X$.

To see so, suppose on contrary that $2 \mid X$, then $X^3 = Y^2 + 37 \equiv 0 \mod 8$, which implies $Y^2 \equiv 3 \mod 8$. However, one can easily check that all square numbers can only have $n^2 \equiv 0, 1, 4 \mod 8$.

On the other hand, if $37 \mid X$, then obviously one also has $37 \mid Y$. Write $X = 37a$ and $Y = 37b$, then

$$(37b)^2 + 37 = (37a)^3 \quad \Rightarrow \quad 37b^2 + 1 = 37^2a^3 \equiv 0 \mod 37$$

But $37b^2 + 1 \equiv 1 \mod 37$, which again gives a contradiction. □

Claim 2. The principal ideals $(Y + \sqrt{-37})$ & $(Y - \sqrt{-37})$ are coprime.

Suppose on contrary there exists a prime ideal $P \mid (Y + \sqrt{-37})$ and $P \mid (Y - \sqrt{-37})$. Thus

$$(Y \pm \sqrt{-37}) \subseteq P \quad \Rightarrow \quad Y \pm \sqrt{-37} \in P \quad \Rightarrow \quad 2\sqrt{-37} = \{Y + \sqrt{-37}\} - \{Y - \sqrt{-37}\} \in P$$

and hence

$$(2\sqrt{-37}) \subseteq P \quad \Rightarrow \quad P \mid (2\sqrt{-37}) = (2)(\sqrt{-37}) \quad \Rightarrow \quad P \mid (2) \text{ or } P \mid (\sqrt{-37})$$

Therefore, $P = (2, 1 + \sqrt{-37})$ or $(\sqrt{-37})$.

Suppose $P = (2, 1 + \sqrt{-37})$, then

$$\begin{aligned} (2) = P^2 | (Y + \sqrt{-37})(Y - \sqrt{-37}) = (Y^2 + 37) = (X^3) \quad &\Rightarrow \quad N((2)) \mid N((X^3)) \\ &\Rightarrow \quad 4 \mid X^6 \\ &\Rightarrow \quad 2 \mid X, \end{aligned}$$

contradicting Claim 1.

On the other hand, suppose $P = (\sqrt{-37})$, then

$$\begin{aligned} Y + \sqrt{-37} \in (\sqrt{-37}) \ &\Rightarrow \ Y = \{Y + \sqrt{-37}\} - \sqrt{-37} \in (\sqrt{-37}) \\ &\Rightarrow \ (\sqrt{-37}) \mid (Y) \\ &\Rightarrow \ 37 = N((\sqrt{-37})) \mid N((Y)) = Y^2 \end{aligned}$$

Hence $37 \mid Y$, and the original equation implies that $37 \mid X$, contradicting Claim 1 again. □

Consider the ideal factorization

$$(Y + \sqrt{-37})(Y - \sqrt{-37}) = (X)^3 = P_1^{3e_1} P_2^{3e_2} \cdots P_r^{3e_r}$$

where $(X) = P_1^{e_1} P_2^{e_2} \dots P_r^{e_r}$ is the prime ideal factorization of (X). By Claim 2 and Dedekind's Theorem, if the prime ideal $P_i \mid (Y + \sqrt{-37})$, then $P_i \nmid (Y - \sqrt{-37})$ and hence $P_i^{3e_i} \mid (Y + \sqrt{-37})$. Therefore,

$$(Y + \sqrt{-37}) = I^3 \qquad (*)$$

where $I = P_{i_1}^{e_{i_1}} \dots P_{i_l}^{e_{i_l}}$.

Now study ideal class group:

$$e = [(Y + \sqrt{-37})] = [I]^3$$

Since $|C_K| = 2$, $[I]^2 = e$ and hence $[I]^3 = [I]$, which gives

$$e = [(Y + \sqrt{-37})] = [I],$$

i.e. $I = (a + b\sqrt{-37})$ itself is a principal ideal. Hence $(*)$ gives

$$\begin{aligned} & (Y + \sqrt{-37}) = (a + b\sqrt{-37})^3 = (\{a^3 - 111ab^2\} + \{3a^2b - 37b^3\}\sqrt{-37}) \\ \Rightarrow \quad & Y + \sqrt{-37} = \pm\{\{a^3 - 111ab^2\} + \{3a^2b - 37b^3\}\sqrt{-37}\} \end{aligned}$$

Since $\{1, \sqrt{-37}\}$ is a $\mathbb{Q}$-basis of K, they are linearly independent and hence

$$Y = \pm a(a^2 - 111b^2), \qquad 1 = \pm b(3a^2 - 37b^2).$$

But the second equation implies that $b = \pm 1$ and hence $3a^2 - 37 = \pm 1$, which implies $3a^2 = 38$ or 36. But this is impossible.

Consequently, the original equation $Y^2 + 37 = X^3$ has no integer solutions. □

Example 8.4.2. *We have seen that for $K = \mathbb{Q}(\sqrt{-6})$ and $\mathcal{O}_K = \mathbb{Z}[\sqrt{-6}]$, $C_K \cong \mathbb{Z}_2$. We will use it to find solutions of the equation:*

$$Y^2 + 54 = X^3 \quad (*)$$

Proof. **Claim 1: $2, 3 \nmid Y$.**

Suppose on contrary $2 \mid Y$, then obviously $2 \mid X$ and hence $8 \mid X^3 = Y^2 + 54$. But $Y^2 + 54 \equiv 2 \mod 4$.

Now suppose $3 \mid Y$. Then $3|X$. Write $Y = 3a$ and $X = 3b$, one has

$$9a^2 + 54 = 27b^3 \quad \Rightarrow \quad a^2 + 6 = 3b^3$$

which forces $3 \mid a \Rightarrow a = 3c$ for some integer c. Thus

$$(3c)^2 + 6 = 3b^3 \quad \Rightarrow \quad 3c^2 + 2 = b^3.$$

But $b^3 \equiv 0, 1, 8 \mod 9$, while $3c^2 + 2 \equiv 2, 5 \mod 9$. □

Claim 2: $(Y + 3\sqrt{-6}), (Y - 3\sqrt{-6})$ **has no common prime factors.**

Suppose on contrary $P \mid (Y \pm 3\sqrt{-6})$, then as before

$$2 \cdot 3\sqrt{-6} \in P \ \Rightarrow \ P \mid (6\sqrt{-6}) = -(\sqrt{-6})^3 \ \Rightarrow \ P \mid (\sqrt{-6}) \ \Rightarrow \ (\sqrt{-6}) \subseteq P \ \Rightarrow \ \sqrt{-6} \in P$$

Thus

$$\sqrt{-6}, Y + 3\sqrt{-6} \in P \ \Rightarrow \ Y \in P \ \Rightarrow \ P \mid (Y) \ \Rightarrow \ N(P) \mid N((Y)) = Y^2$$

But $P \mid (\sqrt{-6})$ implies that $N(P) \mid N((\sqrt{-6})) = 6$. Since P is prime, $N(P)$ is 2 or 3, so either $2 \mid Y$ or $3 \mid Y$, contradicting Claim 1. □

As in Example 1, one can use Claim 2 and the fact that $|C_K| = 2$ to conclude that

$$(Y + 3\sqrt{-6}) = (a + b\sqrt{-6})^3 = (\{a^3 - 18ab^2\} + \{3a^2b - 6b^3\}\sqrt{-6})$$

for some $a, b \in \mathbb{Z}$, which yields

$$Y = \pm a(a^2 - 18b^2) \qquad 3 = \pm 3b(a^2 - 2b^2)$$

So $b = \pm 1$ and $a = \pm 1$ from the second equation, forcing $Y = \pm 17$ in the first equation. Putting it back to the original equation, one has $X = 7$.

In conclusion, the only solutions of the Diophantine equation are

$$(X, Y) = (7, \pm 17).$$

□

Chapter 9

A First Look at Class Field Theory

Throughout the whole course, we studied two main topics:

- Galois extensions $L : K$ and Galois group $\mathrm{Gal}(L/K)$;
- The ideal class group C_K, describing factorization of ideals in $\mathcal{O}_K$.

Class field theory connects these two subjects in the special case when $\mathrm{Gal}(L/K)$ is abelian. Very roughly, it describes finite abelian extensions of K in terms of ideals of $\mathcal{O}_K$ together with congruence conditions on generators of principal ideals.

We first define the Artin map on the free abelian group generated by the unramified prime ideals. We then interpret this group concretely using fractional ideals and recover the ordinary ideal class group as a quotient. This leads naturally to the Hilbert class field, and finally to ray class groups, where congruence conditions account for ramification.

The general proofs of class field theory are beyond the scope of this course. Our goal is to understand the statements, the elementary constructions that lead to them, and several important examples.

Throughout this chapter, we will freely use the analogues over a general number field K of the prime-factorization results proved earlier over $\mathbb{Q}$.

9.1 Decomposition Groups and Frobenius Automorphisms

Let $L : K$ be a finite Galois extension of number fields with rings of integers $\mathcal{O}_L$, $\mathcal{O}_K$. Write

$$G = \mathrm{Gal}(L/K).$$

Let $P \lhd \mathcal{O}_K$ be a nonzero prime ideal and let $\mathfrak{P} \lhd \mathcal{O}_L$ be a prime ideal lying above P, i.e.

$$\mathfrak{P} \cap \mathcal{O}_K = P.$$

The quotient rings

$$k(P) := \mathcal{O}_K/P, \qquad k(\mathfrak{P}) := \mathcal{O}_L/\mathfrak{P}$$

are finite fields. Put

$$q := |k(P)| = N(P).$$

If

$$f := f(\mathfrak{P}/P) = [k(\mathfrak{P}) : k(P)]$$

is the residue degree, then

$$|k(\mathfrak{P})| = N(\mathfrak{P}) = q^f.$$

Definition 9.1.1. *The* **decomposition group** *of* $\mathfrak{P}$ *over* P *is*

$$D(\mathfrak{P}/P) := \{\sigma \in G \mid \sigma(\mathfrak{P}) = \mathfrak{P}\}.$$

Equivalently, it is the stabilizer of $\mathfrak{P}$ *under the natural action of* G *on the prime ideals of* $\mathcal{O}_L$ *lying above* P*.*

Recall from Section 7.2 that G acts transitively on the prime ideals of $\mathcal{O}_L$ lying above P. Let

$$\mathfrak{P}_1 = \mathfrak{P}, \mathfrak{P}_2, \ldots, \mathfrak{P}_g$$

be the distinct primes above P. Then

$$P\mathcal{O}_L = \mathfrak{P}_1^e \cdots \mathfrak{P}_g^e, \qquad N(\mathfrak{P}_i) = N(P)^f.$$

Taking absolute norms gives

$$N(P)^{[L:K]} = N(P)^{efg},$$

so

$$\boxed{[L : K] = efg.}$$

By the orbit–stabilizer theorem,

$$|D(\mathfrak{P}/P)| = \frac{|G|}{g} = \frac{[L : K]}{g} = ef.$$

We now explain the three integers e, f, g directly in terms of the Galois group. The number g comes from the orbit of a prime, e from its inertia group, and f from the induced action on its residue field.

Every $\sigma \in D(\mathfrak{P}/P)$ preserves $\mathfrak{P}$, so it induces an automorphism of the residue field:

$$\overline{\sigma} : k(\mathfrak{P}) \longrightarrow k(\mathfrak{P}), \qquad \alpha + \mathfrak{P} \longmapsto \sigma(\alpha) + \mathfrak{P}.$$

Since σ fixes K, this induced automorphism fixes $k(P)$. Hence we obtain a group homomorphism

$$D(\mathfrak{P}/P) \longrightarrow \mathrm{Gal}(k(\mathfrak{P})/k(P)).$$

Definition 9.1.2. *The kernel*

$$I(\mathfrak{P}/P) := \ker\left(D(\mathfrak{P}/P) \longrightarrow \mathrm{Gal}(k(\mathfrak{P})/k(P))\right)$$

is called the **inertia group** *of* $\mathfrak{P}$ *over* P.

We quote the following standard theorem on prime ideals in a finite Galois extension.

Theorem 9.1.3. *Let* $L : K$ *be a finite Galois extension and let* $\mathfrak{P}$ *lie above* P. *Let* e, f, g *be as given above. Then the following sequence*

$$1 \longrightarrow I(\mathfrak{P}/P) \longrightarrow D(\mathfrak{P}/P) \longrightarrow \mathrm{Gal}(k(\mathfrak{P})/k(P)) \longrightarrow 1$$

is exact. Consequently,

$$|I(\mathfrak{P}/P)| = e,$$

because $|D(\mathfrak{P}/P)| = ef$ *and* $|\mathrm{Gal}(k(\mathfrak{P})/k(P))| = f$.

The important case for us is when P is unramified, i.e. $e = 1$. Then $I(\mathfrak{P}/P) = \{id\}$ and therefore

$$D(\mathfrak{P}/P) \cong \mathrm{Gal}(k(\mathfrak{P})/k(P)).$$

Since $k(\mathfrak{P})$ is a finite extension of the finite field $k(P)$, its Galois group is cyclic of order f and is generated by the finite-field Frobenius

$$x \longmapsto x^q, \qquad q = |k(P)| = N(P).$$

This motivates the following definition.

Definition 9.1.4. *Assume that* P *is unramified in* L. *The* **Frobenius automorphism** *of* $\mathfrak{P}$ *over* P *is the unique element*

$$\mathrm{Frob}_{\mathfrak{P}} \in D(\mathfrak{P}/P)$$

such that

$$\mathrm{Frob}_{\mathfrak{P}}(\alpha) \equiv \alpha^{N(P)} \pmod{\mathfrak{P}}$$

for every $\alpha \in \mathcal{O}_L$.

This convention is sometimes called the **arithmetic Frobenius** convention.

Proposition 9.1.5. *Let* P *be unramified in* L *and let* $\sigma \in G$. *Then*

$$\mathrm{Frob}_{\sigma(\mathfrak{P})} = \sigma\,\mathrm{Frob}_{\mathfrak{P}}\,\sigma^{-1}.$$

Proof. Let $\tau = \mathrm{Frob}_{\mathfrak{P}}$. For every $\alpha \in \mathcal{O}_L$,

$$\tau(\alpha) \equiv \alpha^{N(P)} \pmod{\mathfrak{P}}.$$

Applying σ gives

$$\sigma\tau(\alpha) \equiv \sigma(\alpha)^{N(P)} \pmod{\sigma(\mathfrak{P})}.$$

Replacing α by $\sigma^{-1}(\alpha)$, we obtain

$$\sigma\tau\sigma^{-1}(\alpha) \equiv \alpha^{N(P)} \pmod{\sigma(\mathfrak{P})}.$$

By the uniqueness in Definition 9.1.4, $\sigma\tau\sigma^{-1} = \mathrm{Frob}_{\sigma(\mathfrak{P})}$. □

Thus for a general Galois extension, a prime P determines a conjugacy class of Frobenius elements. If $L : K$ is abelian, conjugation does nothing, and the Frobenius depends only on P.

Definition 9.1.6. *Let $L : K$ be a finite abelian extension and let P be unramified. The element*

$$\left(\frac{L/K}{P}\right) := \mathrm{Frob}_P \in \mathrm{Gal}(L/K)$$

is called the **Artin symbol** *of P.*

Proposition 9.1.7. *Let $L : K$ be finite abelian and let P be unramified. Then*

$$\mathrm{ord}\left(\frac{L/K}{P}\right) = f.$$

In particular,

$$P \text{ splits completely in } L \iff \left(\frac{L/K}{P}\right) = id.$$

Proof. Under the isomorphism

$$D(\mathfrak{P}/P) \cong \mathrm{Gal}(k(\mathfrak{P})/k(P)),$$

the Frobenius corresponds to the generator $x \mapsto x^{N(P)}$ of a cyclic group of order f. Hence its order is f. Since P is unramified, Theorem 9.1.3 gives $[L : K] = fg$. Thus P splits completely if and only if $g = [L : K]$, if and only if $f = 1$, if and only if its Frobenius is the identity. □

9.2 Examples of Frobenius Map

Example 9.2.1 (Quadratic fields)**.** *Let $K = \mathbb{Q}$ and $L = \mathbb{Q}(\sqrt{d})$ where d is squarefree, and let*

$$\mathrm{Gal}(L/\mathbb{Q}) = \{id, \sigma\}, \qquad \sigma(\sqrt{d}) = -\sqrt{d}.$$

Let p be an odd rational prime which is unramified in L. By Theorem 7.1.2,

$$(p) \text{ splits in } \mathcal{O}_L \quad \Leftrightarrow \quad x^2 - d \text{ has a root modulo } p \quad \Leftrightarrow \quad \left(\tfrac{d}{p}\right) = 1.$$

Therefore

$$\boxed{\left(\frac{L/\mathbb{Q}}{p}\right) = \begin{cases} id, & \left(\frac{d}{p}\right) = 1, \\ \sigma, & \left(\frac{d}{p}\right) = -1. \end{cases}}$$

Example 9.2.2 (Cyclotomic fields)**.** *Let $L = \mathbb{Q}(\zeta_m)$, where ζ_m is a primitive m-th root of unity. We have already seen that*

$$\mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^*,$$

where $a \in (\mathbb{Z}/m\mathbb{Z})^*$ corresponds to the automorphism $\sigma_a(\zeta_m) = \zeta_m^a$.

For a rational prime $p \nmid m$, the prime p is unramified in L. Consider $\sigma_p \in \mathrm{Gal}(L/\mathbb{Q})$ given by

$$\sigma_p(\zeta_m) = \zeta_m^p.$$

Since $\mathcal{O}_L = \mathbb{Z}[\zeta_m]$, every $\alpha \in \mathcal{O}_L$ has the form $\alpha = f(\zeta_m)$ with $f(x) \in \mathbb{Z}[x]$. Modulo a prime $\mathfrak{P}$ above p we are in characteristic p, and therefore

$$\sigma_p(\alpha) = f(\zeta_m^p) \equiv f(\zeta_m)^p = \alpha^p \pmod{\mathfrak{P}}.$$

Since $N((p)) = p$, this is exactly the defining Frobenius congruence. Hence

$$\mathrm{Frob}_p(\zeta_m) = \zeta_m^p.$$

and hence

$$\boxed{\left(\frac{L/\mathbb{Q}}{p}\right) = p \pmod{m}}$$

in $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^*$.

By Proposition 9.1.7, the residue degree of a prime above p is

$$f = \mathrm{ord}_m(p),$$

the multiplicative order of p in $(\mathbb{Z}/m\mathbb{Z})^*$. In particular,

$$p \text{ splits completely in } \mathbb{Q}(\zeta_m) \quad \Longleftrightarrow \quad p \equiv 1 \pmod{m}.$$

Example 9.2.3. *Take $m = 5$. Then*

$$\mathrm{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong (\mathbb{Z}/5\mathbb{Z})^* \cong \mathbb{Z}_4.$$

For a rational prime $p \neq 5$:

- *if $p \equiv 1 \pmod 5$, then $f = 1$ and p splits completely;*
- *if $p \equiv 4 \pmod 5$, then $f = 2$, so (p) is a product of two prime ideals, each with residue degree 2;*
- *if $p \equiv 2$ or $3 \pmod 5$, then $f = 4$, so (p) remains prime.*

Thus the congruence class of p modulo 5 determines the factorization of (p) in $\mathbb{Z}[\zeta_5]$.

We end with an example where $L : K$ is unramified for all prime ideals $P \lhd \mathcal{O}_K$. This will play an important role in Hilbert Class Field Theory:

Example 9.2.4. *Let $K = \mathbb{Q}(\sqrt{-5})$, $L = K(i)$, and write*

$$\mathrm{Gal}(L/K) = \{1, \sigma\}, \qquad \sigma(i) = -i.$$

We first check that $L : K$ is unramified. Since $m_i(x) = x^2 + 1$, whose discriminant is -4, every prime $P \nmid 2$ is unramified in $L : K$. Thus the only prime that still needs to be considered is

$$P_2 = (2, 1 + \sqrt{-5}),$$

the unique prime of K above 2.

To study P_2, observe that

$$\sqrt{5} = i\sqrt{-5} \in L,$$

so

$$L = K(\sqrt{5}) = K\left(\frac{1 + \sqrt{5}}{2}\right).$$

Put

$$\eta = \frac{1 + \sqrt{5}}{2}.$$

Then η satisfies

$$f(x) = x^2 - x - 1,$$

whose discriminant is 5. Since $P_2 \nmid 5$, the prime P_2 is unramified in $L : K$.

Moreover,

$$\mathcal{O}_K/P_2 \cong \mathbb{F}_2,$$

and modulo P_2 we have

$$f(x) \equiv x^2 + x + 1.$$

This polynomial has no root in $\mathbb{F}_2$, so it is irreducible. Hence P_2 is inert in the quadratic extension $L : K$. Thus $L : K$ is unramified at every finite prime.

Now we compute $\left(\frac{L/K}{P}\right)$ for all such P: For $P = P_2$, since it is inert in L (and does not split), one has

$$\left(\frac{L/K}{P_2}\right) = \sigma.$$

Now let $P \nmid 2$, and let $\mathfrak{P}$ be a prime of L above P. By the defining property of the Frobenius element,

$$\mathrm{Frob}_P(i) \equiv i^{N(P)} \pmod{\mathfrak{P}}.$$

Since $P \nmid 2$, the elements i and $-i$ are distinct modulo $\mathfrak{P}$. Also $N(P)$ is odd, and therefore

$$i^{N(P)} = \begin{cases} i, & N(P) \equiv 1 \pmod 4, \\ -i, & N(P) \equiv 3 \pmod 4. \end{cases}$$

It follows that

$$\boxed{\left(\frac{L/K}{P}\right) = \begin{cases} \sigma, & P = P_2 \text{ or } N(P) \equiv 3 \pmod 4, \\ 1, & N(P) \equiv 1 \pmod 4. \end{cases}}$$

Thus we have computed the Frobenius element for every prime of K.

9.3 The Artin Map

Let $L : K$ be a finite abelian extension, and let S be the finite set of finite prime ideals of $\mathcal{O}_K$ which ramify in L. For every $P \notin S$, the Artin symbol

$$\left(\frac{L/K}{P}\right) = \mathrm{Frob}_P$$

is defined.

At this stage, we do not yet need fractional ideals. Define $I_K(S)$ to be the free abelian group generated by the prime ideals $P \notin S$. Thus an element of $I_K(S)$ is a formal finite product

$$\mathfrak{a} = \prod_{P \notin S} P^{n_P}, \qquad n_P \in \mathbb{Z},$$

with all but finitely many n_P equal to zero. Multiplication is given by adding the exponents. Because the factorization is unique, we may extend the Artin symbol multiplicatively from prime ideals to all elements of $I_K(S)$.

Definition 9.3.1. *The* **Artin map** *for $L : K$ is the homomorphism*

$$\Theta_{L/K} : I_K(S) \longrightarrow \mathrm{Gal}(L/K)$$

defined by

$$\boxed{\Theta_{L/K}\left(\prod_{P\notin S} P^{n_P}\right) = \prod_{P\notin S}\left(\frac{L/K}{P}\right)^{n_P}.}$$

The product is finite, and it is unambiguous because $\mathrm{Gal}(L/K)$ is abelian. Thus the Artin map packages all Frobenius elements into one homomorphism.

A main problem of class field theory is to (1) understand the domain $I_K(S)$ of $\Theta_{L/K}$ in terms of ideal class group C_K; and (2) describe the kernel of $\Theta_{L/K}$.

Example 9.3.2. *Let*

$$L = \mathbb{Q}(i), \qquad K = \mathbb{Q}.$$

Then

$$\mathrm{Gal}(L/K) = \{1, \sigma\}, \qquad \sigma(i) = -i.$$

The only ramified rational prime is 2, so $I_{\mathbb{Q}}(\{2\})$ is the free abelian group generated by the odd rational prime ideals $(p) \lhd K$. We have seen in Section 9.2 that

$$\mathrm{Frob}_p = \begin{cases} 1, & p \equiv 1 \pmod 4, \\ \sigma, & p \equiv 3 \pmod 4. \end{cases}$$

Hence, if n is an odd positive integer with prime factorization $n = \prod_{p|n} p^{v_p(n)}$, then

$$\boxed{\Theta_{L/K}\left((n)\right) = \Theta_{L/K}\left(\prod_{p|n}(p)^{v_p(n)}\right) = \prod_{p|n}\mathrm{Frob}_p^{v_p(n)} = \begin{cases} 1, & n \equiv 1 \pmod 4, \\ \sigma, & n \equiv 3 \pmod 4. \end{cases}}$$

In particular,

$$\Theta_{L/K}((3)) = \sigma \neq 1.$$

Here is an example where $L : K$ is unramified:

Example 9.3.3. *Let $K = \mathbb{Q}(\sqrt{-5})$, $L = \mathbb{Q}(i, \sqrt{5})$ with $\mathrm{Gal}(L/K) = \{1, \sigma\}$, and*

$$\mathrm{Frob}_P = \begin{cases} \sigma, & P = P_2 \text{ or } N(P) \equiv 3 \pmod 4, \\ 1, & \text{otherwise}, \end{cases}$$

We claim that every nonzero principal integral ideal of K maps to the identity under the Artin map. Let

$$\alpha = a + b\sqrt{-5} \in \mathcal{O}_K, \qquad \alpha \neq 0,$$

and put

$$n = N((\alpha)) = |N_{K/\mathbb{Q}}(\alpha)| = a^2 + 5b^2.$$

Write

$$(\alpha) = P_2^{e_2} \prod_{P \neq P_2} P^{e_P}.$$

By the Frobenius computation above,

$$\Theta_{L/K}((\alpha)) = \sigma^E, \qquad E \equiv e_2 + \sum_{\substack{P \neq P_2 \\ N(P) \equiv 3 \pmod 4}} e_P \pmod 2.$$

We want to show that E is even. To do so, write

$$a = 2^r a_0, \qquad b = 2^r b_0,$$

where a_0, b_0 are not both even. Since

$$(\alpha) = (2)^r (a_0 + b_0\sqrt{-5}) = P_2^{2r}(a_0 + b_0\sqrt{-5}),$$

the factor P_2^{2r} does not affect E modulo 2. Thus we may assume that a and b are not both even.

Suppose first that one of a, b is even and the other is odd. Then

$$n = a^2 + 5b^2 \equiv 1 \pmod 4.$$

In particular $e_2 = 0$. Since

$$n = \prod_{P \neq P_2} N(P)^{e_P},$$

and every odd $N(P)$ is congruent to 1 or 3 modulo 4, the congruence $n \equiv 1 \pmod 4$ implies that

$$\sum_{\substack{P \neq P_2 \\ N(P) \equiv 3 \pmod 4}} e_P$$

is even. Hence E is even.

Now suppose that a and b are both odd. Then

$$n = a^2 + 5b^2 \equiv 6 \pmod 8.$$

Thus

$$e_2 = 1, \qquad \frac{n}{2} \equiv 3 \pmod 4.$$

But

$$\frac{n}{2} = \prod_{P \neq P_2} N(P)^{e_P},$$

so the number of factors $N(P) \equiv 3 \pmod 4$*, counted with multiplicity, is odd. Therefore*

$$E \equiv 1 + 1 \equiv 0 \pmod 2.$$

Thus in every case

$$\boxed{\Theta_{L/K}((\alpha)) = 1.}$$

i.e. every principal ideal lies in the kernel of the Artin map. Later we will see that this is precisely what should happen: L *is the Hilbert class field of* $K = \mathbb{Q}(\sqrt{-5})$*.*

9.4 Fractional Ideals and the Ideal Class Group

To relate the Artin map $\Theta_{L/K}$ to the ideal class group C_K, we now give a concrete interpretation of the group $I_K(S)$ when $S = \emptyset$, i.e. all ideals $\mathfrak{P} \lhd \mathcal{O}_L$ over a prime ideal $P \lhd \mathcal{O}_K$ are unramified. In such a case, we call $L : K$ an **unramified** extension.

Recall that in Chapter 6, C_K was defined using equivalence classes of nonzero integral ideals $I, J \lhd \mathcal{O}_K$, where

$$I \sim J$$

if there exists nonzero $\alpha, \beta \in \mathcal{O}_K$ such that

$$(\alpha)I = (\beta)J.$$

We now introduce fractional ideals.

Definition 9.4.1. *A* **fractional ideal** *of* K *is a nonzero* $\mathcal{O}_K$*-submodule*

$$\mathfrak{a} \subseteq K$$

such that there exists a nonzero $d \in \mathcal{O}_K$ *with*

$$d\mathfrak{a} \subseteq \mathcal{O}_K.$$

Thus every nonzero ideal $I \lhd \mathcal{O}_K$ is a fractional ideal. Conversely, if $\mathfrak{a}$ is a fractional ideal and $d\mathfrak{a} \subseteq \mathcal{O}_K$, then

$$I := d\mathfrak{a}$$

is an integral ideal and

$$\mathfrak{a} = \frac{1}{d} I.$$

For a nonzero ideal $J \subseteq \mathcal{O}_K$, define its **inverse fractional ideal** by

$$J^{-1} := \{x \in K : xJ \subseteq \mathcal{O}_K\}.$$

In particular, for a principal ideal (d),

$$(d)^{-1} = \frac{1}{d}\mathcal{O}_K.$$

Thus, if $\mathfrak{a} = \frac{1}{d}I$, then

$$\mathfrak{a} = (d)^{-1}I.$$

Now factor the two integral ideals:

$$I = \prod_P P^{a_P}, \qquad (d) = \prod_P P^{b_P},$$

with $a_P, b_P \geq 0$. Hence

$$\mathfrak{a} = (d)^{-1}I = \prod_P P^{a_P - b_P}.$$

Therefore every nonzero fractional ideal has a unique expression

$$\mathfrak{a} = \prod_P P^{n_P}, \qquad n_P \in \mathbb{Z},$$

where all but finitely many n_P are zero.

Definition 9.4.2. *Let*

$$I_K := \left\{ \prod_P P^{n_P} \;\middle|\; n_P \in \mathbb{Z},\ n_P = 0 \text{ for all but finitely many } P \right\}$$

denote the abelian group of all nonzero fractional ideals of K, with multiplication of ideals as the group operation.

Recall the definition of $I_K(S)$. In the case when $L : K$ is unramified, one therefore has

$$I_K(S) = I_K(\emptyset) = I_K.$$

We next single out the principal fractional ideals.

Definition 9.4.3. *Let*

$$P_K := \{(\alpha) = \alpha\mathcal{O}_K \mid \alpha \in K^\times\}.$$

Then P_K is a subgroup of I_K, called the group of **principal fractional ideals**.

The quotient I_K/P_K is exactly the ideal class group defined in Chapter 6.

Proposition 9.4.4. *There is a natural isomorphism*

$$\boxed{C_K \cong I_K/P_K.}$$

Proof. Define

$$\Phi : C_K \longrightarrow I_K/P_K$$

by

$$\Phi([I]) = IP_K,$$

where $I \lhd \mathcal{O}_K$ is a nonzero integral ideal.

We first check that Φ is well-defined. Suppose

$$I \sim J.$$

By definition, there exist nonzero $\alpha, \beta \in \mathcal{O}_K$ such that

$$(\alpha)I = (\beta)J.$$

Therefore, as fractional ideals,

$$I = \left(\frac{\beta}{\alpha}\right) J.$$

Hence

$$IP_K = JP_K.$$

The map Φ is surjective. Indeed, let $\mathfrak{a}$ be any fractional ideal. Choose $d \in \mathcal{O}_K \setminus \{0\}$ such that

$$I := d\mathfrak{a} \subseteq \mathcal{O}_K.$$

Then I is an integral ideal and

$$\mathfrak{a} = \frac{1}{d} I.$$

Since $(1/d)$ is principal,

$$\mathfrak{a}P_K = IP_K.$$

Finally, suppose

$$\Phi([I]) = \Phi([J]).$$

Then

$$IP_K = JP_K,$$

so

$$I = (\gamma)J$$

for some $\gamma \in K^\times$. Write

$$\gamma = \frac{\alpha}{\beta}$$

for some nonzero $\alpha, \beta \in \mathcal{O}_K$. Then

$$(\beta)I = (\alpha)J.$$

Hence $I \sim J$, so

$$[I] = [J].$$

Thus Φ is injective.

Therefore $C_K \cong I_K/P_K$. □

9.5 The Hilbert Class Field

We now consider the first case in class field theory - we construct an unramified field extension $H_K : K$ (Hilbert class field), so that $I_K(S) = I_K$, and the kernel of the Artin map is precisely P_K.

Theorem 9.5.1 (Hilbert Class Field Theorem)**.** *For every number field K, there exists a finite abelian Galois extension*

$$H_K : K,$$

called the **Hilbert class field** *of K, with the following properties:*

1. *every nonzero prime ideal $P \subseteq \mathcal{O}_K$ is unramified in H_K (and hence $I_K(S) = I_K$);*
2. *for each real embedding $\sigma : K \hookrightarrow \mathbb{R}$, (see Section 8.2), there are exactly $[H_K : K]$ embeddings $\widetilde{\sigma} : H_K \hookrightarrow \mathbb{R}$ extending σ;*
3. *every finite abelian extension of K satisfying (1) and (2) is contained in H_K;*
4. *the Artin map induces an isomorphism*

$$\boxed{C_K \;\cong\; \mathrm{Gal}(H_K/K)}, \qquad [P] \longmapsto \mathrm{Frob}_P\,.$$

We will not prove the Hilbert class field theorem. Its proof is one of the main results of global class field theory. What matters here is the precise relation between the theorem and the quotient $C_K = I_K/P_K$.

Corollary 9.5.2. *For the Hilbert class field,*

$$\boxed{\ker \Theta_{H_K/K} = P_K.}$$

Equivalently, a fractional ideal has trivial Artin symbol in H_K if and only if it is principal.

Proof. The Artin map

$$\Theta_{H_K/K} : I_K \longrightarrow \mathrm{Gal}(H_K/K)$$

induces, by Theorem 9.5.1, an isomorphism

$$\theta_{H_K/K} : I_K/P_K \xrightarrow{\sim} \mathrm{Gal}(H_K/K).$$

Therefore its kernel is exactly P_K. □

Corollary 9.5.3. *The degree of the Hilbert class field is the class number:*

$$\boxed{[H_K : K] = h_K.}$$

Proof. By Theorem 9.5.1,

$$\mathrm{Gal}(H_K/K) \cong C_K.$$

Hence $[H_K : K] = |\,\mathrm{Gal}(H_K/K)| = |C_K| = h_K$. □

Thus the class number has the interpretation

$$\boxed{h_K = \text{degree of the maximal everywhere-unramified abelian extension of } K.}$$

Corollary 9.5.4. *Let P be a prime ideal of $\mathcal{O}_K$. Then*

$$\boxed{P \text{ is principal} \iff P \text{ splits completely in } H_K.}$$

Proof. Under the Artin isomorphism,

$$[P] \longmapsto \mathrm{Frob}_P\,.$$

Now

$$P \text{ is principal} \iff [P] = e,$$

while Proposition 9.1.7 gives

$$P \text{ splits completely in } H_K \iff \mathrm{Frob}_P = id.$$

The two conditions are therefore equivalent. □

Corollary 9.5.5. *The following are equivalent:*

1. $C_K = \{e\}$;
2. $h_K = 1$;
3. $\mathcal{O}_K$ *is a PID;*
4. $H_K = K$.

Example 9.5.6. *Let $K = \mathbb{Q}(\sqrt{-5})$, $L = \mathbb{Q}(i, \sqrt{5})$. Earlier we computed*

$$C_K \cong \mathbb{Z}_2, \qquad h_K = 2.$$

In Example 9.2.4 we showed directly that $L : K$ is a quadratic extension unramified at every finite prime of K. Since K is imaginary quadratic, it has no real places, so $L : K$ is unramified everywhere. Hence, by the Hilbert class field theorem,

$$L \subseteq H_K.$$

But $[L : K] = 2 = h_K = [H_K : K]$, *and therefore*

$$\boxed{H_K = L = \mathbb{Q}(i, \sqrt{5}).}$$

Consequently, the Artin map induces an isomorphism

$$\theta_{L/K} : I_K/P_K \cong C_K \xrightarrow{\sim} \mathrm{Gal}(H_K/K).$$

(In Example 9.3.3, we checked directly that principal ideals lie in $\ker \Theta_{L/K}$*.) More precisely, the nonprincipal prime ideal* $[P_2] = [(2, 1 + \sqrt{-5})]$ *represents the nontrivial class in* C_K*, and one has*

$$\theta_{L/K}([P_2]) = \sigma.$$

The Fundamental Theorem of Galois Theory gives a further interpretation of the subgroup structure of C_K. If

$$K \subseteq M \subseteq H_K$$

and $A \leq C_K$ corresponds to M under $C_K \cong \mathrm{Gal}(H_K/K)$, then

$$\mathrm{Gal}(M/K) \cong C_K/A.$$

Thus subgroups of the ideal class group describe the intermediate abelian extensions inside the Hilbert class field.

Remark 9.5.7. *Sections 9.1–9.5 form the unramified part of the story. The remaining sections explain how congruence conditions enlarge this picture to arbitrary finite abelian extensions.*

9.6 Ray Class Groups

The Hilbert class field is governed by the quotient I_K/P_K. For a ramified abelian extension, the example $\mathbb{Q}(i) : \mathbb{Q}$ above shows that P_K is too large to be the kernel of the Artin map. The correction is to keep only those principal ideals which admit generators satisfying specified congruence and sign conditions.

Definition 9.6.1. *A* **modulus** *of* K *is a formal product*

$$\mathfrak{m} = \mathfrak{m}_0 \mathfrak{m}_\infty,$$

where

$$\mathfrak{m}_0 = \prod_P P^{n_P}, \qquad n_P \geq 1,$$

is a nonzero integral ideal with only finitely many primes occurring, and $\mathfrak{m}_\infty$ *is a product of distinct real places of* K *(here real place means an embedding* $K \hookrightarrow \mathbb{R}$*).*

The finite part $\mathfrak{m}_0$ records finite primes where ramification is allowed. The infinite part $\mathfrak{m}_\infty$ records real places which are allowed to become complex.

Let

$$I_K(\mathfrak{m}) = \left\{ \prod_P P^{n_P} \in I_K \;\middle|\; n_P = 0 \text{ whenever } P \mid \mathfrak{m}_0 \right\}.$$

Thus $I_K(\mathfrak{m})$ is the group of fractional ideals prime to the finite part $\mathfrak{m}_0$.

Definition 9.6.2. *Let $\alpha \in K^*$. We write*

$$\alpha \equiv 1 \pmod{\mathfrak{m}}$$

if the following two conditions hold:

1. *for every $P^{n_P} \mid \mathfrak{m}_0$, one has*
$$v_P(\alpha - 1) \geq n_P;$$
2. *for every real place σ occurring in $\mathfrak{m}_\infty$, one has*
$$\sigma(\alpha) > 0.$$

Here v_P denotes the exponent of P in the prime-ideal factorization of a principal fractional ideal. Equivalently, when $\alpha = a/b$ with $a, b \in \mathcal{O}_K$ both coprime to $\mathfrak{m}_0$, the first condition says

$$a \equiv b \pmod{P^{n_P}}$$

for every $P^{n_P} \mid \mathfrak{m}_0$.

Let $P_{K,\mathfrak{m},1}$ be the subgroup of $I_K(\mathfrak{m})$ consisting of principal fractional ideals (α) with

$$\alpha \equiv 1 \pmod{\mathfrak{m}}.$$

Notice that the congruence condition forces $v_P(\alpha) = 0$ for every $P \mid \mathfrak{m}_0$, so (α) is indeed prime to $\mathfrak{m}_0$.

Definition 9.6.3. *The* **ray class group modulo $\mathfrak{m}$** *is*

$$\boxed{C_K(\mathfrak{m}) := I_K(\mathfrak{m})/P_{K,\mathfrak{m},1}.}$$

Example 9.6.4. *Let $\mathfrak{m} = 1$ be the trivial modulus. Then*

$$I_K(1) = I_K, \qquad P_{K,1,1} = P_K,$$

and therefore

$$\boxed{C_K(1) = C_K.}$$

Thus the ordinary ideal class group is the first ray class group, with no congruence or sign conditions imposed.

Example 9.6.5 (Ray class groups of $\mathbb{Q}$)**.** *Let $K = \mathbb{Q}$ and take the modulus*

$$\mathfrak{m} = m\infty,$$

where ∞ denotes the unique real place of $\mathbb{Q}$. Every fractional ideal of $\mathbb{Z}$ is principal. If $\mathfrak{a} \in I_{\mathbb{Q}}(m\infty)$, then among the two generators $\pm r$ of $\mathfrak{a}$ there is a unique positive generator $r > 0$. Write

$$r = \frac{a}{b}, \qquad a, b \in \mathbb{Z}_{>0}, \qquad \gcd(ab, m) = 1.$$

Define

$$\Phi : I_{\mathbb{Q}}(m\infty) \longrightarrow (\mathbb{Z}/m\mathbb{Z})^*$$

by

$$\Phi(\mathfrak{a}) = ab^{-1} \pmod{m}.$$

This is well-defined. Indeed, if $a/b = a'/b'$ and all four integers are coprime to m, then $ab' = a'b$, so after reducing modulo m and multiplying by $b^{-1}(b')^{-1}$ we get

$$ab^{-1} \equiv a'(b')^{-1} \pmod{m}.$$

It is clearly a surjective homomorphism.

Now $\mathfrak{a} = (a/b)$ lies in the kernel precisely when $a \equiv b \pmod{m}$. Because $a/b > 0$, this is exactly the condition

$$\frac{a}{b} \equiv 1 \pmod{m\infty}.$$

Hence

$$\ker \Phi = P_{\mathbb{Q},m\infty,1}.$$

By the First Isomorphism Theorem,

$$\boxed{C_{\mathbb{Q}}(m\infty) \cong (\mathbb{Z}/m\mathbb{Z})^*.}$$

Note that the infinite place is essential in the above example. Without the positivity condition, the same ideal has generators r and $-r$, whose residue classes differ by -1 modulo m; one would instead identify these two classes.

9.7 Artin Reciprocity and the Existence Theorem

By the example in the previous section,

$$C_{\mathbb{Q}}(m\infty) \cong (\mathbb{Z}/m\mathbb{Z})^* \cong \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}).$$

This is no coincidence – indeed this is a special case of the following:

Theorem 9.7.1 (Artin Reciprocity)**.** *Let $L : K$ be a finite abelian extension. Then there exists a modulus $\mathfrak{m}$ containing all finite and infinite places which ramify in L such that the Artin map*

$$\Theta_{L/K} : I_K(\mathfrak{m}) \longrightarrow \mathrm{Gal}(L/K)$$

is surjective and

$$P_{K,\mathfrak{m},1} \subseteq \ker \Theta_{L/K}.$$

Consequently, $\Theta_{L/K}$ factors through the ray class group and gives a surjection

$$\boxed{C_K(\mathfrak{m}) \twoheadrightarrow \mathrm{Gal}(L/K).}$$

Among all moduli through which the Artin map factors, there is a unique minimal one under divisibility. It is called the **conductor** of $L : K$.

The converse to Artin reciprocity is the existence theorem. It is most useful in the following subgroup form.

Theorem 9.7.2 (Existence Theorem)**.** *Let $\mathfrak{m}$ be a modulus of K, and let H be a subgroup satisfying*

$$P_{K,\mathfrak{m},1} \subseteq H \subseteq I_K(\mathfrak{m}).$$

Then there exists a unique finite abelian extension $L_H : K$ inside a fixed algebraic closure, with conductor dividing $\mathfrak{m}$, for which the Artin map induces an isomorphism

$$\boxed{I_K(\mathfrak{m})/H \;\cong\; \mathrm{Gal}(L_H/K).}$$

In particular, if one takes

$$H = P_{K,\mathfrak{m},1}$$

then $K_{\mathfrak{m}} := L_{P_{K,\mathfrak{m},1}}$ is called the **ray class field***, and one has:*

$$C_K(\mathfrak{m}) \cong \mathrm{Gal}(K_{\mathfrak{m}}/K).$$

Example 9.7.3. *For the trivial modulus $\mathfrak{m} = 1$, the subgroup on the ideal side is $P_{K,1,1} = P_K$. Hence*

$$K_1 = H_K, \qquad C_K = C_K(1) \cong \mathrm{Gal}(H_K/K).$$

Thus the Hilbert class field theorem is the unramified special case of the general theory.

Example 9.7.4. *Recall by Example 9.6.5 that*

$$C_{\mathbb{Q}}(m\infty) \cong (\mathbb{Z}/m\mathbb{Z})^* \cong \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$$

We study $C_{\mathbb{Q}}(m\infty) \to \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ directly – recall that for $\mathfrak{a} = (a/b) \in I_{\mathbb{Q}}(m\infty)$, the Artin map corresponds to

$$\mathfrak{a} \longmapsto ab^{-1} \pmod{m}.$$

This is exactly the isomorphism constructed in Example 9.6.5, whose kernel is $P_{\mathbb{Q},m\infty,1}$. Therefore, this verifies the isomorphism

$$\boxed{K_{m\infty} = \mathbb{Q}(\zeta_m).}$$

in Theorem 9.7.2 in this setup.

As a consequence, the class-field description over $\mathbb{Q}$ becomes the following classical theorem.

Theorem 9.7.5 (Kronecker–Weber Theorem)**.** *Every finite abelian extension $L : \mathbb{Q}$ is contained in a cyclotomic field:*

$$L \subseteq \mathbb{Q}(\zeta_m)$$

for some $m \geq 1$.

Proof. Let $L : \mathbb{Q}$ be a finite abelian extension. By Artin reciprocity (Theorem 9.7.1), there exists a modulus $\mathfrak{m}$ such that

$$P_{\mathbb{Q},\mathfrak{m},1} \subseteq \ker \Theta_{L/\mathbb{Q}}.$$

Since $\mathbb{Q}$ has only one real place, after enlarging the modulus (which makes $P_{\mathbb{Q},\mathfrak{m},1}$ smaller) if necessary we may take

$$\mathfrak{m} = m\infty$$

for some positive integer m. Hence

$$P_{\mathbb{Q},m\infty,1} \subseteq H := \ker \Theta_{L/\mathbb{Q}}.$$

By the existence theorem (Theorem 9.7.2), the subgroup H corresponds to the extension $L : \mathbb{Q}$, while $P_{\mathbb{Q},m\infty,1}$ corresponds to the full ray class field $K_{m\infty}$. Since the correspondence between subgroups and class fields is inclusion-reversing,

$$P_{\mathbb{Q},m\infty,1} \subseteq H \qquad \Longrightarrow \qquad L \subseteq K_{m\infty}.$$

By the preceding example, $K_{m\infty} = \mathbb{Q}(\zeta_m)$. Therefore

$$L \subseteq \mathbb{Q}(\zeta_m)$$

as stated in the theorem. □

Theorem 9.7.6 (Quadratic Reciprocity)**.** *Let p and q be distinct odd rational primes. Then*

$$\boxed{\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.}$$

Proof. Put

$$q^* = (-1)^{\frac{q-1}{2}} q, \qquad L = \mathbb{Q}(\sqrt{q^*}).$$

For $p \neq q$, Example 9.2.1 gives

$$\mathrm{Frob}_p(\sqrt{q^*}) = \left(\frac{q^*}{p}\right)\sqrt{q^*}. \tag{1}$$

Now we apply Artin reciprocity. Note that

$$\mathrm{disc}(L) = q^*,$$

so the conductor of $L : \mathbb{Q}$ divides $q\infty$. Hence Theorem 9.7.1 implies that

$$P_{\mathbb{Q},q\infty,1} \subseteq \ker \Theta_{L/\mathbb{Q}},$$

where

$$\Theta_{L/\mathbb{Q}} : I_{\mathbb{Q}}(q\infty) \longrightarrow \mathrm{Gal}(L/\mathbb{Q})$$

is the Artin map. Therefore $\Theta_{L/\mathbb{Q}}$ factors through the quotient

$$C_{\mathbb{Q}}(q\infty) = I_{\mathbb{Q}}(q\infty)/P_{\mathbb{Q},q\infty,1}.$$

That is, there is a surjective homomorphism

$$\theta_{L/\mathbb{Q}} : C_{\mathbb{Q}}(q\infty) \cong (\mathbb{Z}/q\mathbb{Z})^* \longrightarrow \mathrm{Gal}(L/\mathbb{Q}) \cong \{\pm 1\}.$$

Thus the action of Frob_p on $\sqrt{q^*}$ is given by a nontrivial quadratic character of $(\mathbb{Z}/q\mathbb{Z})^*$.

Since $(\mathbb{Z}/q\mathbb{Z})^*$ is cyclic, it has a unique nontrivial quadratic character, namely

$$a \longmapsto \left(\frac{a}{q}\right).$$

Therefore

$$\mathrm{Frob}_p(\sqrt{q^*}) = \left(\frac{p}{q}\right)\sqrt{q^*}. \tag{2}$$

Comparing (1) and (2),

$$\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right).$$

Since

$$q^* = (-1)^{\frac{q-1}{2}} q \qquad \text{and} \qquad \left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}},$$

we obtain

$$\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}} \left(\frac{q}{p}\right),$$

which is quadratic reciprocity. □